Introduction to IoT

By

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Asst Prof, Dept of ECE

IIoT-Syllabus

R25 B.Tech. IT Syllabus ACEEC

IT305PC: INTRODUCTION TO IOT

B.Tech. II Year I Sem.

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Course Objectives:

1. Understand the concepts of Internet of Things and able to build IoT applications
2. Learn the programming and use of Arduino and Raspberry Pi boards.
3. Know about data handling and analytics in SDN.

Course Outcomes: Upon completing this course, the student will be able to:

1. Understand key characteristics, layers, protocols, sensor networks and their role in IoT.
2. Differentiate between IoT and M2M communications and achieve interoperability using Arduino Programming.
3. Apply Python constructs on Raspberry Pi interfacing models and their use cases.
4. Design and implement SDN-based solutions for IoT deployments using Raspberry Pi
5. Apply the knowledge of cloud storage models to select an appropriate model for any IoT application.
6. Analyze different sensor technologies for real world applications through IoT.

IIoT-Syllabus

UNIT- I

Introduction to IoT and Sensor Networks

Introduction to Internet of Things (IoT), Characteristics and Applications of IoT, IoT Architecture and Reference Models (IETF, ITU-T)
Physical Design of IoT: Devices, Gateways, and Data Centers
Functional Blocks of IoT: Sensing, Actuation, Communication, Enabling Technologies: RFID, Wireless Sensor Networks, Nanotechnology
Basics of Networking and Communication Protocols: MQTT, CoAP, ZigBee, HTTP Sensor Networks: Types, Topologies, and Protocols
Introduction to IoT Security and Privacy Fundamentals.

UNIT- II

Machine to Machine (M2M) and Embedded Programming for IoT

Machine-to-Machine Communications Overview, Difference between IoT and M2M, Interoperability in IoT: Standards and Protocols.
Introduction to Arduino Programming for IoT, Integration of Sensors and Actuators with Arduino. Hands-on Exercises: Sensor Data Acquisition and Actuator Control, Basic Communication Protocols Implementation on Arduino, IoT Device Interoperability Challenges and Solutions.

UNIT- III

Raspberry Pi with Python Programming for IoT

Introduction to Python Programming Basics
Overview of Raspberry Pi and Its Role in IoT, Interfacing Raspberry Pi with Sensors and Actuators (UART, SPI, I2C).
Data Acquisition and Local Processing on Raspberry Pi, Sending Data to Cloud Platforms.
Implementation of IoT Projects Using Raspberry Pi
Case Studies: Smart Home Automation, Healthcare Monitoring, Environmental Sensing

UNIT- IV

Network Management and Software Defined Networking (SDN) for IoT

Introduction to Software Defined Networking (SDN), SDN Architecture: Controller, Data Plane, Control Plane, Benefits of SDN in IoT: Flexibility, Centralized Control, Security, Traffic Management, SDN Models: Open SDN, API SDN, Overlay, Hybrid
Data Handling, Analytics, and Security Considerations in SDN for IoT

Network Management Challenges in IoT, Integration of SDN with IoT Networks

UNIT- V

Cloud and Edge Computing Models with IoT Use Cases

Introduction to Cloud Computing and Cloud Storage Models, Edge and Fog Computing Concepts for IoT, Web Servers and Cloud Platforms for IoT (AWS IoT, Azure IoT, etc.), Sensor-Cloud Architecture Data Analytics and Visualization Techniques for IoT Data
IoT Use Cases: Smart Cities, Smart Homes, Connected Vehicles, Smart Grid, Industrial IoT Case Studies: Agriculture, Healthcare, Activity Monitoring.

TEXT BOOKS

1. Pethuru Raj and Anupama C. Raman "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", CRC Press.
2. Vijay Madisetti, Arshdeep Bahga, "Internet of Things: A Hands-On Approach"

REFERENCE BOOKS

1. Terokarvinen, kemo, karvinen and villeyvaltokari, "Make sensors": 1st edition,2014, Maker media.
2. Waltenegus Dargie, Christian Poellabauer, Fundamentals of Wireless Sensor Networks: Theory and Practice.
3. Charles Bell, Beginning Sensor networks with Arduino and Raspberry Pi, 2013, Apress.
4. Fei Hu, Security and Privacy in Internet of Things (IoTs), CRC Press, Taylor & Francis Group.
5. S. Sahoo, S. Sahoo, S. Mishra, Software-Defined Networking for Future Internet Technology: Concepts and Applications, Routledge.
6. Honbo Zhou, The Internet of Things in the Cloud: A Middleware Perspective. 2012, CRC Press (Taylor & Francis).

UNIT- I
Introduction to IoT and Sensor Networks

- Introduction to Internet of Things (IoT),
- Characteristics and Applications of IoT,
- IoT Architecture and Reference Models (IETF, ITU-T)
- Physical Design of IoT: Devices, Gateways, and Data Centers
- Functional Blocks of IoT: Sensing, Actuation, Communication,
- Enabling Technologies: RFID, Wireless Sensor Networks, Nanotechnology
- Basics of Networking and
- Communication Protocols: MQTT, CoAP, ZigBee, HTTP
- Sensor Networks: Types, Topologies, and Protocols
- Introduction to IoT Security and Privacy Fundamentals.

Evolution of IoT
or
A Brief History of IoT



Evolution of IoT



Internet of Things—a comprehensive timeline



1989
The first IoT device is created



1993
The first online webcam is used at Cambridge



1994
Steve Mann creates WearCam



1999
Kevin Ashton coins the term Internet of Things



2008
The number of connected devices overtakes the number of people in the world and IoT is 'born'



2007
First iPhone is launched



2005
UN publishes its first report on the Internet of Things



2000
LG announces the first smart refrigerator



2009
The original Fitbit activity tracker is released



2011
IIoT comes into being



2014
Seoul becomes the world's first smart city



2015
IoT goes Mobile with smartphones



2022
World Economic Forum names IoT as one of the three most impactful technological advancements



2021
More than 10 billion active IoT devices active



2020
The number of IoT device connections increased more than 50% of the active connected devices



2016
AWS IoT core is launched

Factors for IoT Growth

- an expansion of networking capabilities
- the development of large-scale data analytics tools to aid in managing and interpreting data from the connected devices
- the establishment of new standards and protocols that simplify the interaction of hardware and software from multiple vendors
- the tremendous growth of the Cloud storage system
  - low-cost
  - always-on storage
  - highly available
  - easy computations from IoT devices to Cloud servers

Technical definition of IoT

A Dynamic global network infrastructure with self configurable capabilities based on standard / interoperable communication protocols, **where physical/virtual things have identities and use intelligent interfaces and integrated into information network**, **Often communicate data associated with users and their environments**.

Characteristics of IoT

- Dynamic & Self Adopting
- Self Configuring
- Interoperable Communication Protocols
- Unique Identity
- Integrated into Information Network

1. Dynamic and self adapting

- IoT devices and systems may have the capability to **dynamically adapt** with the **changing** **contexts and take actions**.
- Example – Surveillance system – can adapt normal (day) and night time modes .

1. Dynamic and self adapting

- Cameras should switch from lower resolution to higher resolution modes when any motion is detected and alert nearby by cameras.
- In this surveillance system is adapting itself based on the context and changing (dynamic) conditions .



1. Dynamic and self adapting

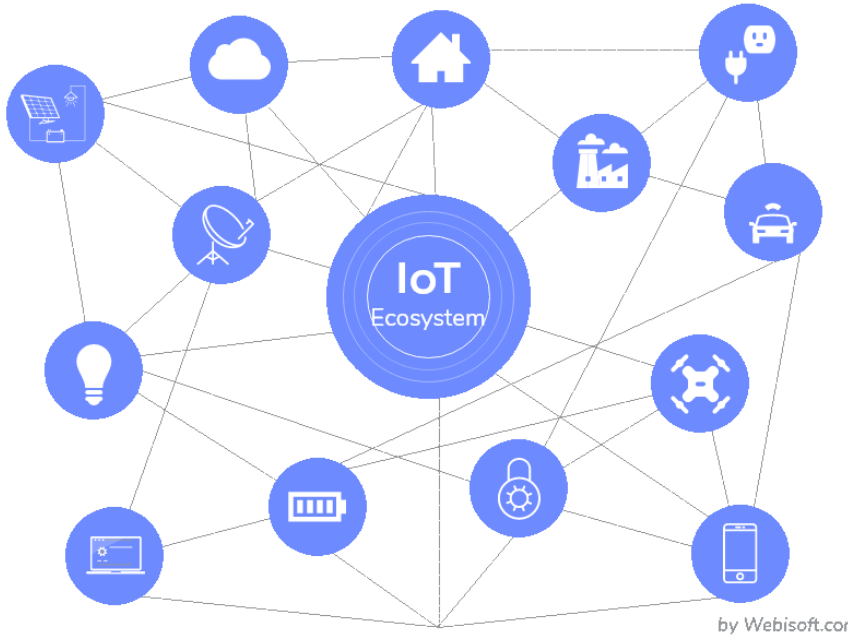


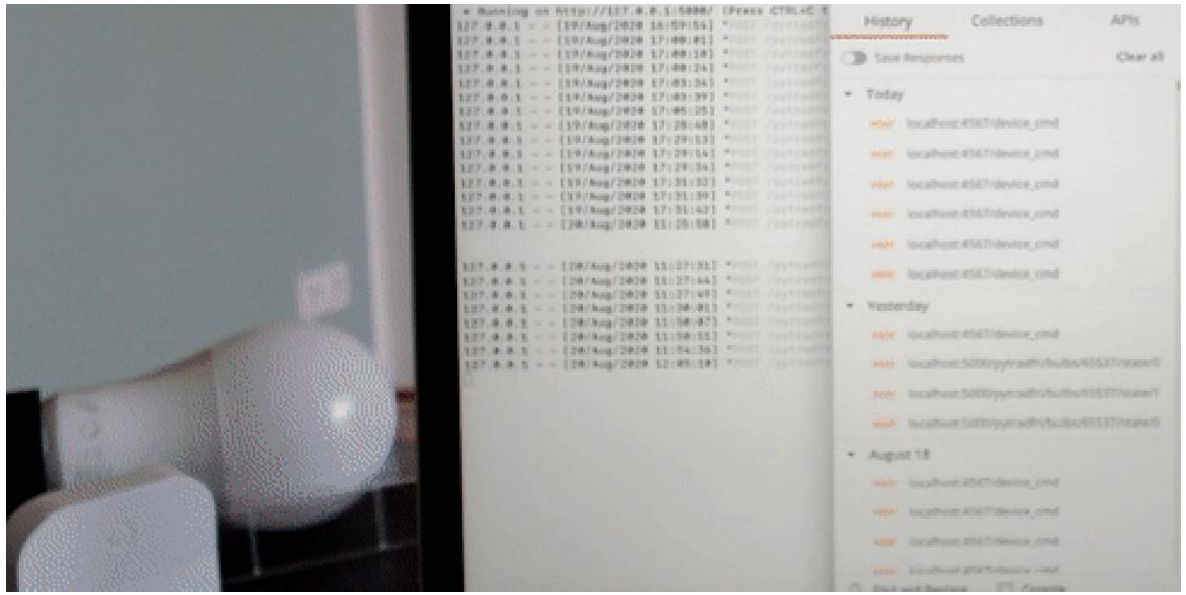
2. Self configuring

- IoT devices may have the **ability** to **configure** themselves
- Allowing a large no of devices to work together to provide certain functionality, Ex: Weather monitoring

2. Self configuring

- Setup networking
- Fetch latest software upgrades with minimal user intervention.

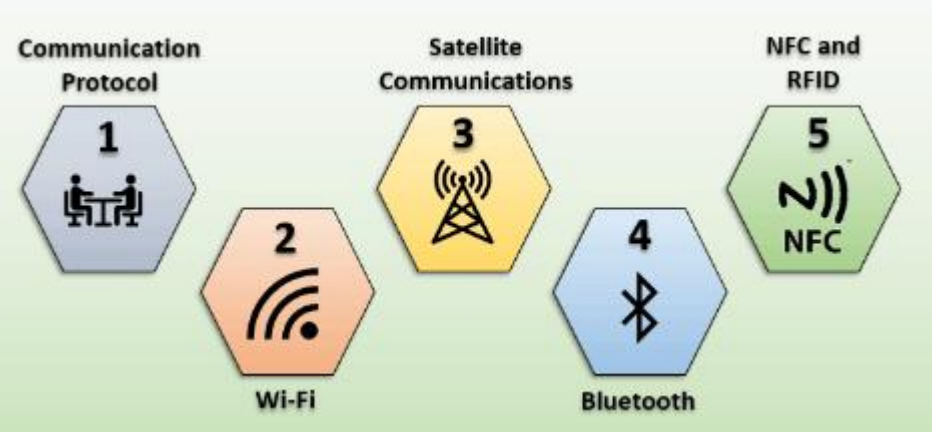




3. Interoperable Communication Protocols

-Speaking the same language

- IoT Devices may support a number of interoperable communication protocols
- Heterogeneous devices, systems, platforms or networks





4.Unique Identity

- Giving an identity to the device is an essential aspect of loT.

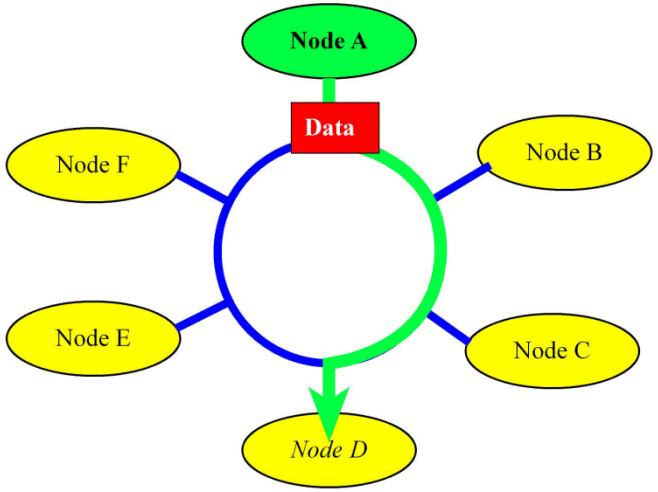
- **Identity helps** to **differentiate** between various internet devices and select the device we want to send the command.

- **It is essential** to give **a unique identity** to every device so **that we can set up passwords or other security means**.

4.Unique Identity

- Each IoT device has a unique and a unique identifier as an IP address or URL .

- For example, fingerprints, face recognition IP addresses, and Face lock systems are several means of security given to the different identified devices to protect them.



5. Integrated into Information Network

- IoT Devices are usually integrated into the information network that allows them to communicate and exchange data with other devices and systems.
- IoT devices can be dynamically discovered in the network by other devices and /or network, and have capability to describe themselves to other devices or users.

5. Integrated into Information Network

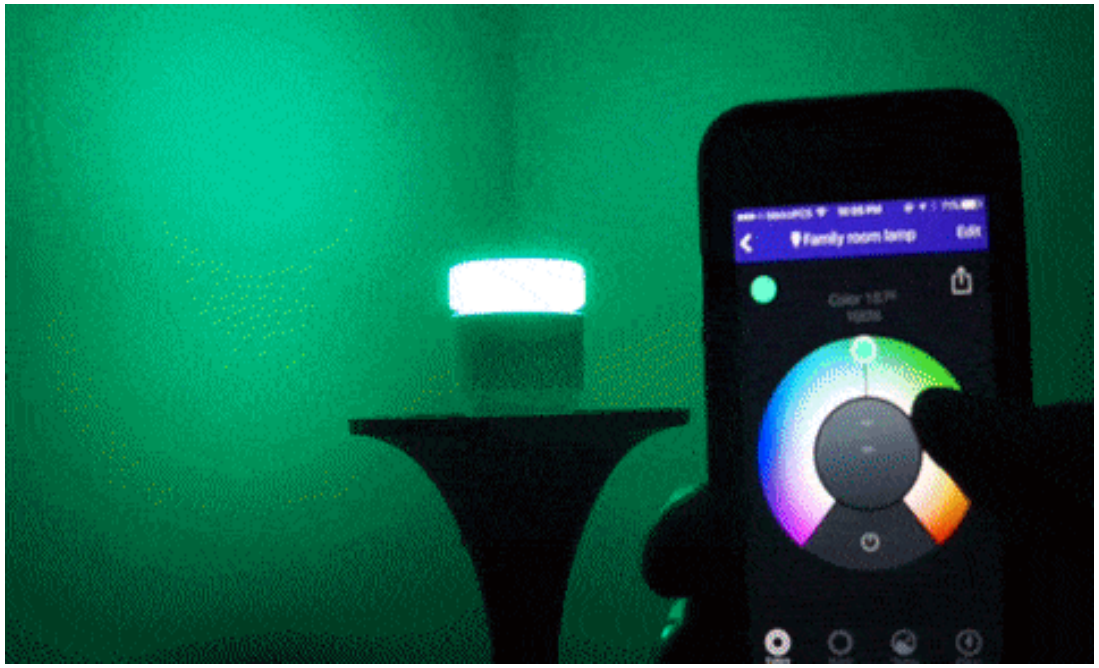
- Example: A weather monitoring node can describe its monitoring capabilities to another connected node.
- They can communicate and exchange data
- Data from large number of connected weather monitoring IoT nodes can be aggregated and analyzed to predict the weather.

Application areas of IoT

- **Home automation**
- **Smart Cities**
- **Environment**
- **Energy**
- **Retail**
- **Logistics**
- **Agriculture**
- **Industry**
- **Health & Lifestyle**

Home automation

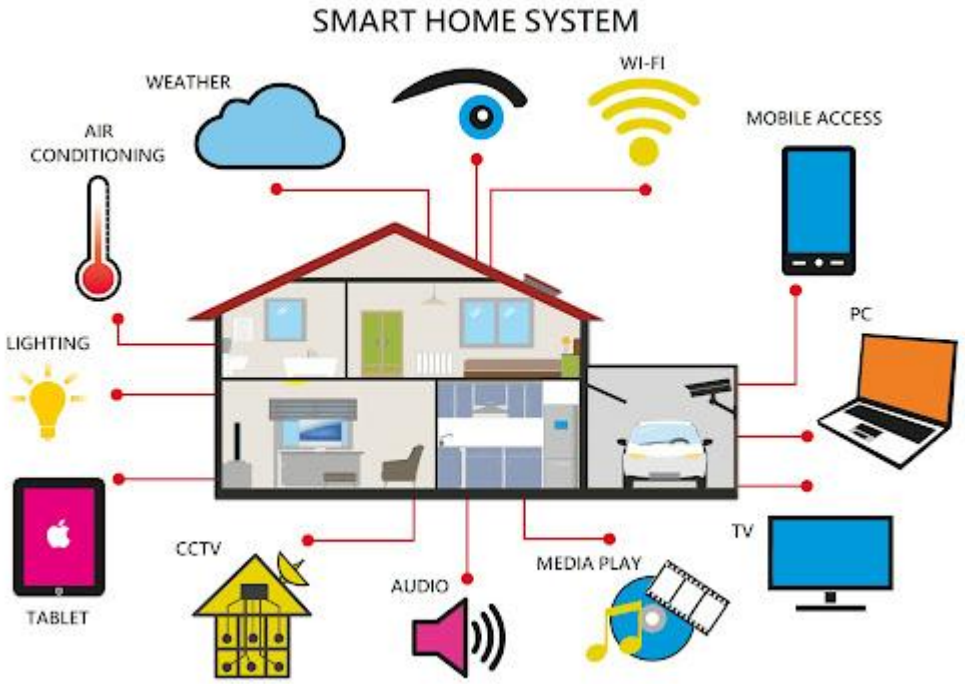
- Smart Lighting

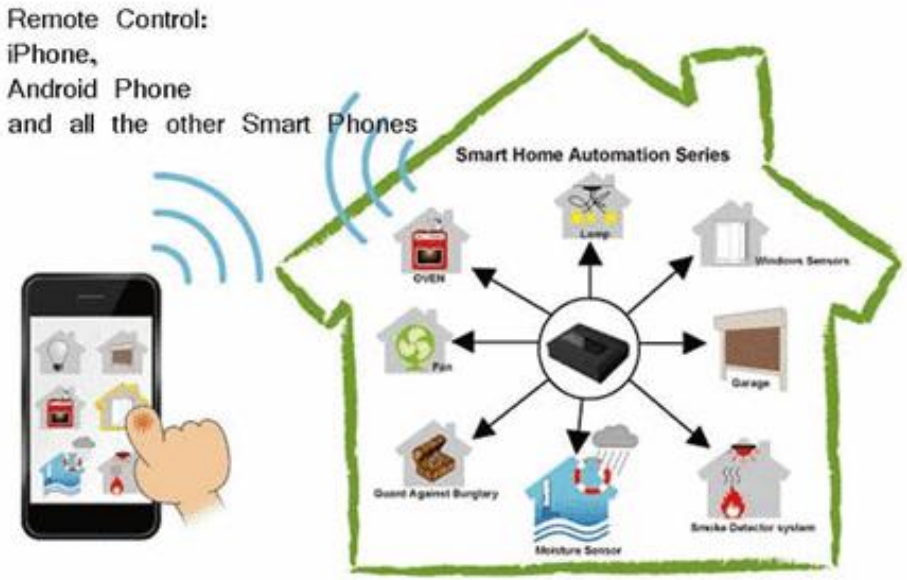




Home automation

- Smart Appliances





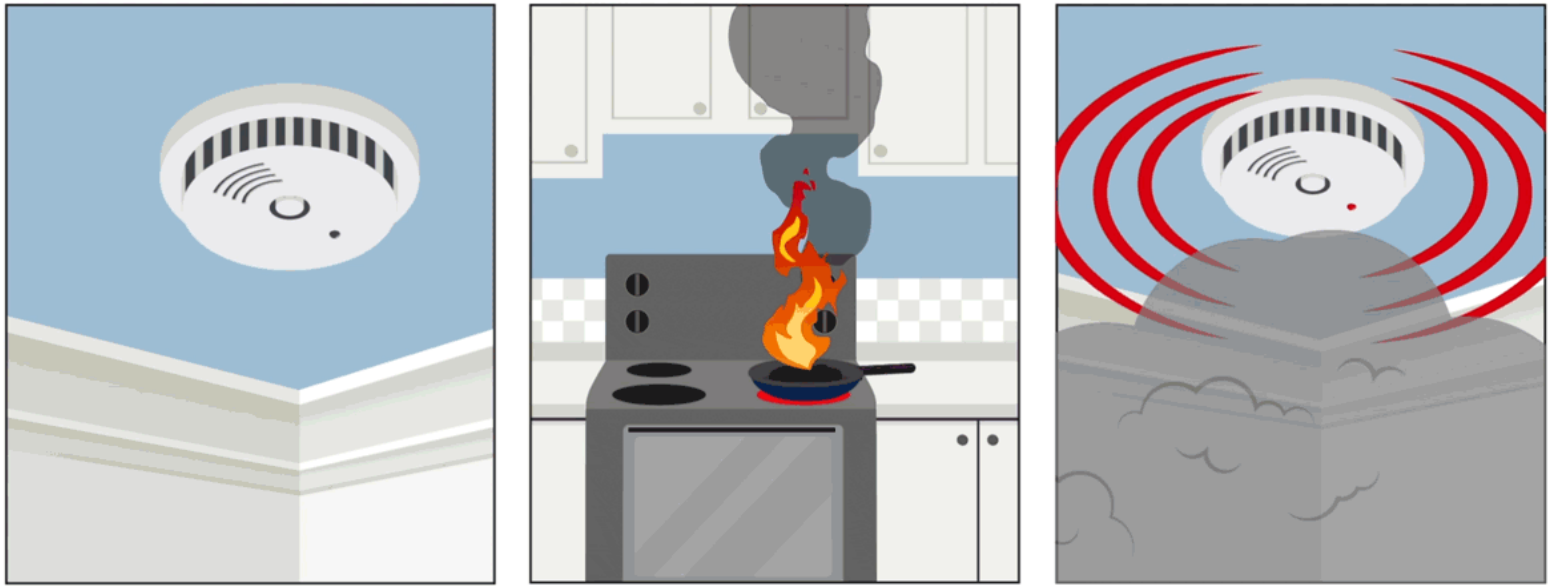
Home automation

- Intrusion Detection



Home automation

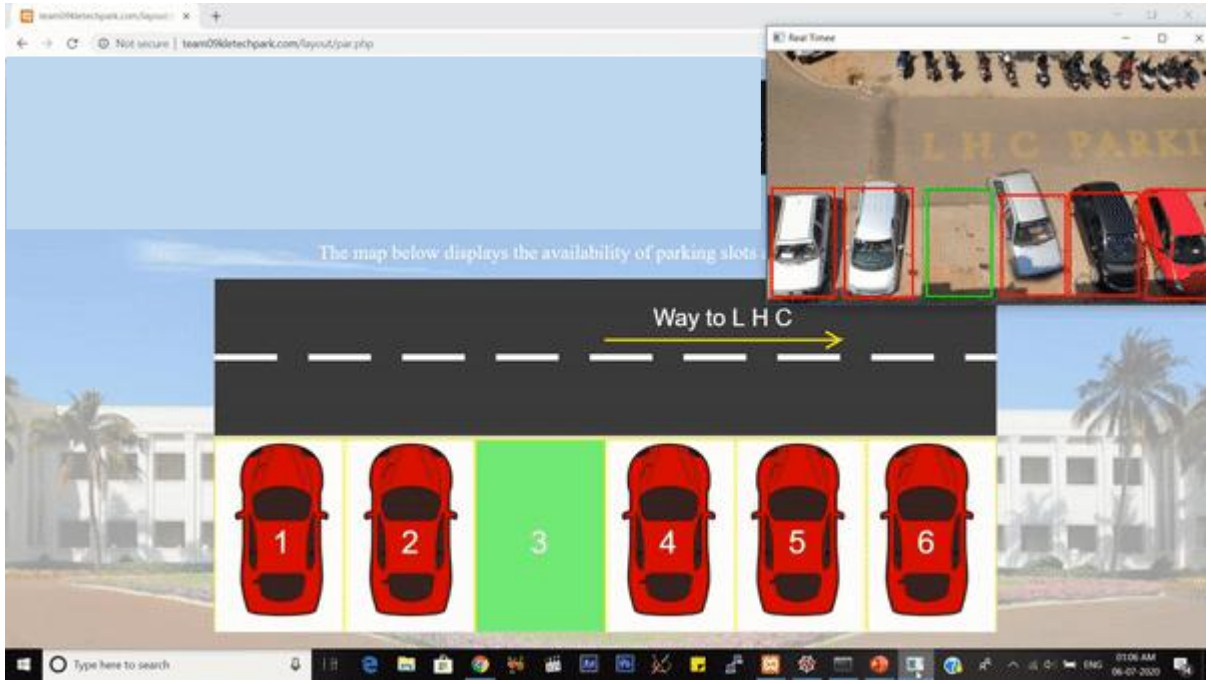
- Smoke/Gas Detectors

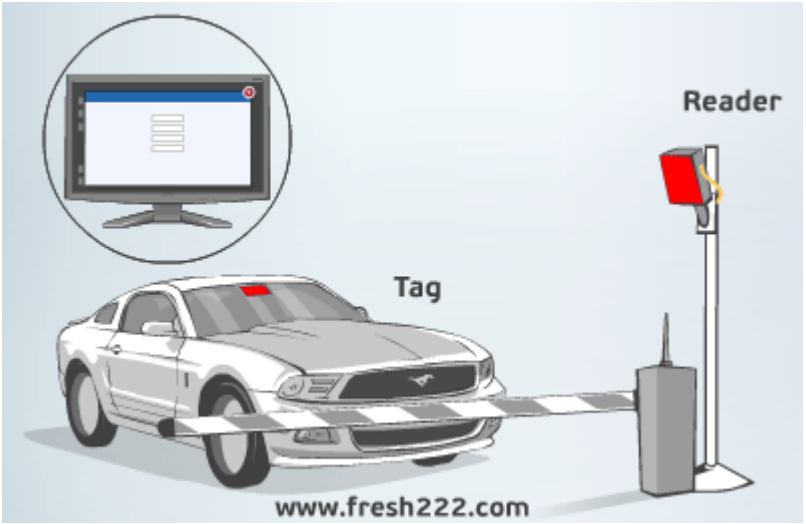


Smoke alarms beep loudly when there is a fire

Smart Cities

- Smart Parking





Smart Cities

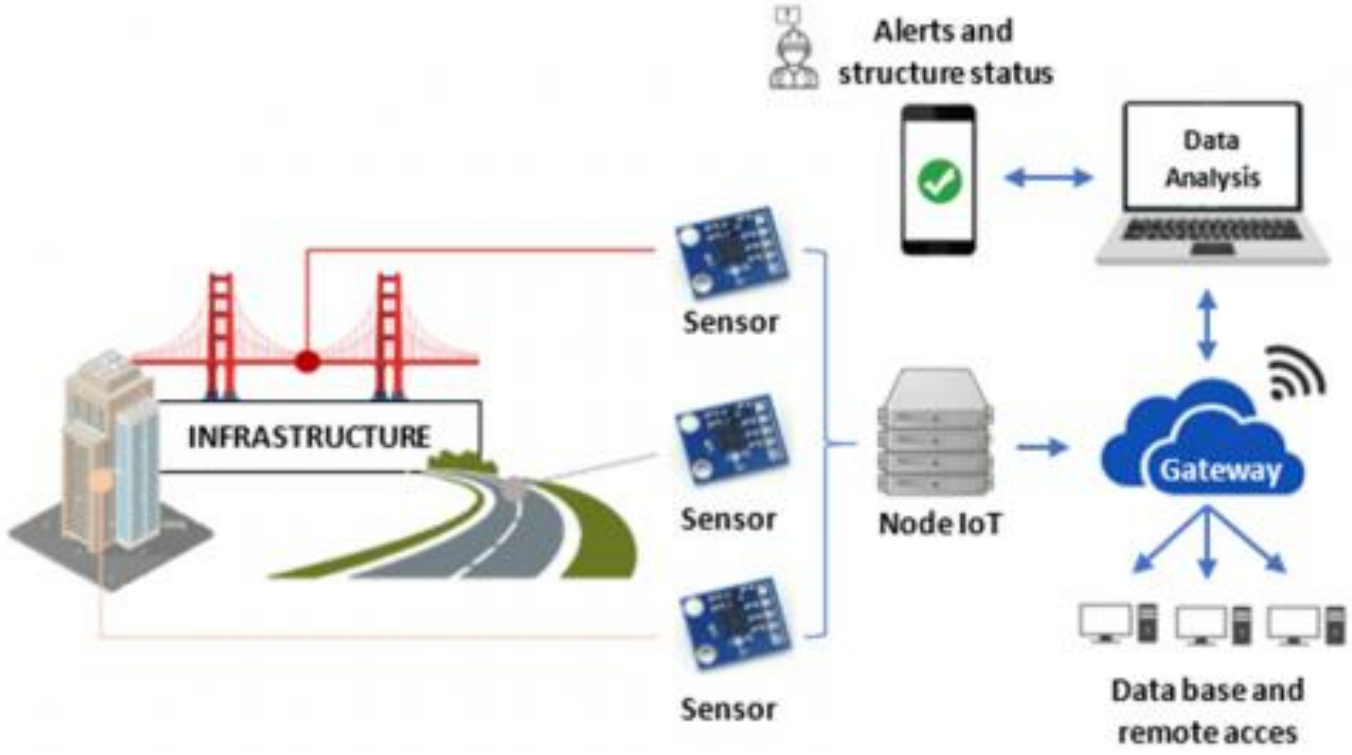
- Smart Roads





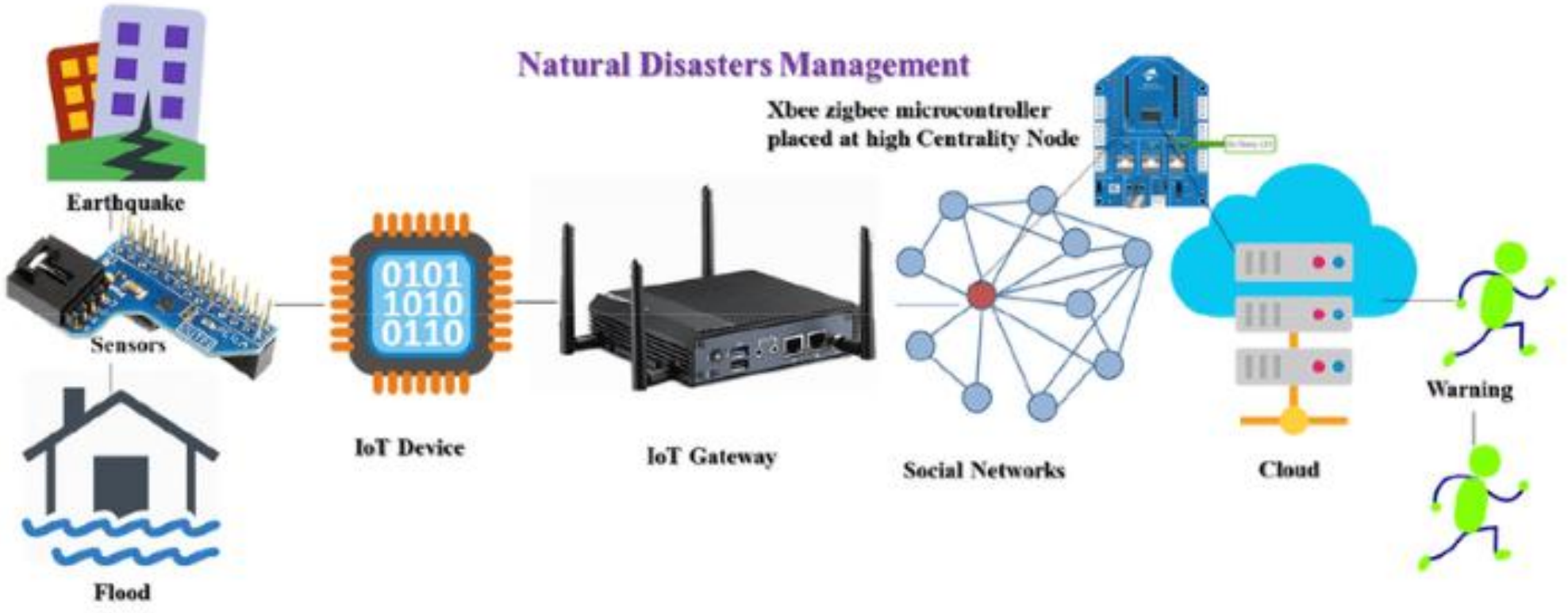
Smart Cities

- Structural Health Monitoring



Smart Cities

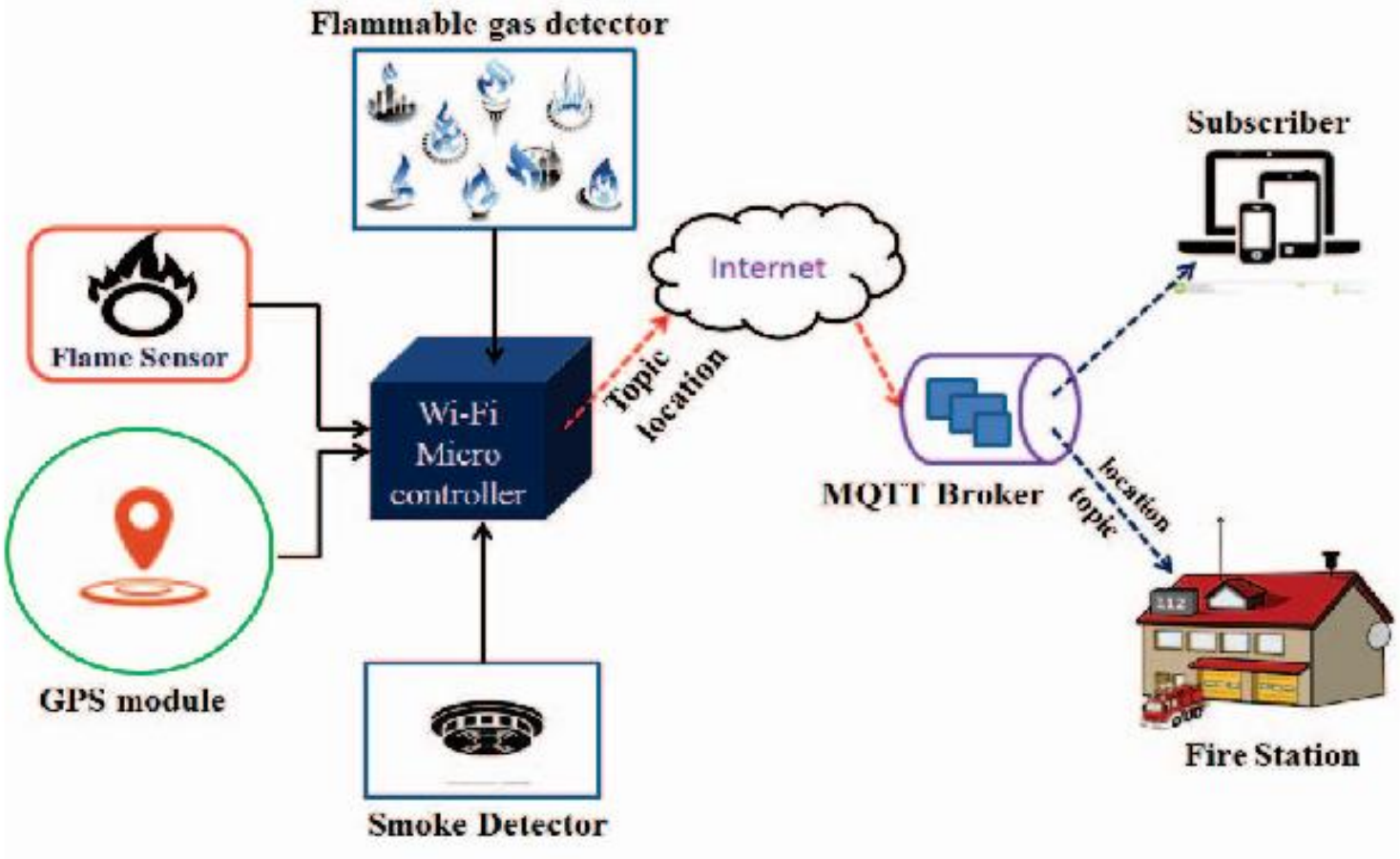
- Emergency Response



Natural disaster management by integrating internet of things with social networks

Smart Cities

- Emergency Response



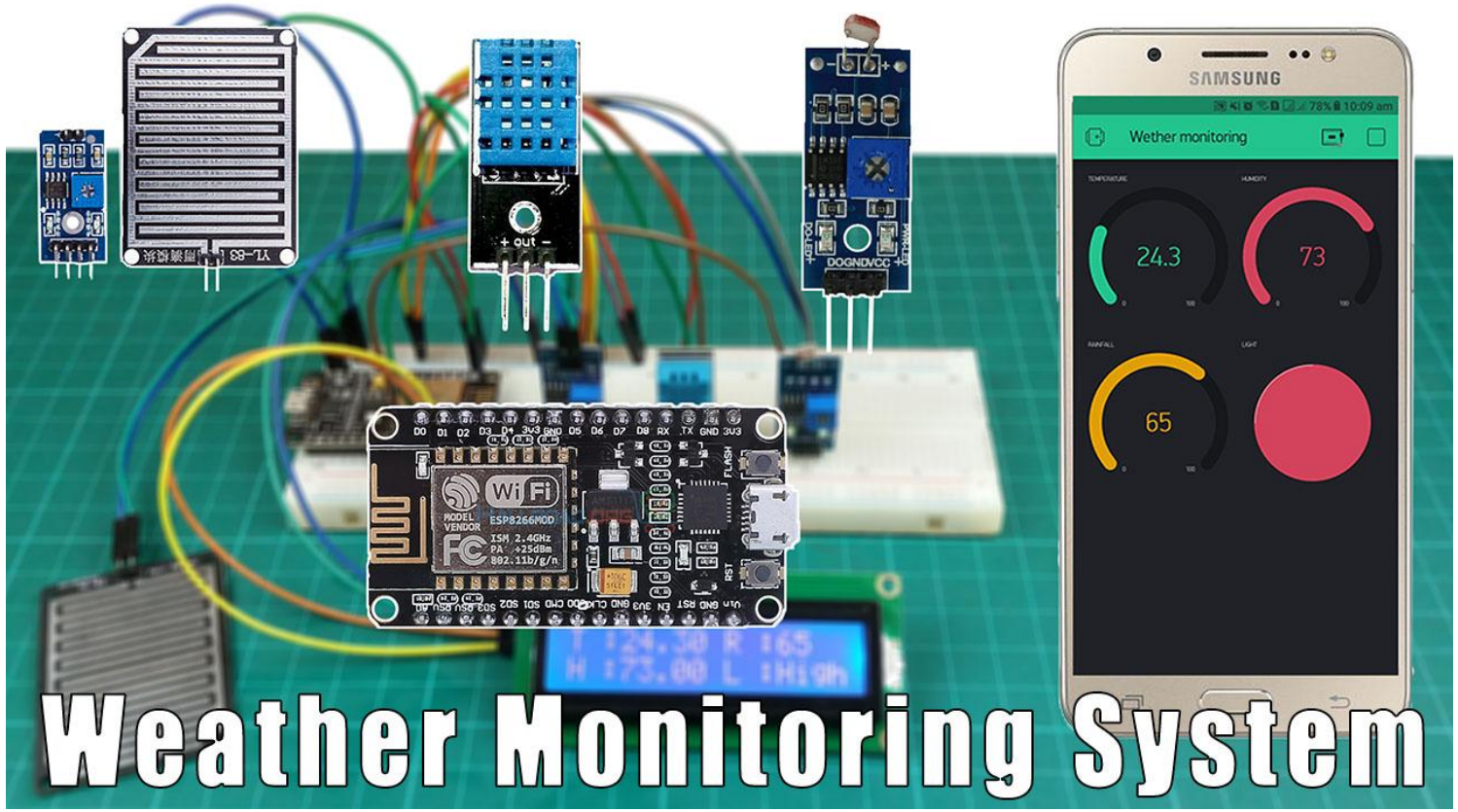
Environment

- Weather Monitoring



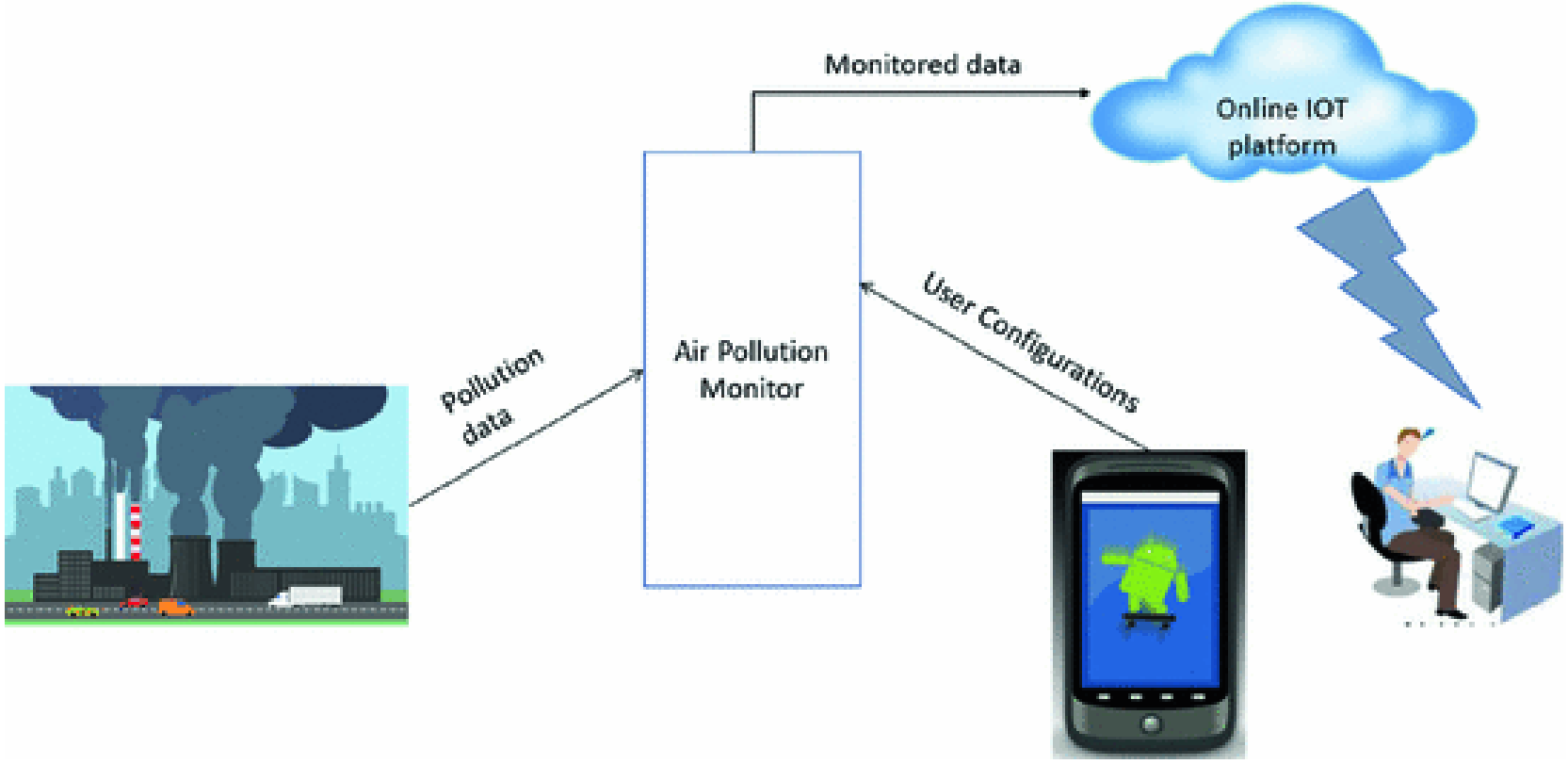
Environment

- Weather Monitoring



Environment

- Air Pollution Monitoring



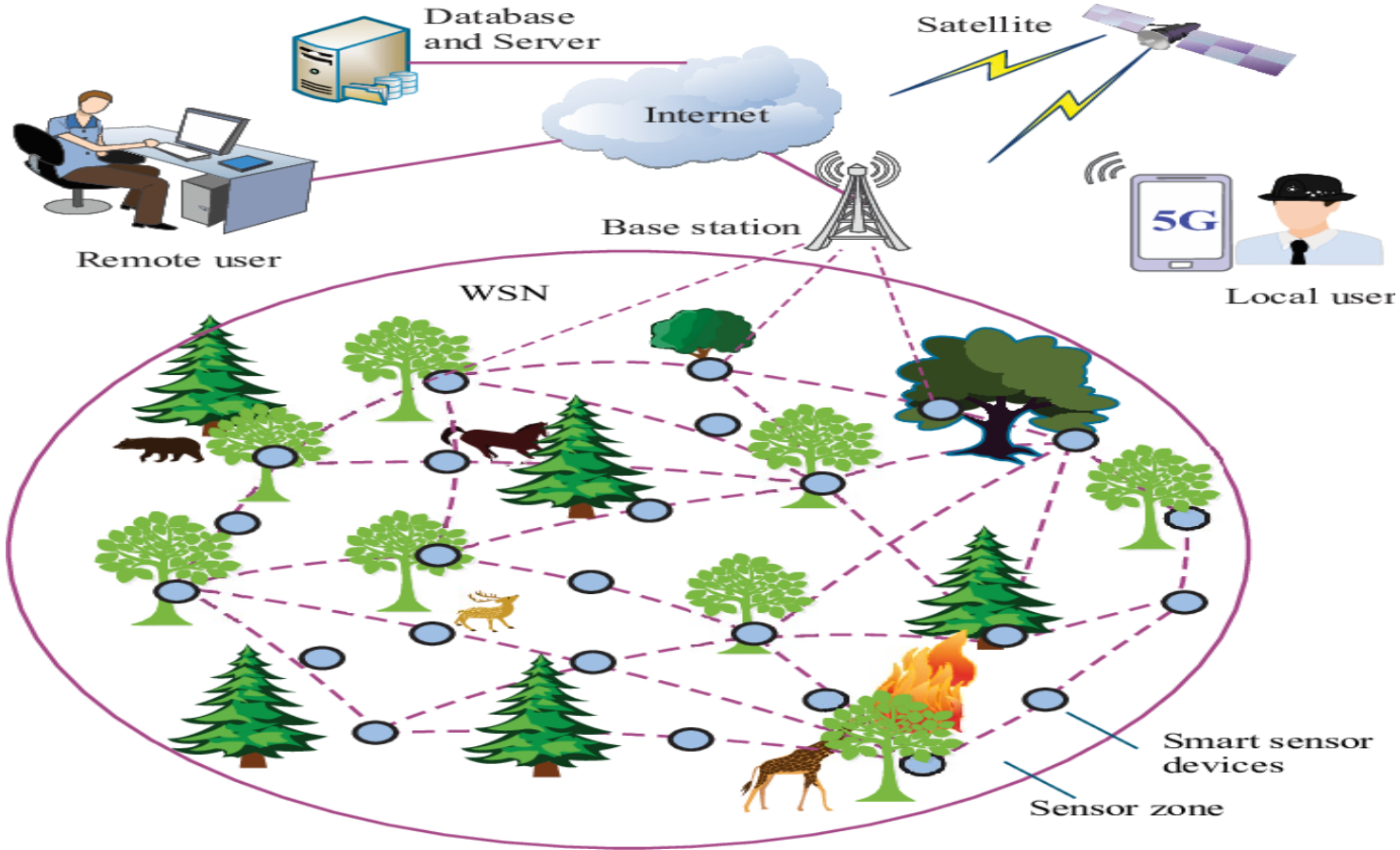
Environment

- Noise Pollution Monitoring



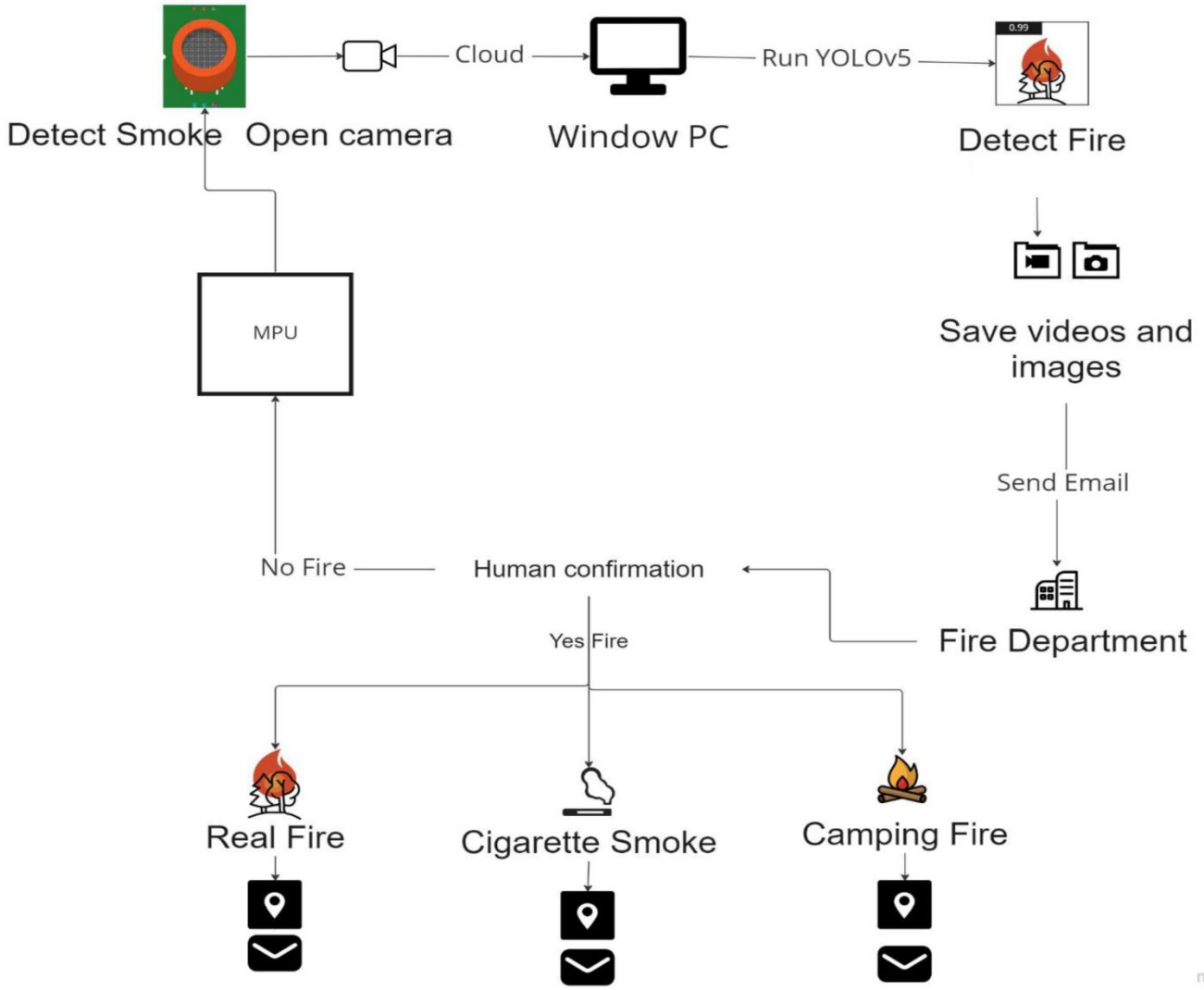
Environment

- Forest Fire Detection



Environment

- Forest Fire Detection



Energy

- Smart Grids



Energy

- Smart Grids



Energy

- Smart Grids



Energy

- Renewable Energy Systems



Energy

- Prognostics
  - An engineering discipline focused on predicting the time at which a system or a component will no longer perform its intended function.



Retail

- Inventory Management





Retail

- Smart Payments





Retail

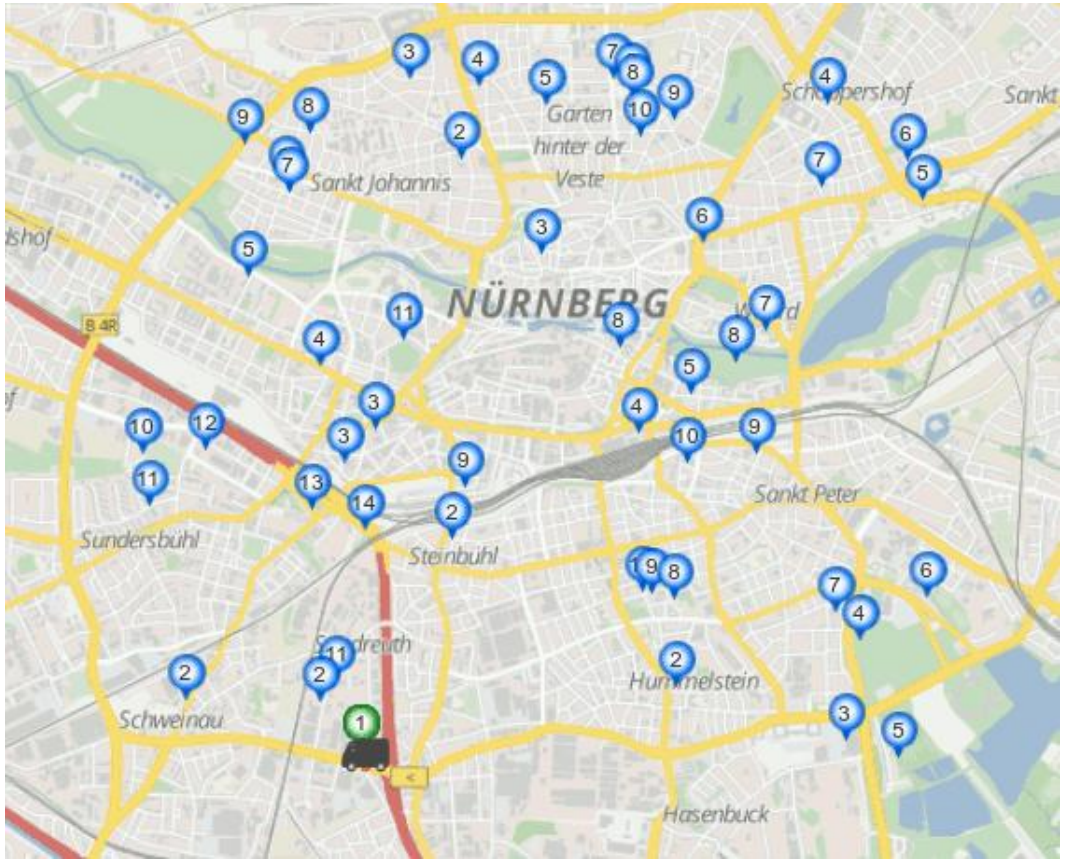
- Smart Vending Machines

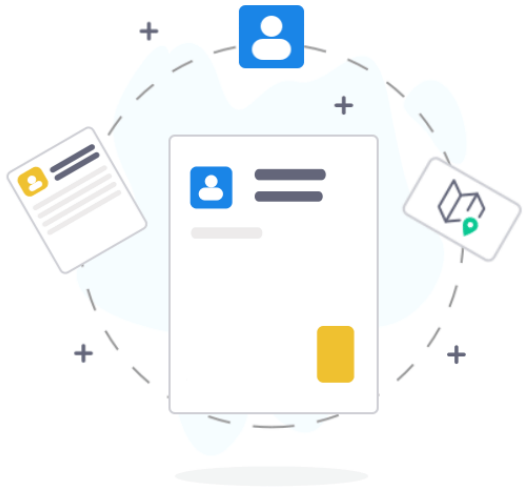




Logistics

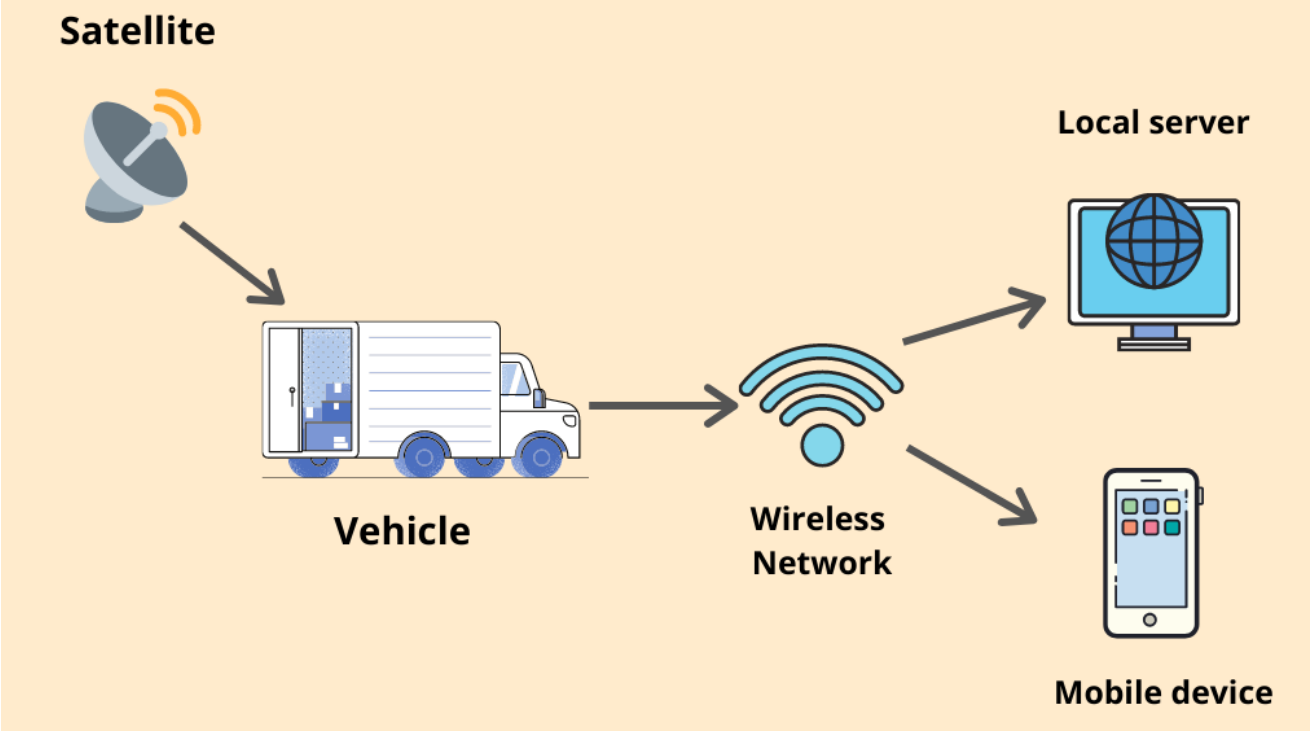
- Route Generation & Scheduling





Logistics

- Fleet Tracking

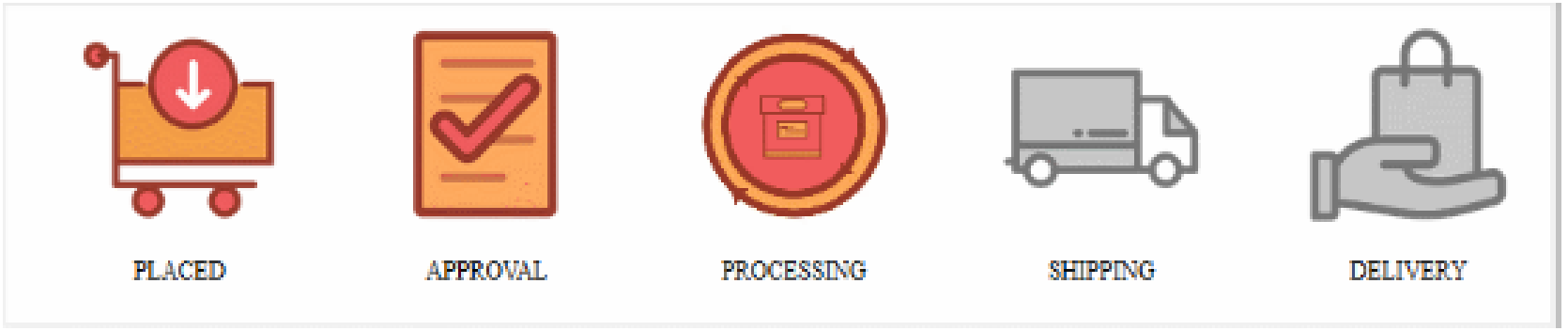




Logistics

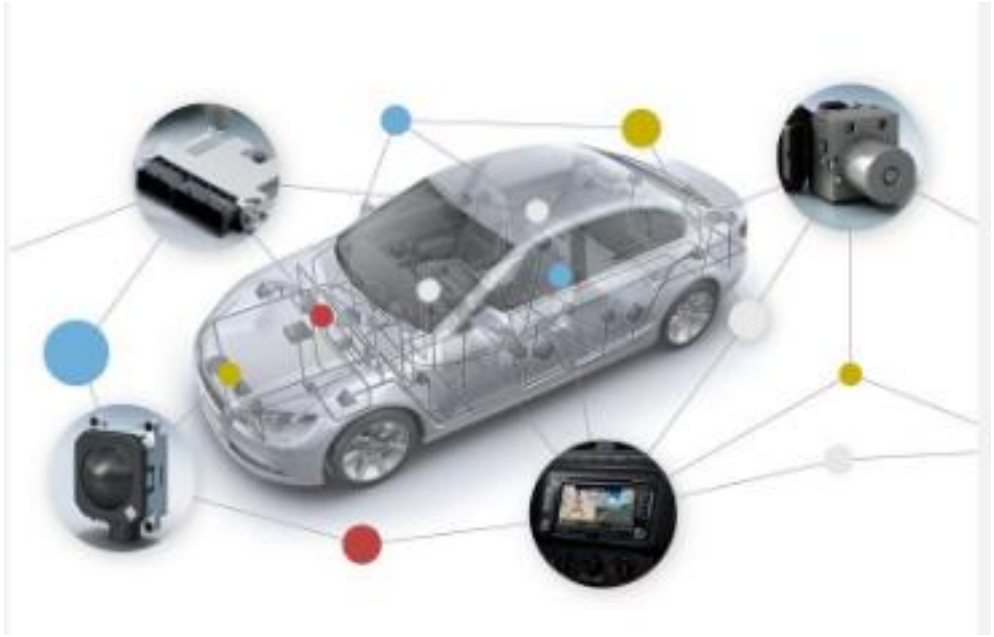
- Shipment Monitoring





Logistics

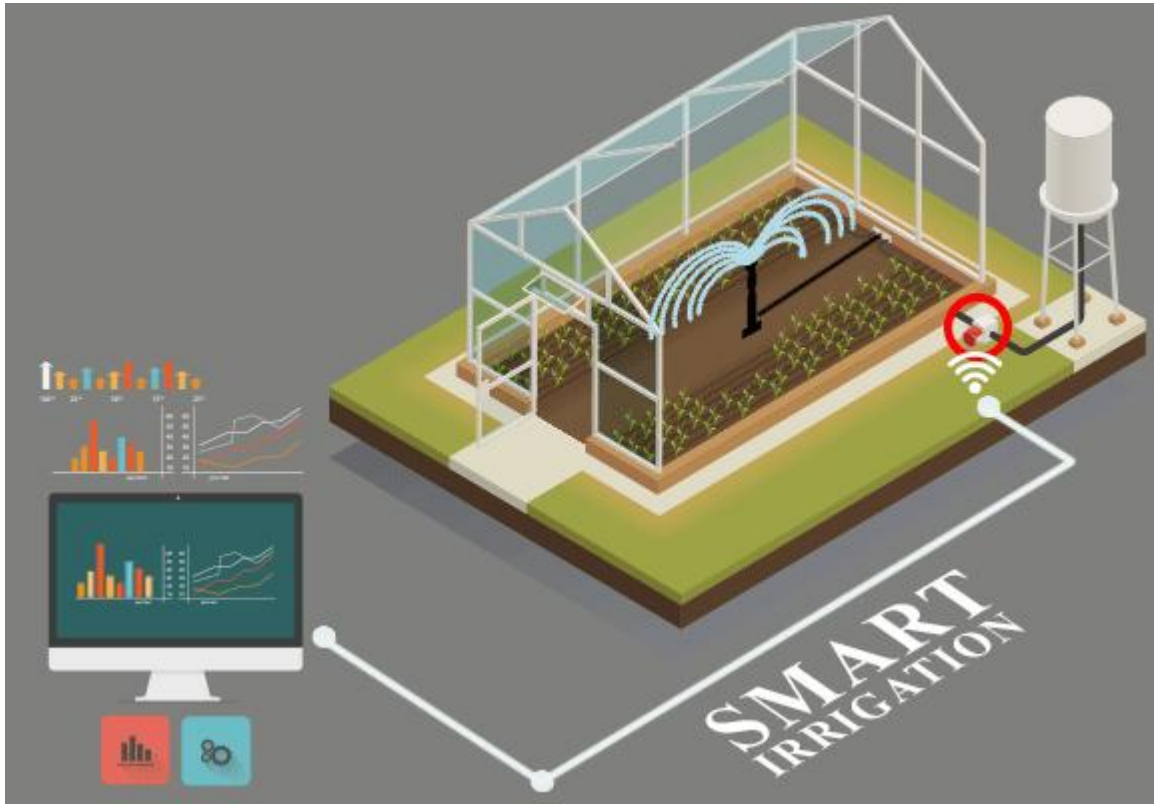
- Remote Vehicle Diagnostics





Agriculture

- Smart Irrigation



Agriculture

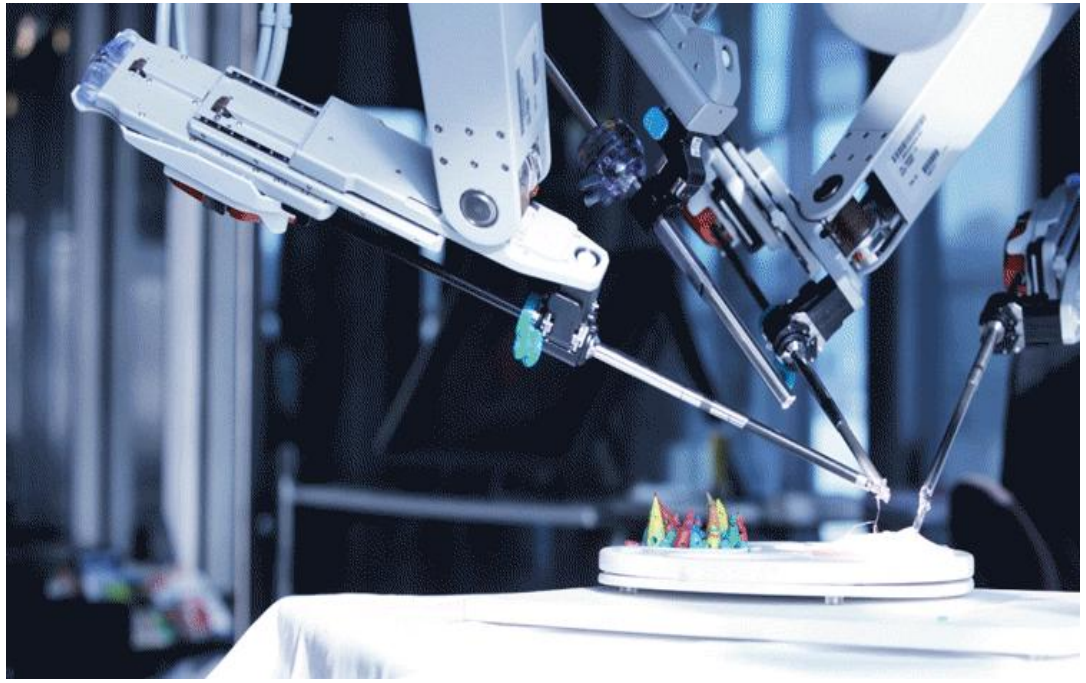
- Green House Control

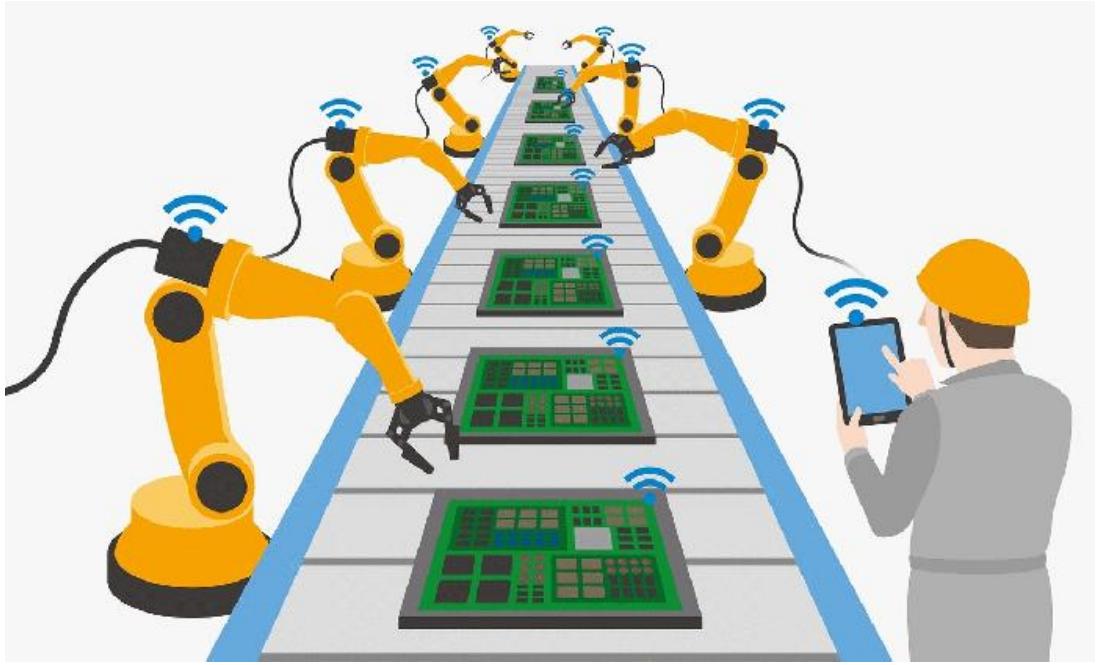




Industry

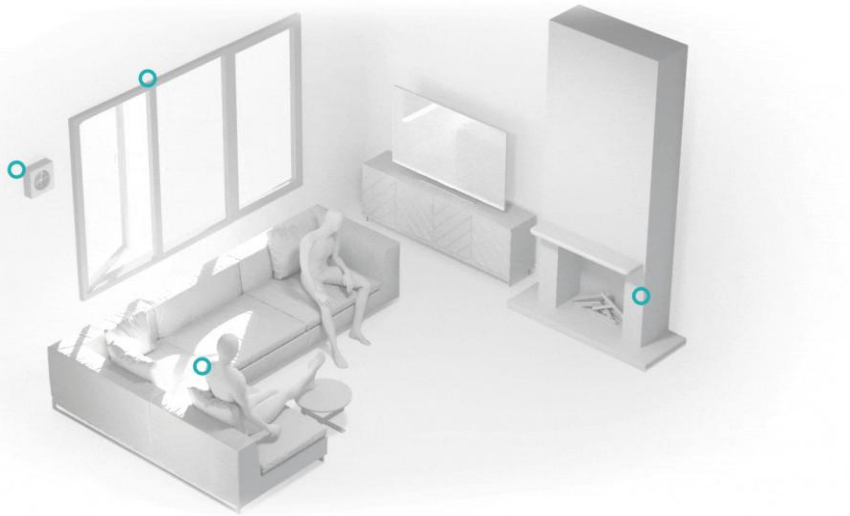
- Machine Diagnosis & Prognosis

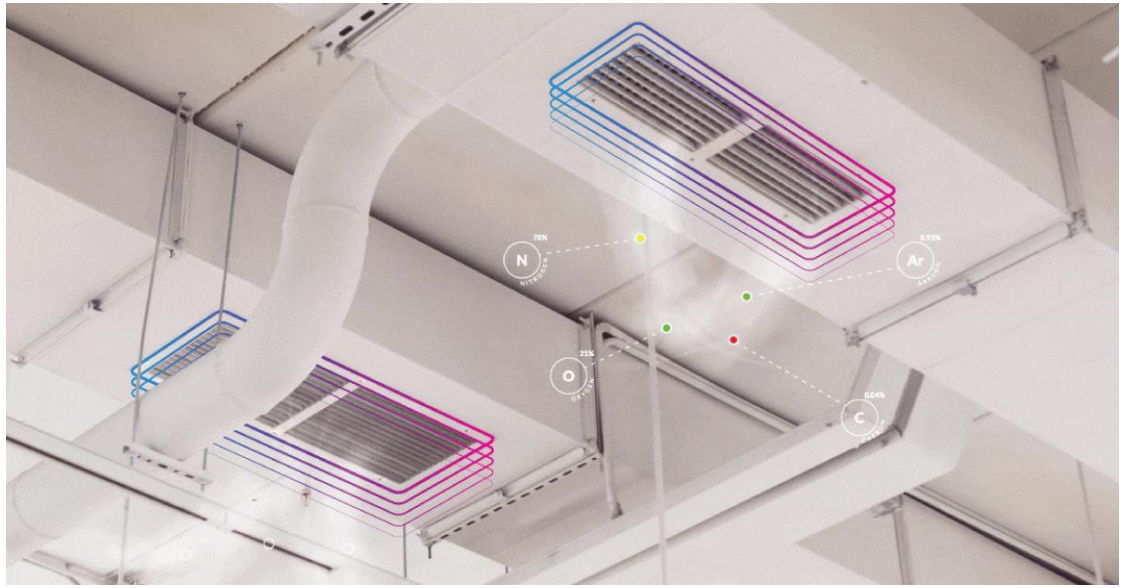




Industry

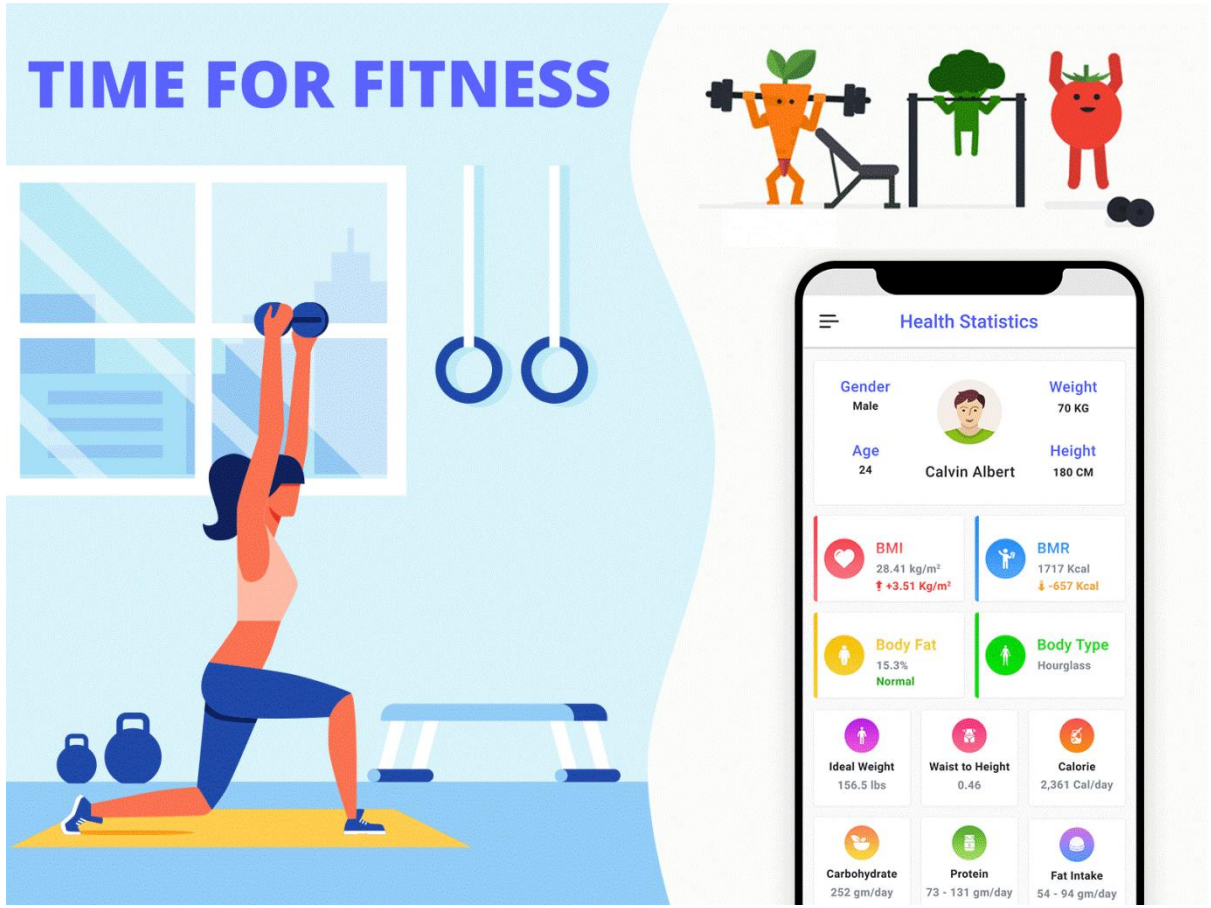
- Indoor Air Quality Monitoring





Health & Lifestyle

- Health & Fitness Monitoring





Health & Lifestyle

- Wearable Electronics





Health & Lifestyle

- Wearable Electronics



Sound Activated LED Bracelet Wristband



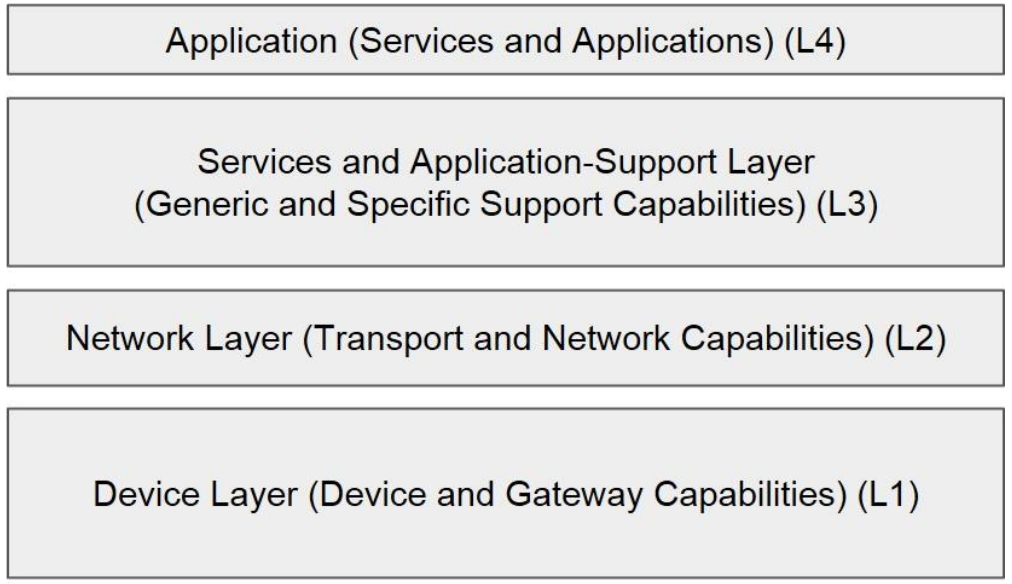
Portable Air Purifier Necklace

IoT Architecture and Reference Models (IETF, ITU-T)

Both IETF and ITU-T have contributed foundational reference models for IoT, but they approach the problem from different angles.
— IETF focuses on protocol-level architecture (how devices communicate), while
— ITU-T focuses on a broader service/network layered model (how IoT fits into telecom infrastructure).

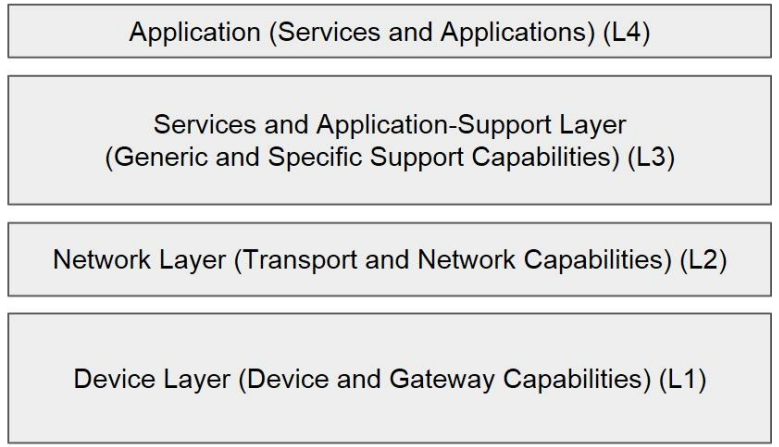
ITU-T IoT Reference Model

1. Device Layer
•**Devices**: sensing, actuation, data capture, data storage, data processing
•**Gateways**: protocol conversion between device networks and the network layer (e.g., ZigBee/Bluetooth to IP)



IoT Architecture and Reference Models (IETF, ITU-T)

ITU-T IoT Reference Model



2. Network Layer

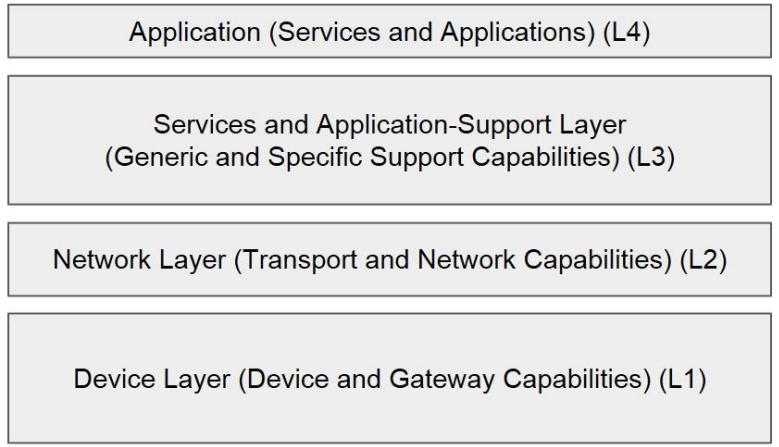
- **Networking capabilities**: routing/transport of IoT data (fixed, mobile, wireless networks)
- **Transport capabilities**: connects device-layer traffic to service/application layer, and supports IoT-related management/security signaling

3. Service Support and Application Support Layer

- **Generic support capabilities**: shared across applications (data processing, storage)
- **Specific support capabilities**: tailored capabilities different application types need

IoT Architecture and Reference Models (IETF, ITU-T)

ITU-T IoT Reference Model



4. Application Layer

•Contains the actual IoT applications (smart grid, e-health, smart transport, etc.)

Cross-layer capabilities (span all layers):

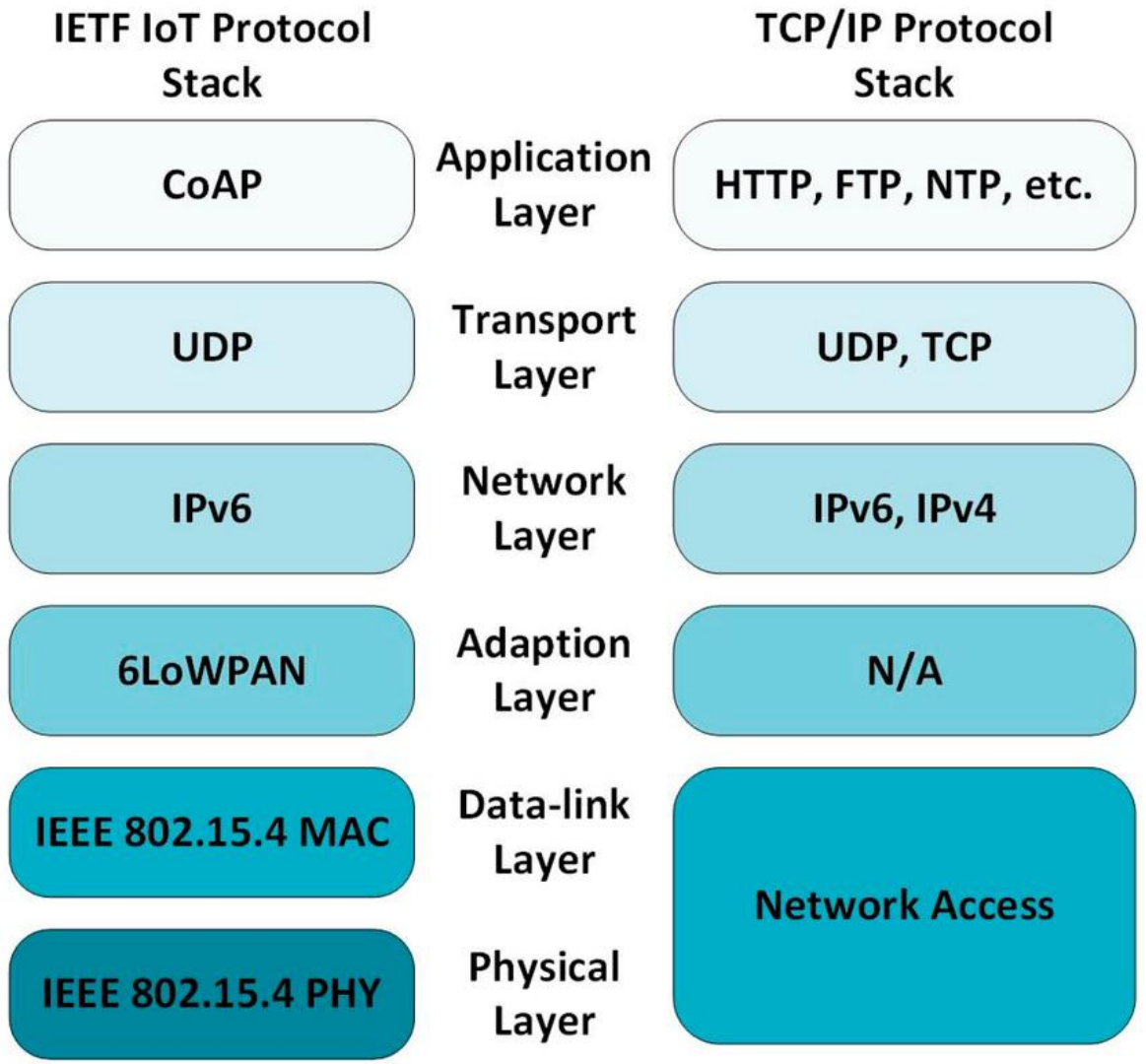
•**Management capabilities**: fault management, configuration, performance, device management

•**Security capabilities**: authorization, authentication, data confidentiality/integrity, privacy protection

IoT Architecture and Reference Models (IETF, ITU-T)

IETF IoT Reference Model

The **IETF IoT Reference Model** modifies the standard 7-layer OSI model into a specialized **6-layer stack** explicitly designed for resource-constrained smart devices and lossy networks



IoT Architecture and Reference Models (IETF, ITU-T)

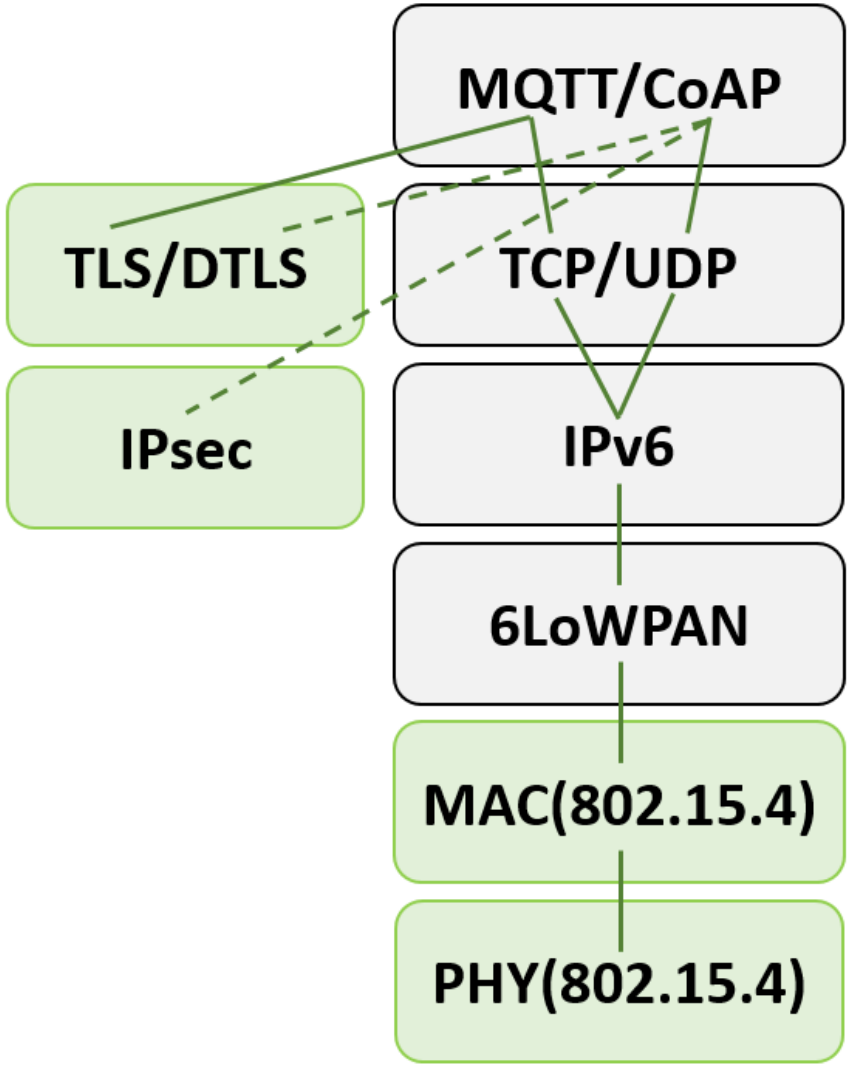
IETF IoT Reference Model

•**Layer 6: Application Layer** – Defines how software applications interact.

•It uses lightweight, web-like architectures (RESTful) to communicate state transitions via small packets.

•*Core Protocol:* **CoAP** (Constrained Application Protocol — RFC 7252).

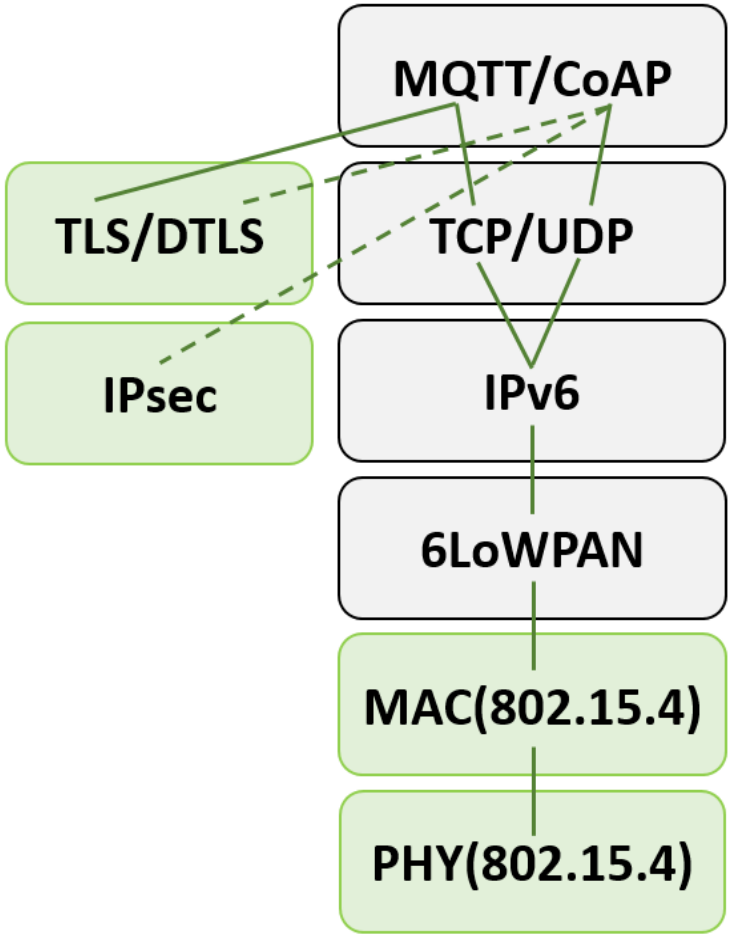
RFC is an **official, highly technical blueprint document** that defines the rules, specifications, and protocols that make the internet work.



IoT Architecture and Reference Models (IETF, ITU-T)

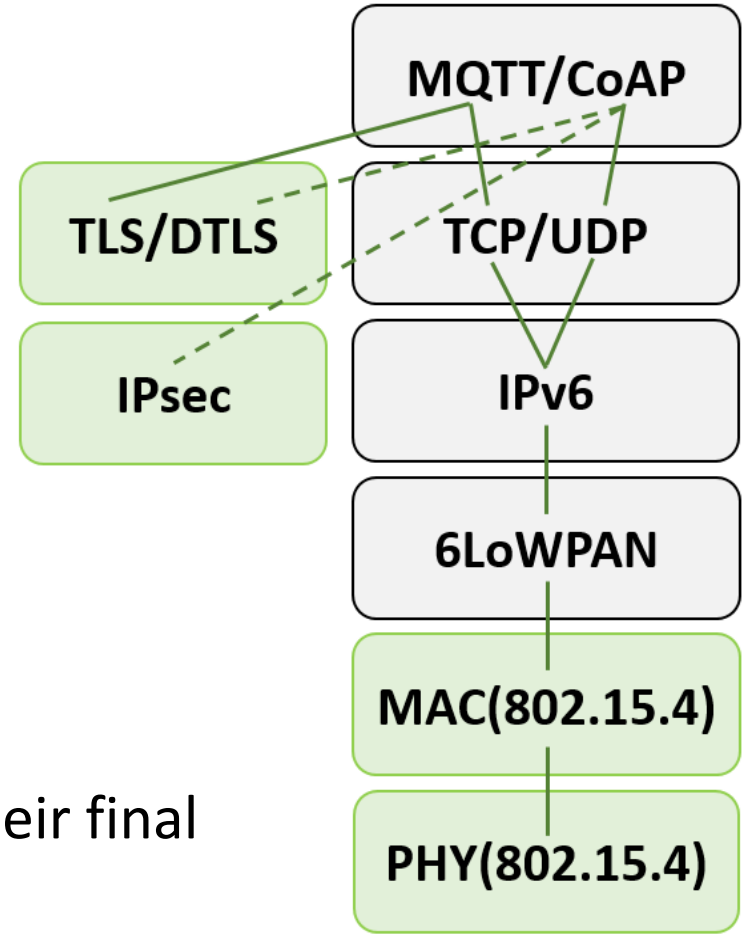
IETF IoT Reference Model

- **Layer 5: Application Support Layer** – Consolidates the traditional OSI *Session* and *Presentation* layers into a single thin profile.
- It manages connection persistence, formatting, data security, and basic data analytics structures.
- *Core Protocols:* **DTLS** (Datagram Transport Layer Security), **Object Security (OSCORE)**



IoT Architecture and Reference Models (IETF, ITU-T)

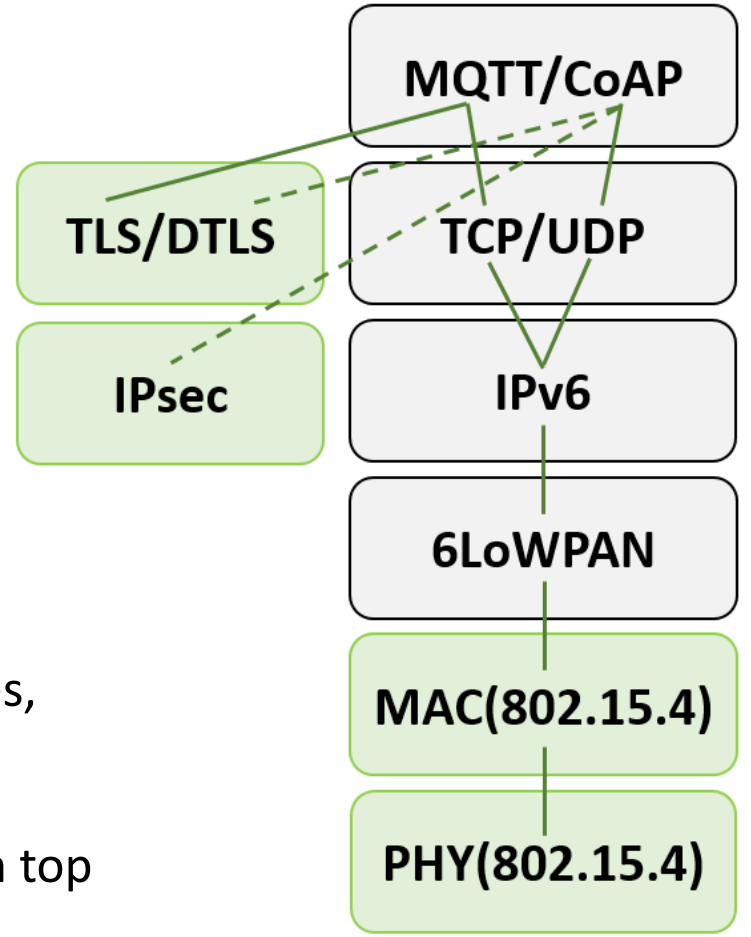
IETF IoT Reference Model



- **Layer 4: Transport Layer** – Handles end-to-end communication data control.
- It heavily favors connectionless, low-overhead pathways over bulky standard validation loops.
- *Core Protocols:* **UDP** (primarily used for minimal header size) and optimized **TCP** (via RFC 9006 metrics)
- **Layer 3: Network Layer** – Performs endpoint addressing and determines how packets hop through a mesh network toward their final cloud destination.
- *Core Protocols:* **IPv6**, **RPL** (Routing Protocol for Low-Power and Lossy Networks)

IoT Architecture and Reference Models (IETF, ITU-T)

IETF IoT Reference Model



- **Layer 2: Data Adaptation Layer** – The **most critical inclusion** in the IETF model.

- This layer functions directly between the hardware and the network layer, compressing huge 128-bit IPv6 headers and handling fragmentation so they can fit inside tiny, low-power radio frames.

- *Core Protocols:* **6LoWPAN** (IPv6 over Low-Power Wireless Personal Area Networks) and **6lo**

- **Layer 1: Physical / Data Link Layer** – The physical hardware, radio frequencies, and media access control rules.

- IETF does not write these standards, but designs its stack to seamlessly sit on top of them.

- *Core Hardware Standards:* **IEEE 802.15.4** (Zigbee-like radios), **Bluetooth Low Energy (BLE)**, and **DECT ULE**.

Physical design of IoT: Devices, Gateways, and Data centers

The physical design of an IoT system comprises the tangible hardware components that collect, process, and store data.
It is structured into three primary tiers:

- **Devices (Things)** that gather physical data,
- **Gateways** that aggregate and translate that data, and
- **Data Centers (Cloud)** that provide large-scale processing and storage.

Physical design of IoT: Devices, Gateways, and Data centers

Devices (Things)

IoT devices are the endpoints of the network responsible for interacting with the physical environment. They vary from small microcontrollers to complex industrial sensors.

Sensors: Collect physical data (temperature, humidity, motion, pressure) and convert it into digital signals.

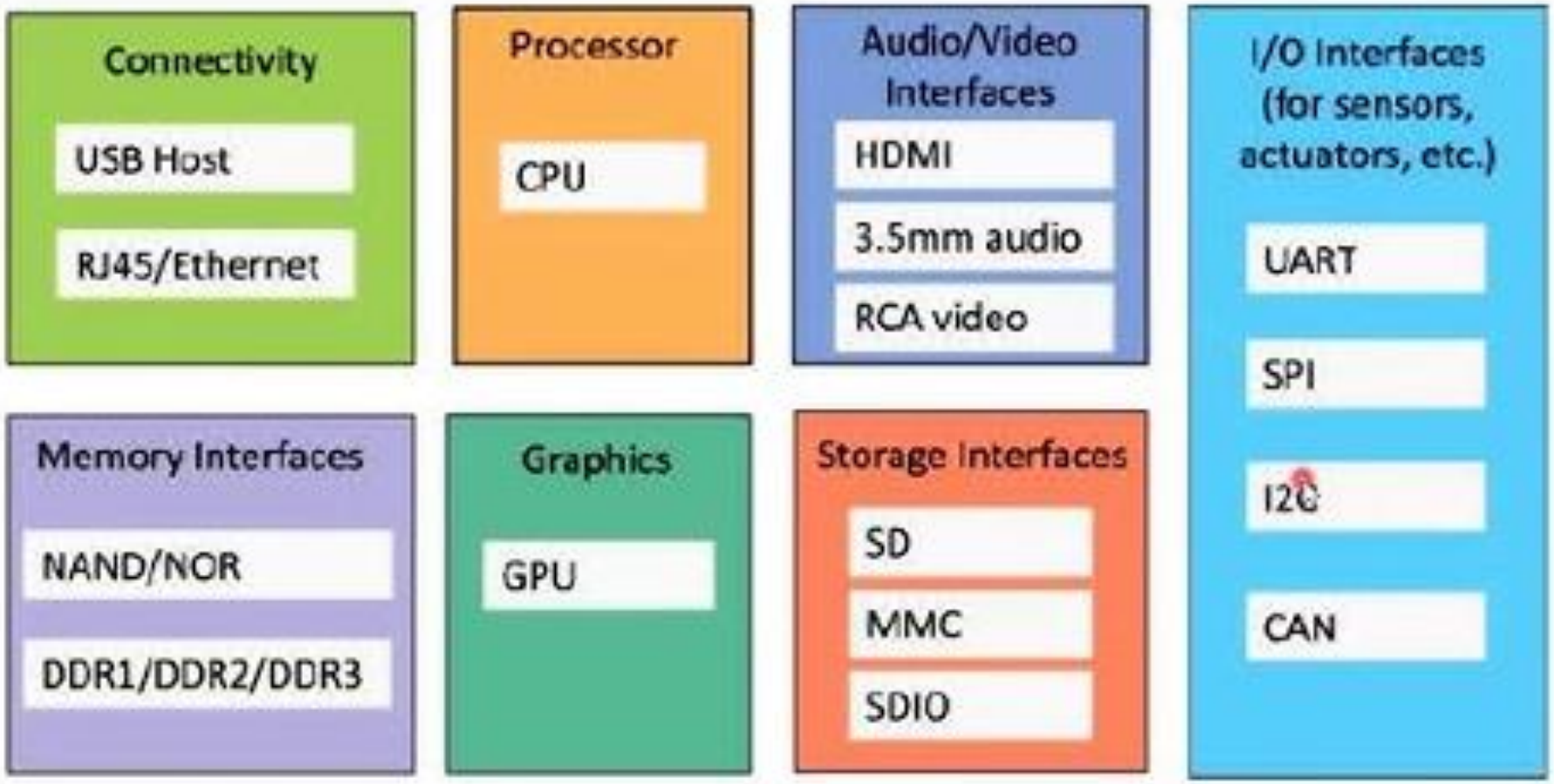
Actuators: Perform physical actions in response to commands (e.g., turning on a motor, opening a valve).

Processing Units: Embedded systems (like CPUs or microcontrollers) that process data locally, manage power, and execute local algorithms.

Connectivity Modules: Hardware chips and antennas that allow devices to connect to networks via Wi-Fi, Bluetooth, Zigbee, or cellular (4G/5G).

Power Supplies: Can be mains-powered or battery-dependent, often requiring energy-harvesting hardware for prolonged use

Generic Block Diagram of an IoT Device



Connectivity interfaces

- USB Host





USB 3.0

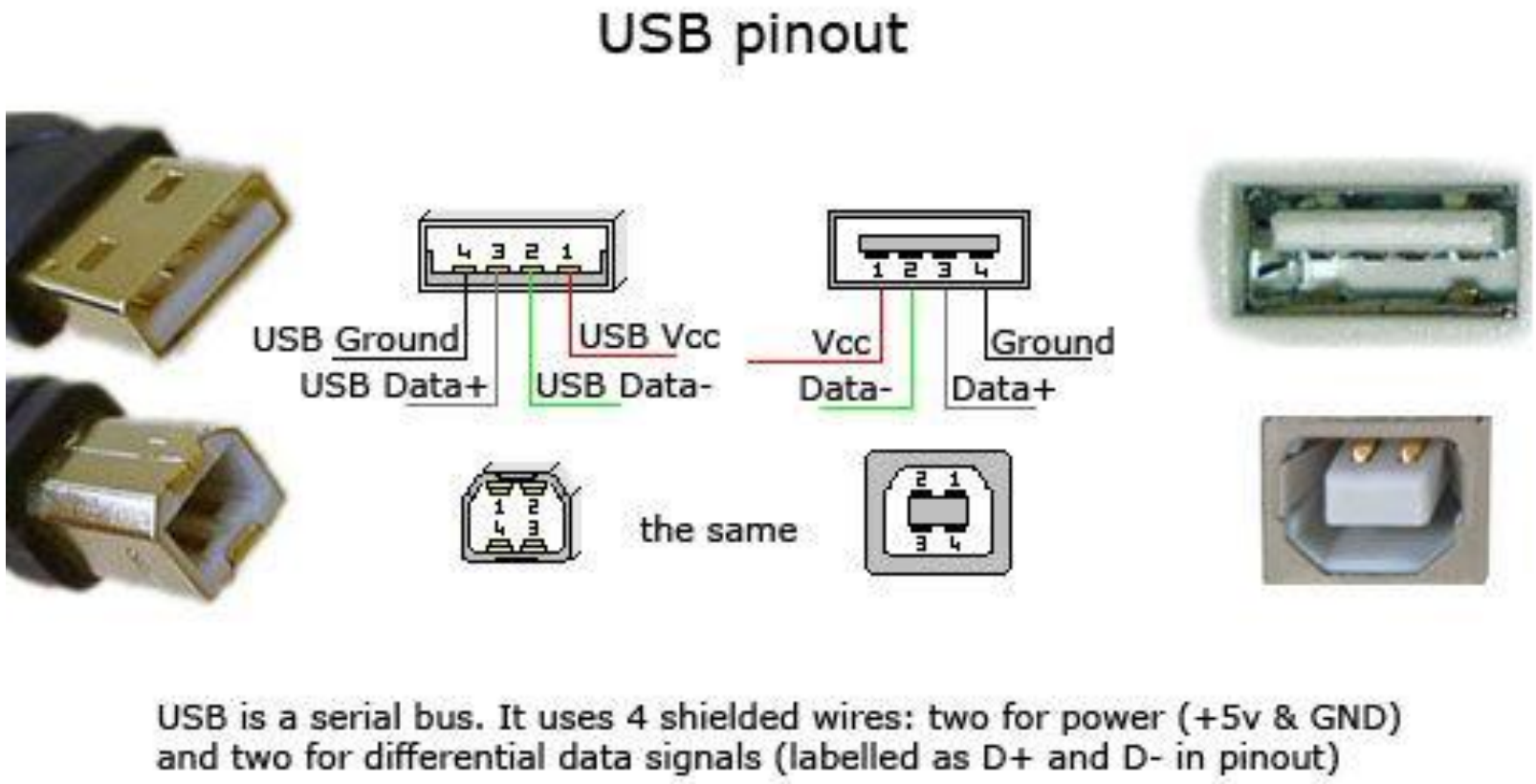


USB 2.0



Connectivity interfaces

- USB Host



USB is a serial bus. It uses 4 shielded wires: two for power (+5v & GND) and two for differential data signals (labelled as D+ and D- in pinout)

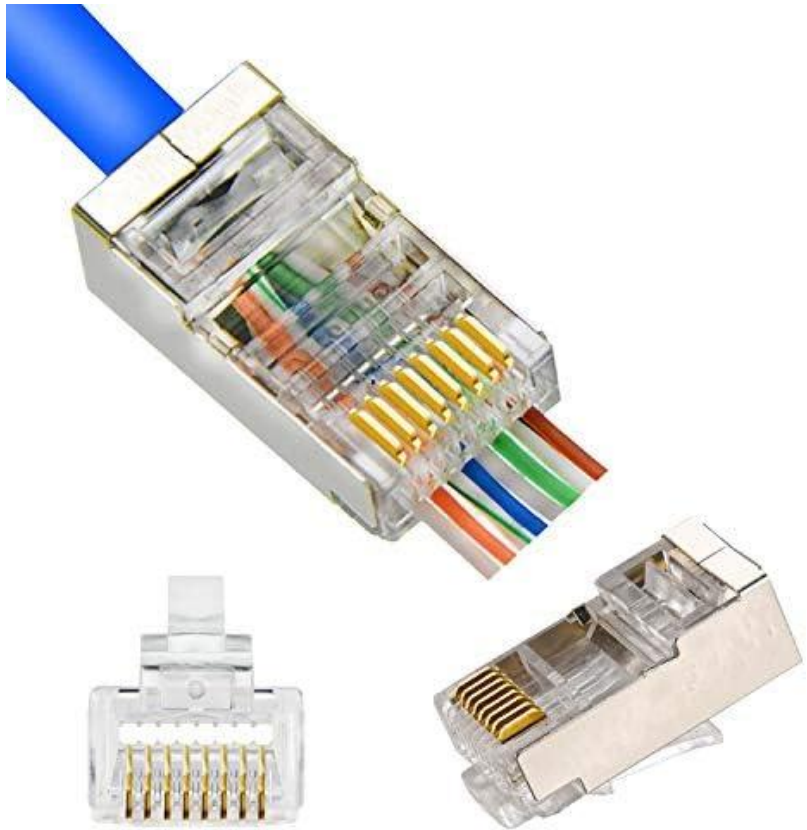
Connectivity interfaces

- RJ45/ Ethernet- IEEE 802.3

RJ stands for registered jack while the “45” refers to the listing number

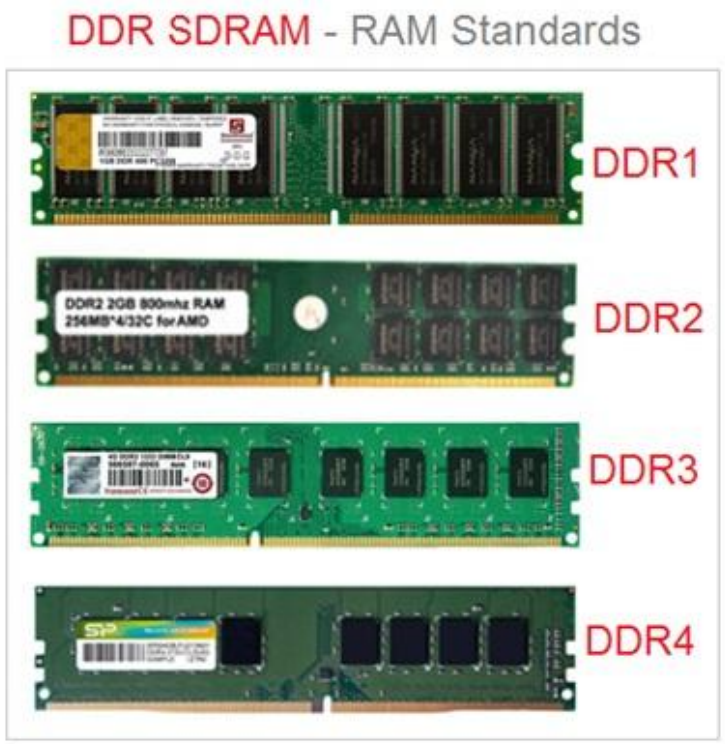


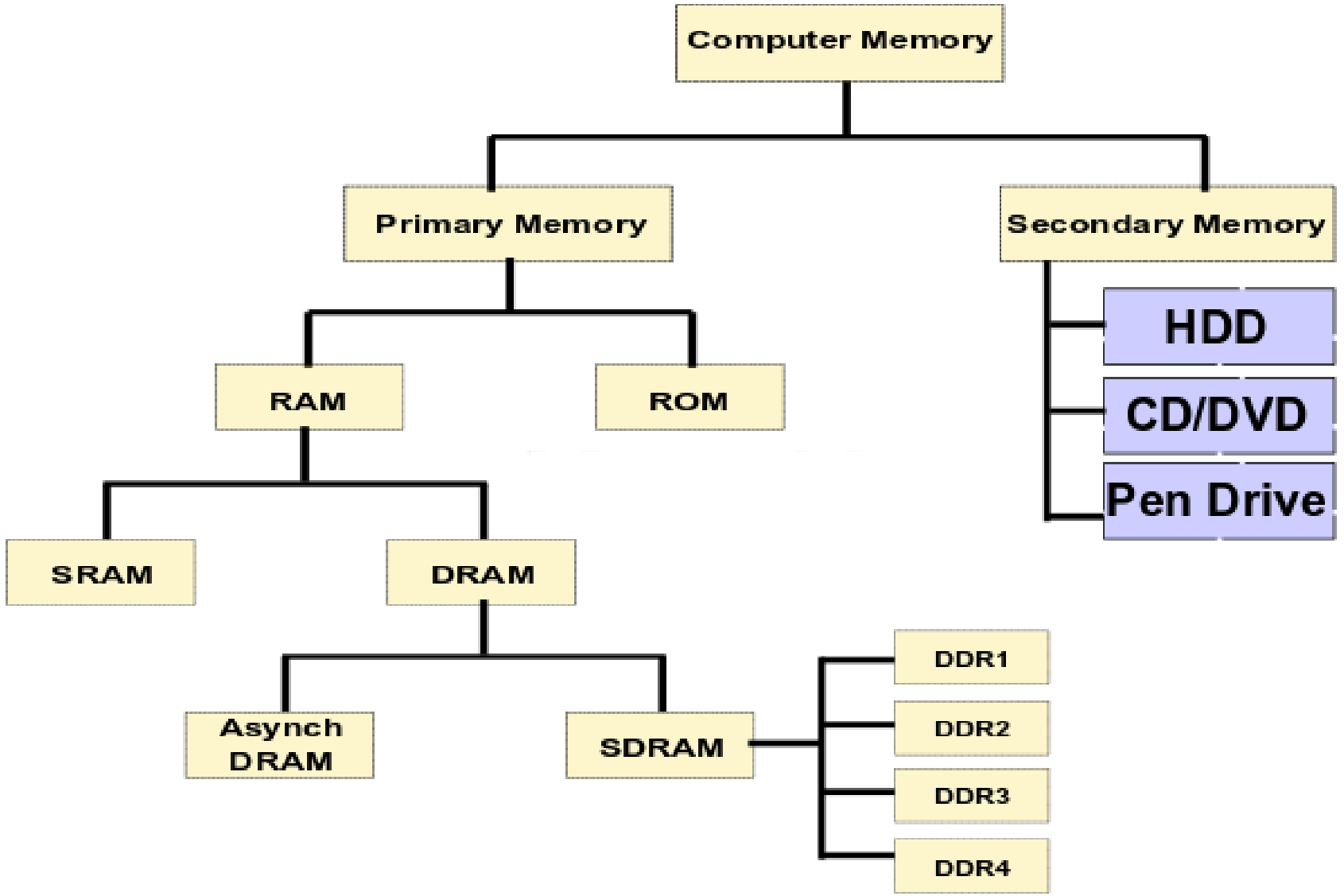




Memory Interfaces

- NAND/NOR
- DDR1/DDR2/DDR3





Processing interfaces

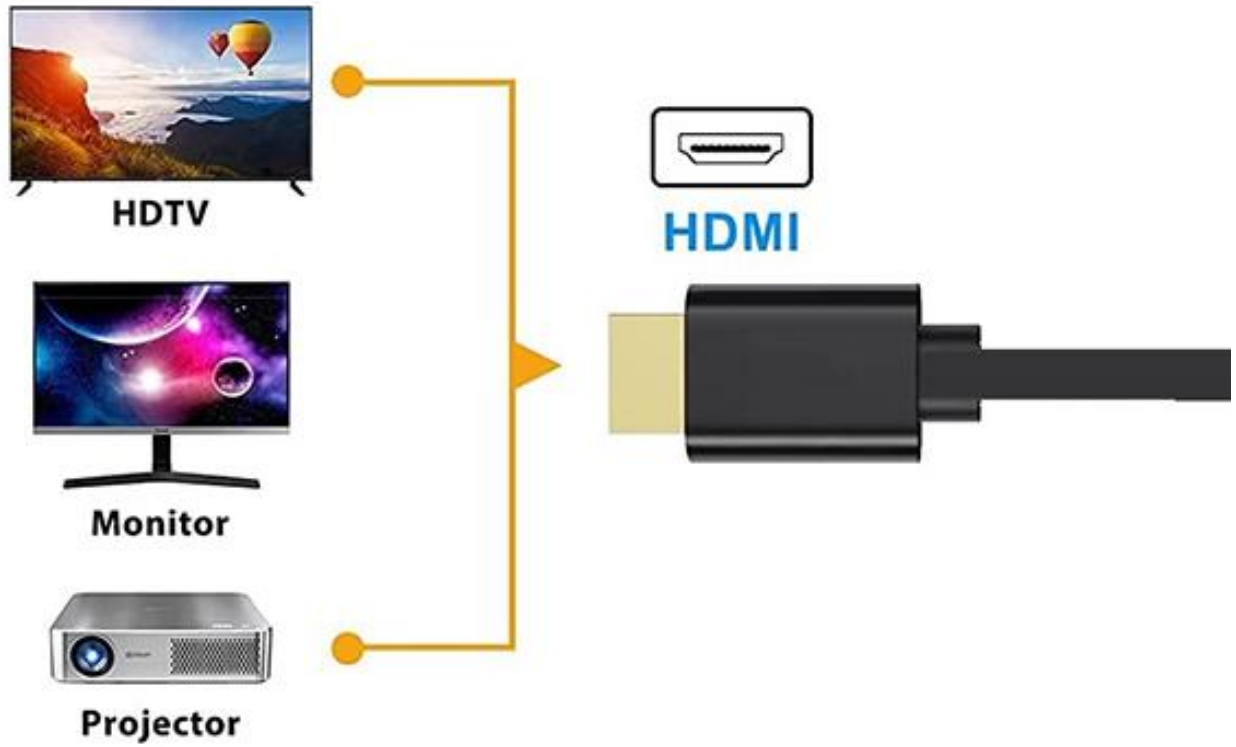
- CPU -Central Processing Unit
  - -purpose
    - **Data Processing**
    - **Instruction Execution**
    - **Device Control**
    - **Connectivity Management**
    - **Power Management**

Graphics interfaces

- GPU- Graphics Processing Unit
  - GPUs are also known as video cards or graphics cards
  - In order to display pictures, videos, and 2D or 3D animations, each device uses a GPU

Audio/ Video Interfaces

- HDMI
- 3.5mm audio
- RCA video

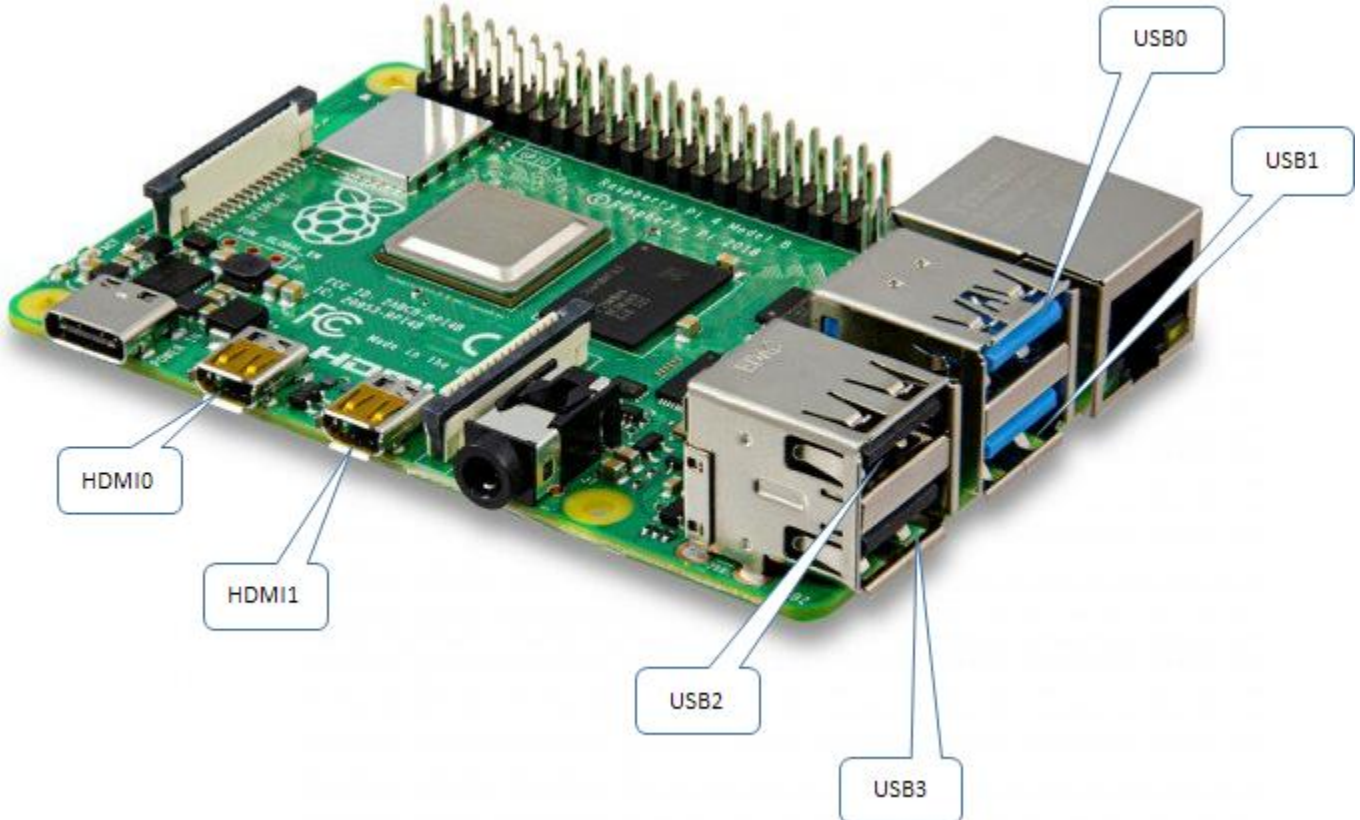


HDMI

Audio/ Video Interfaces

- Types of HDMI





Audio/ Video Interfaces

- 3.5mm audio

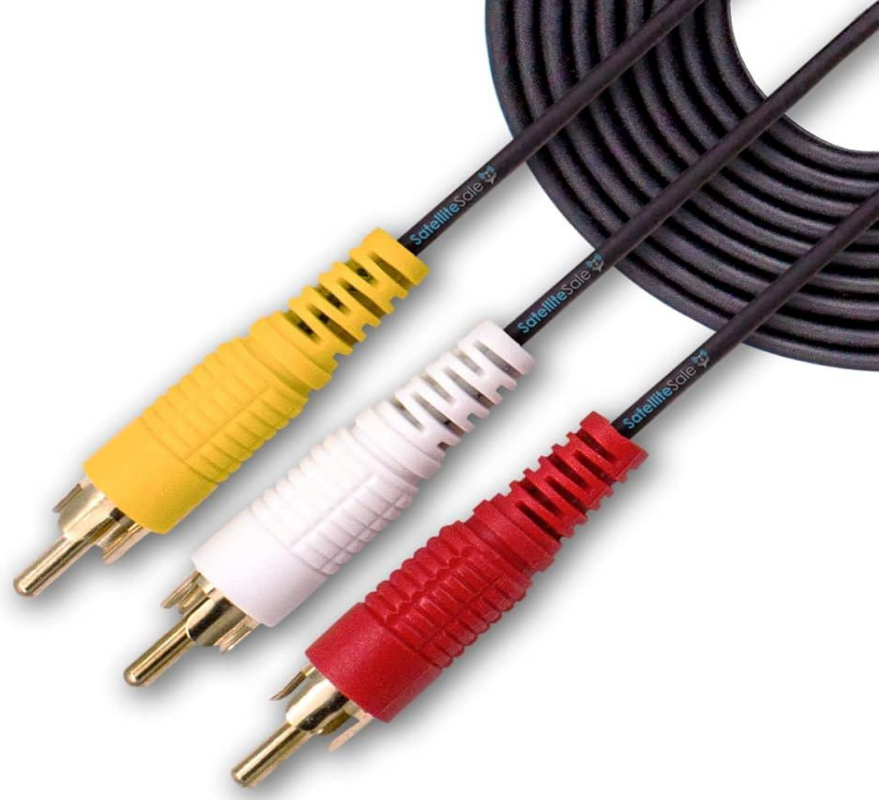






Audio/ Video Interfaces

- RCA video

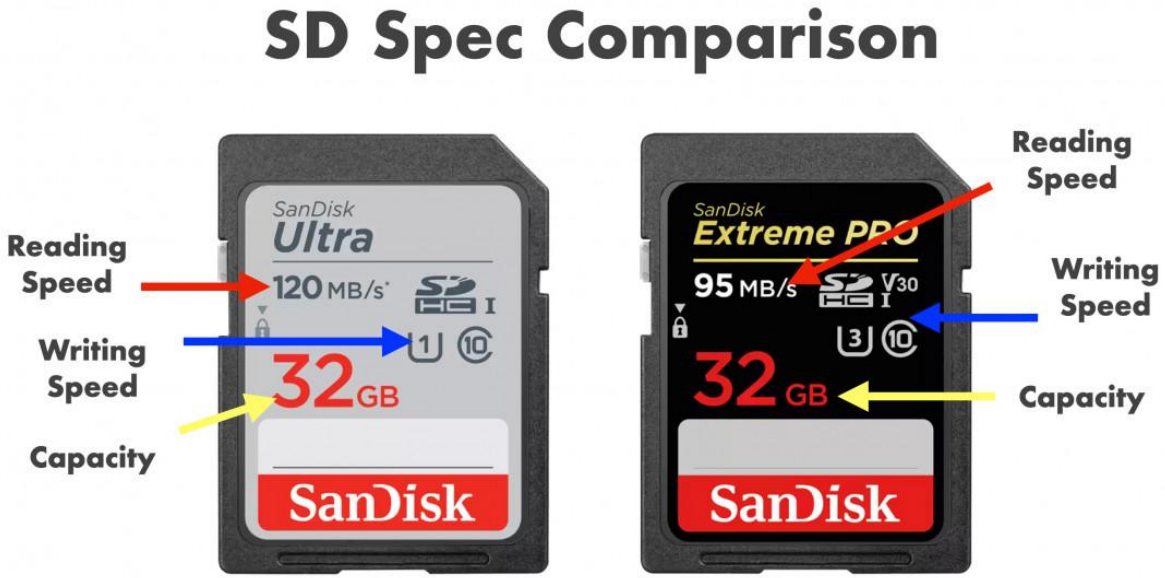






Storage Interfaces

- SD
- MMC
- SDIO







Storage Interfaces

- MMC
- The **Multi Media Card**
- Unveiled in 1997

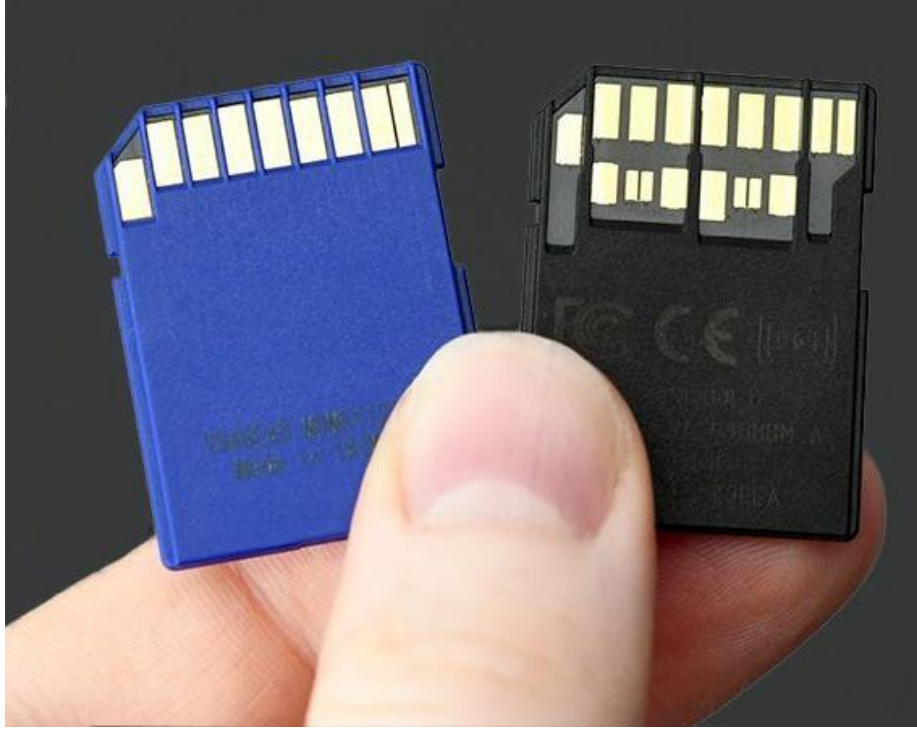
  by SanDisk and Siemens





Storage Interfaces

- SDIO
  - physical SD slot to support peripherals like Wi-Fi, Bluetooth, and GPS
  - Input/output (sends and receives data streams)

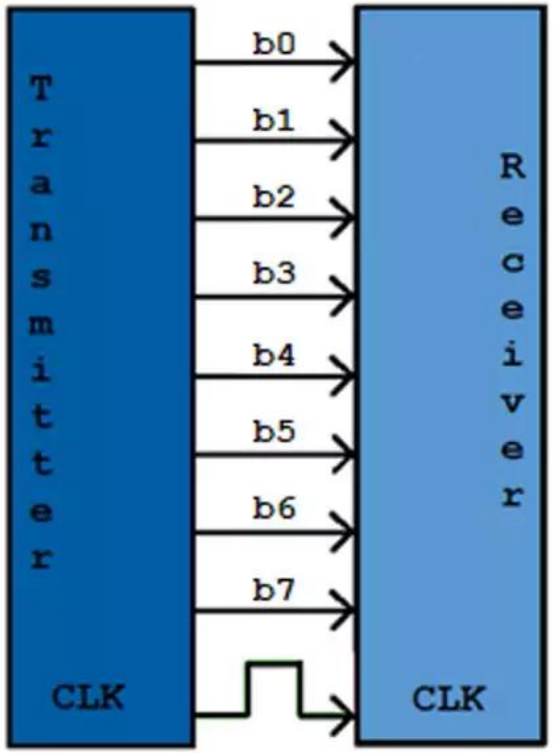


I/O Interfaces

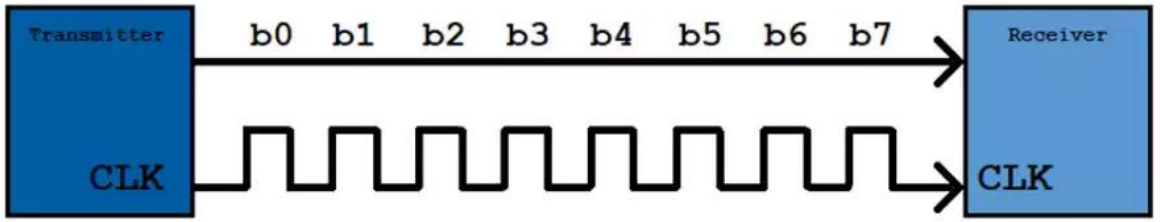
- UART
- SPI
- I2C
- CAN

I/O Interfaces

- Serial Vs Parallel



8-bit Parallel Interface



8-bit Serial Interface

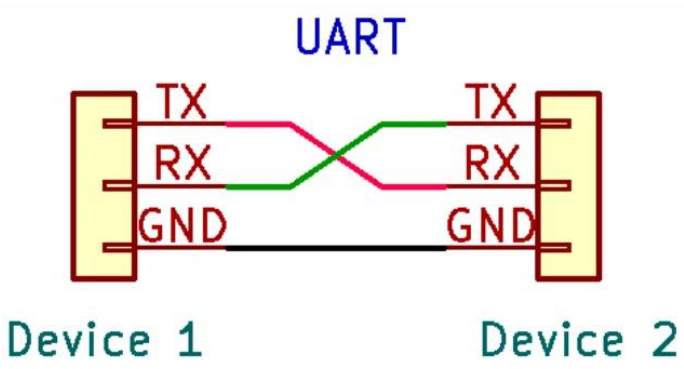
I/O Interfaces

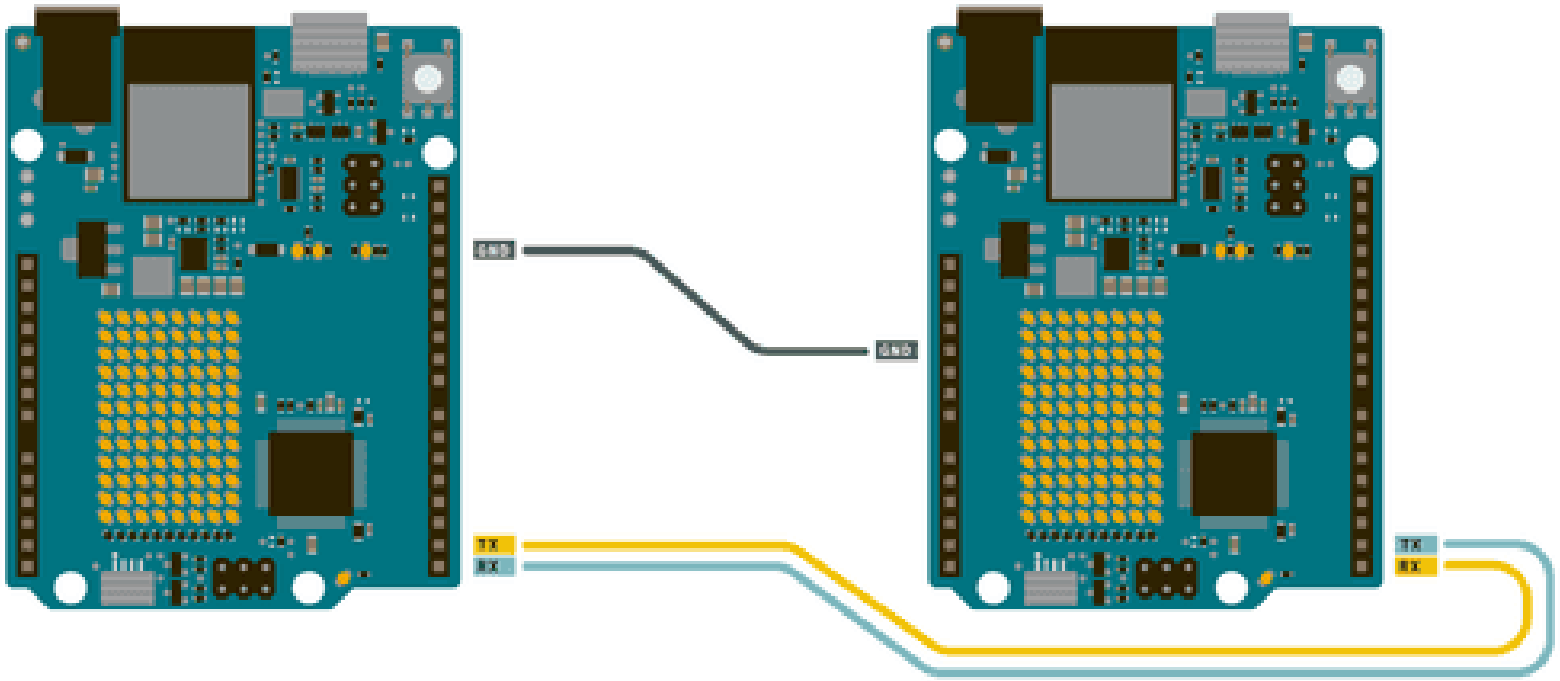
- Synchronous Vs Asynchronous



I/O Interfaces

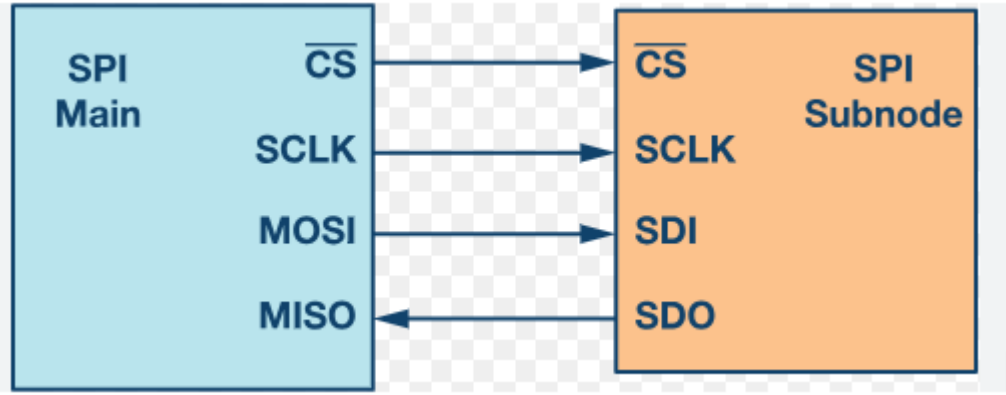
- UART- Universal Asynchronous Receiver Transmitter



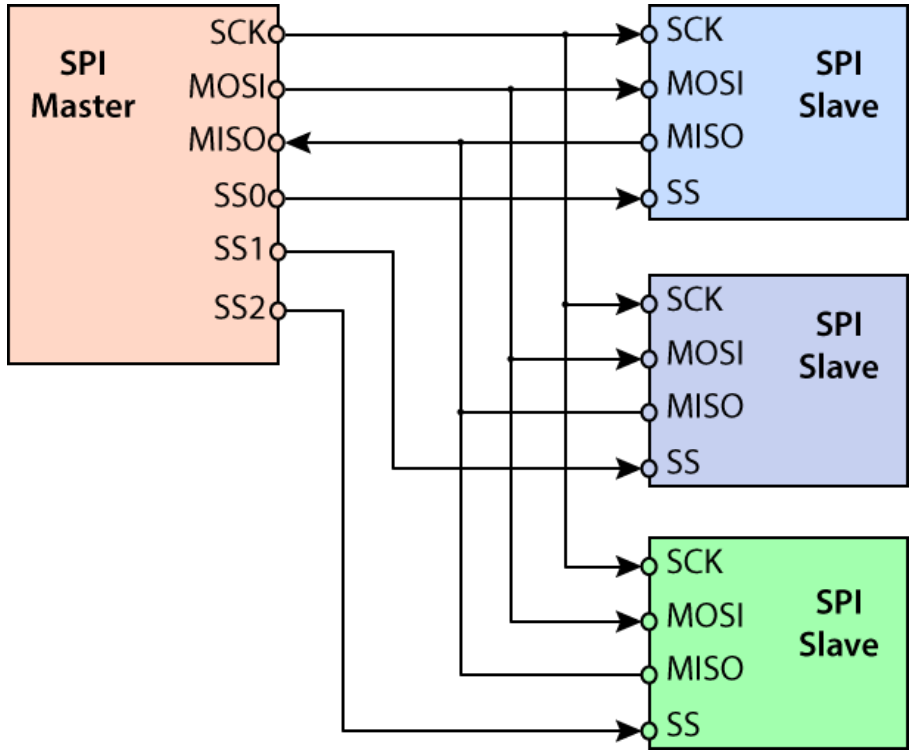


I/O Interfaces

- **SPI**



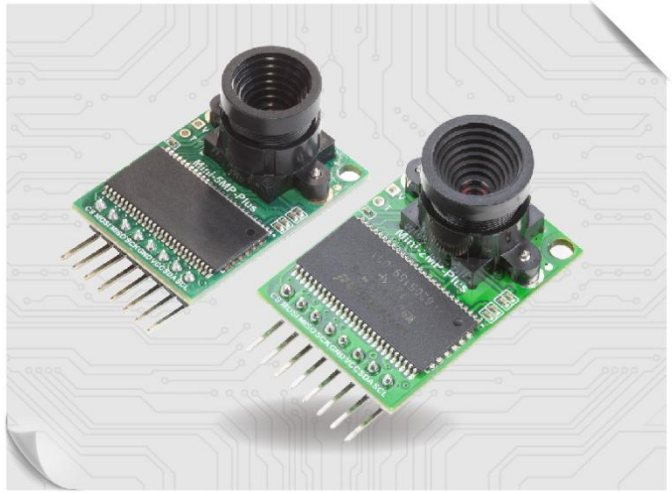
Single Master & single slave
Interface with **SPI**



Single Master & Multi slave interfaced
With SPI

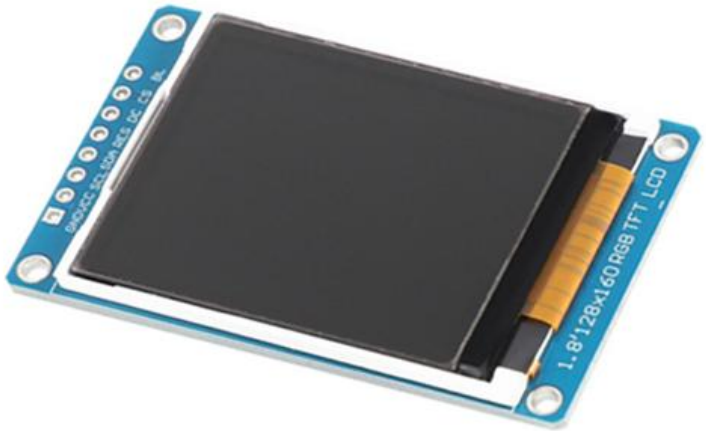
I/O Interfaces

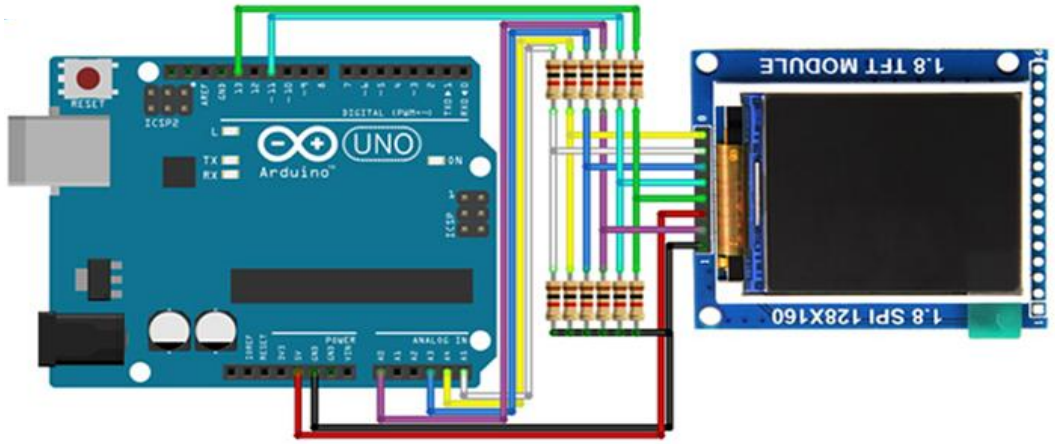
- **SPI**



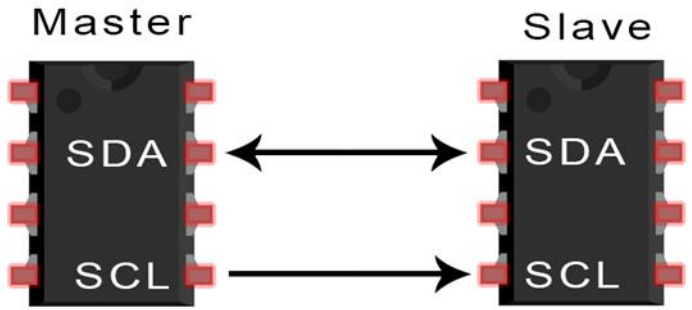
SPI Camera

1.8” SPI TFT Color Display Arduino





I/O Interfaces

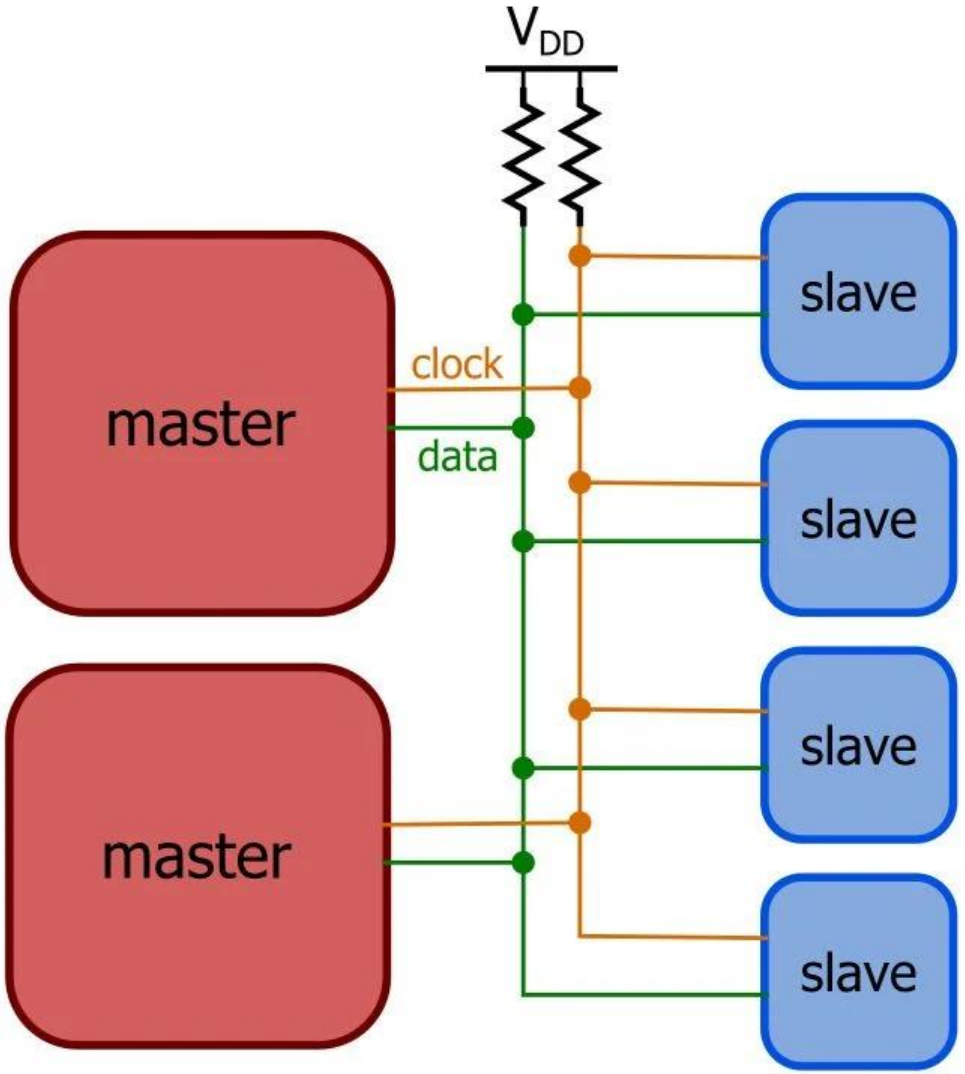


- IIC or I2C-**Inter Integrated Circuit**
- It is a bus interface connection protocol incorporated into devices for serial communication.
- It was originally designed by **Philips Semiconductor** in 1982.
- Recently, it is a widely used protocol **for short-distance** communication. It is also known as **Two Wired** Interface(TWI).

I/O Interfaces

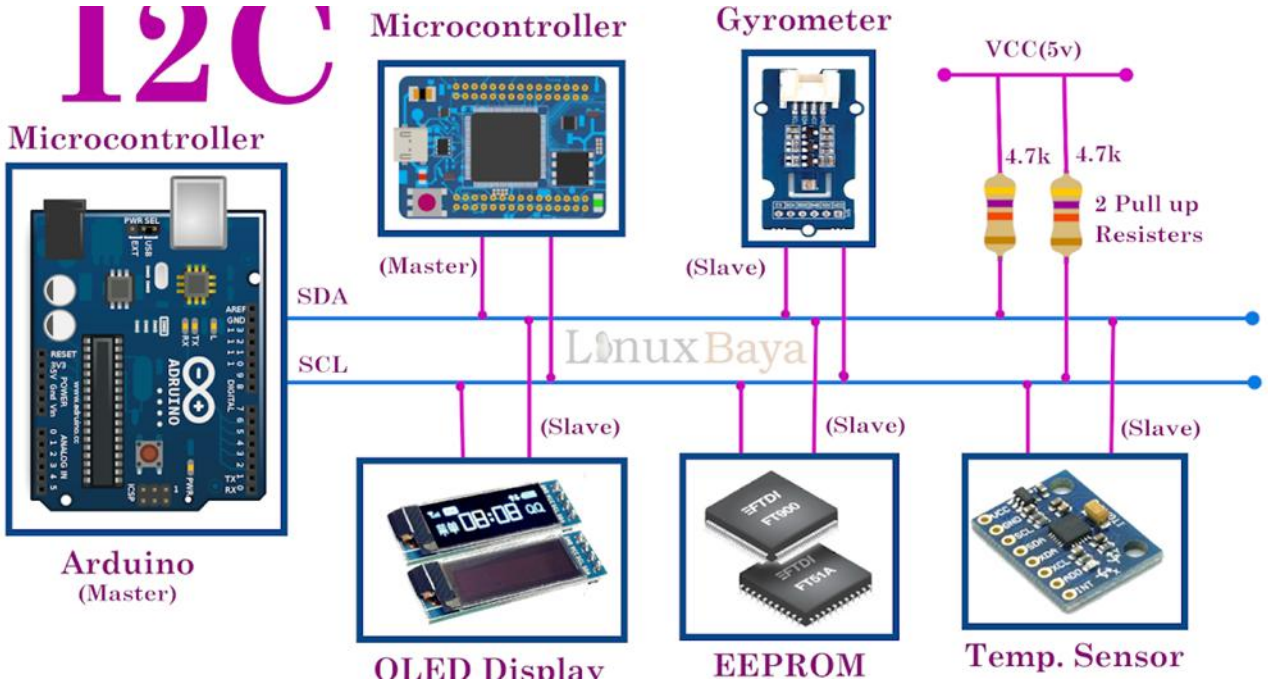
- IIC or I2C-**Inter Integrated Circuit**

Multi Master & Multi slave
Interface with **I2C**



I/O Interfaces

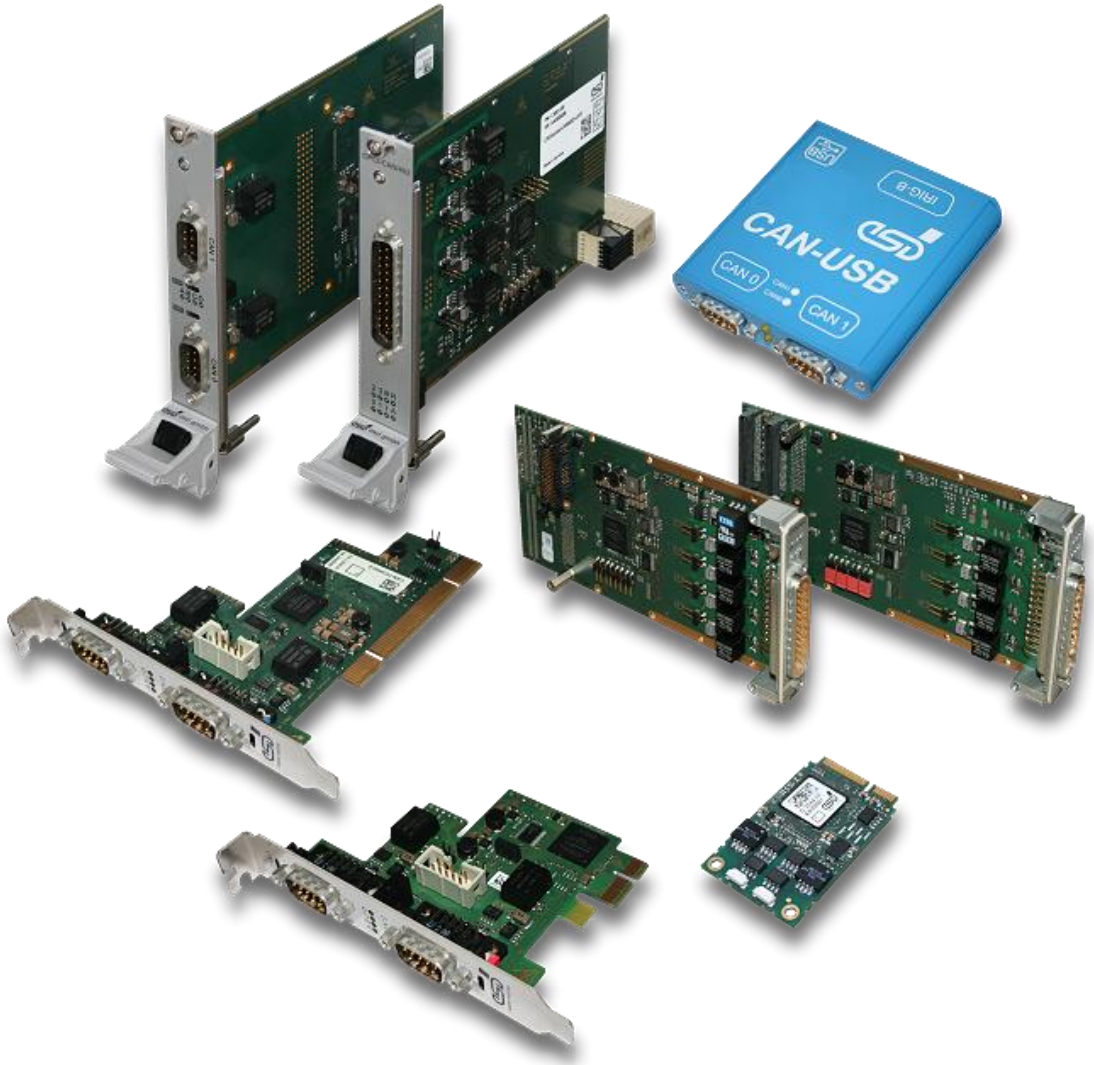
- IIC or I2C-**Inter Integrated Circuit**



Multi Master & Multi slave Interface with **I2C**

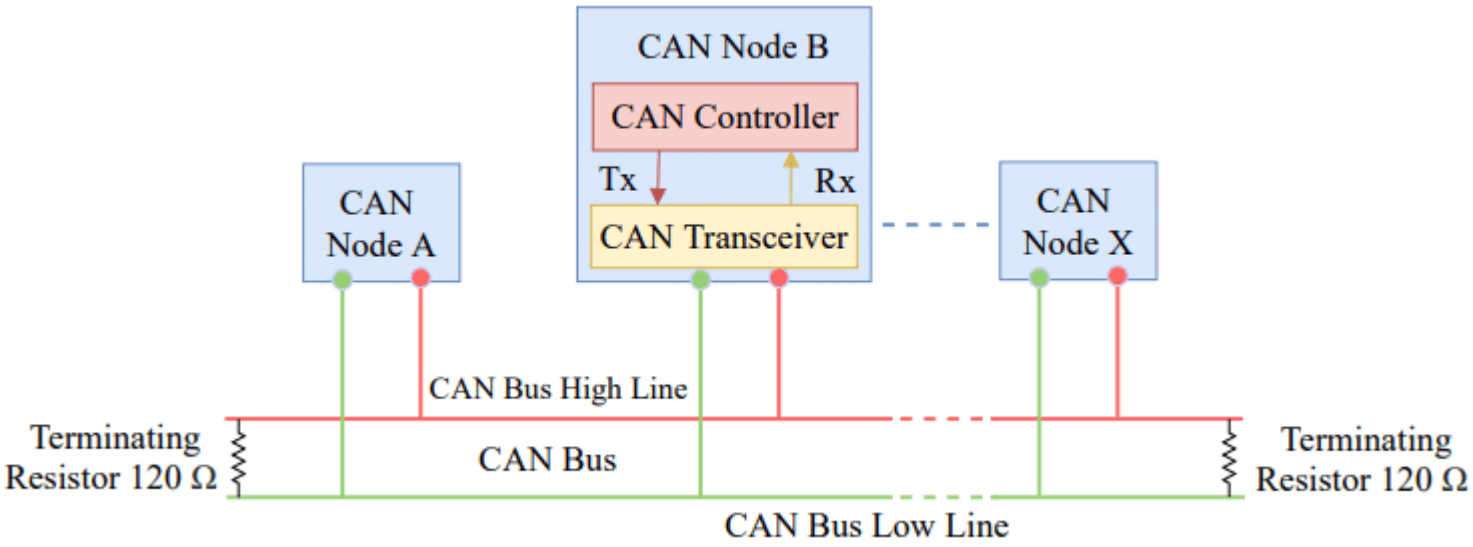
I/O Interfaces

- CAN- Controller area network or
    - Campus area network



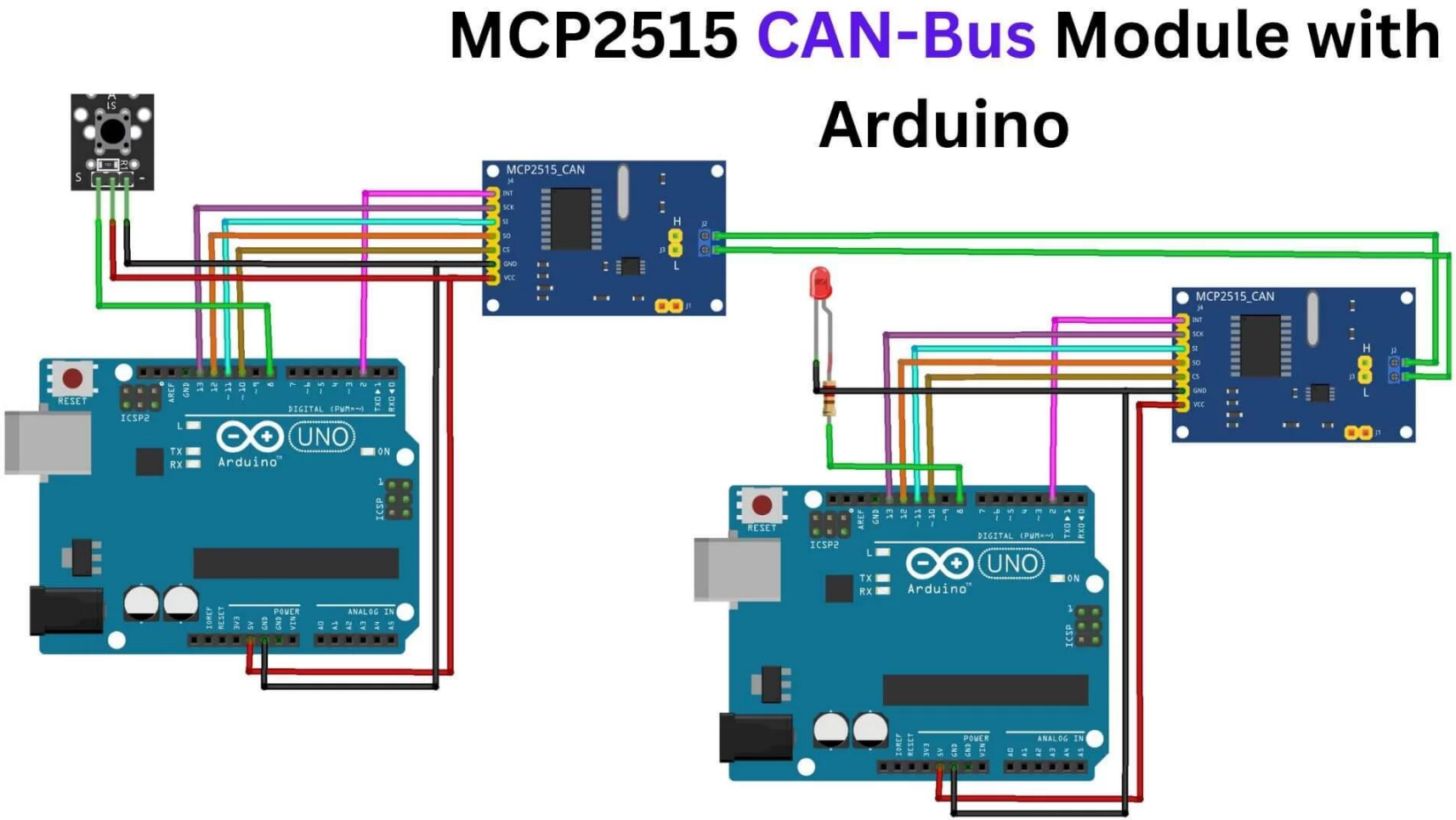
I/O Interfaces

- CAN



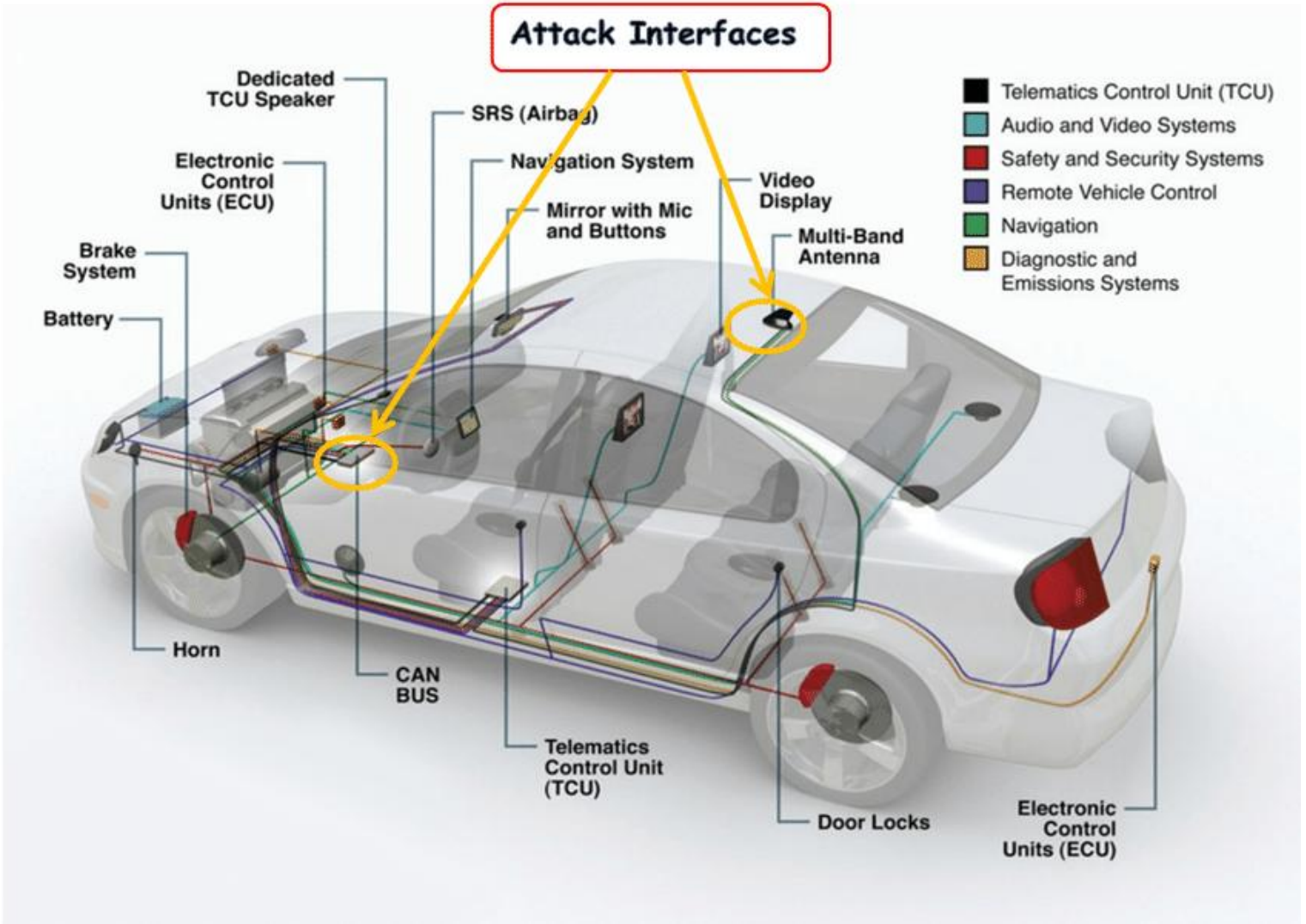
I/O Interfaces

- CAN



I/O Interfaces

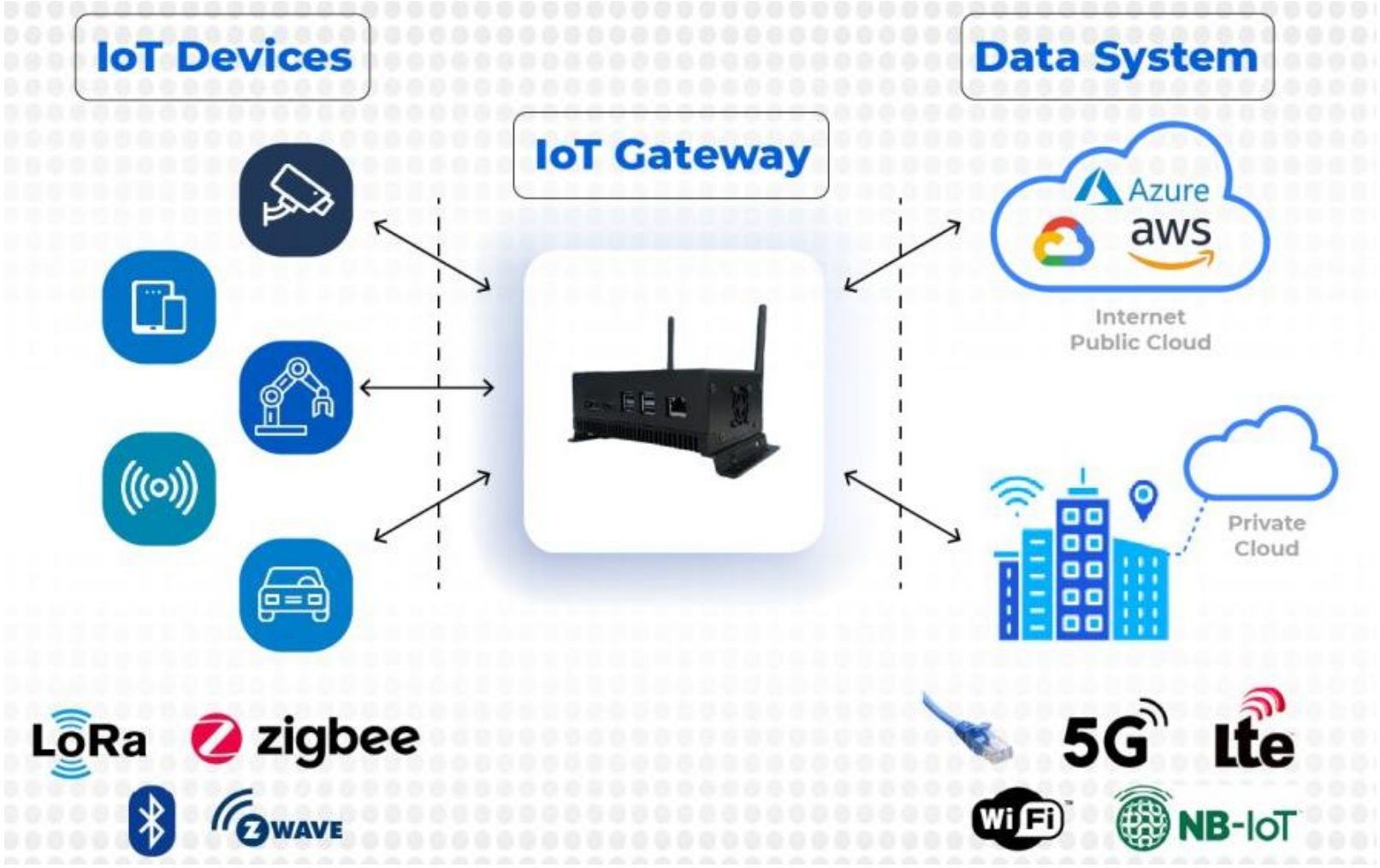
- CAN-



Physical design of IoT: Devices, Gateways, and Data centers

2. IoT Gateway

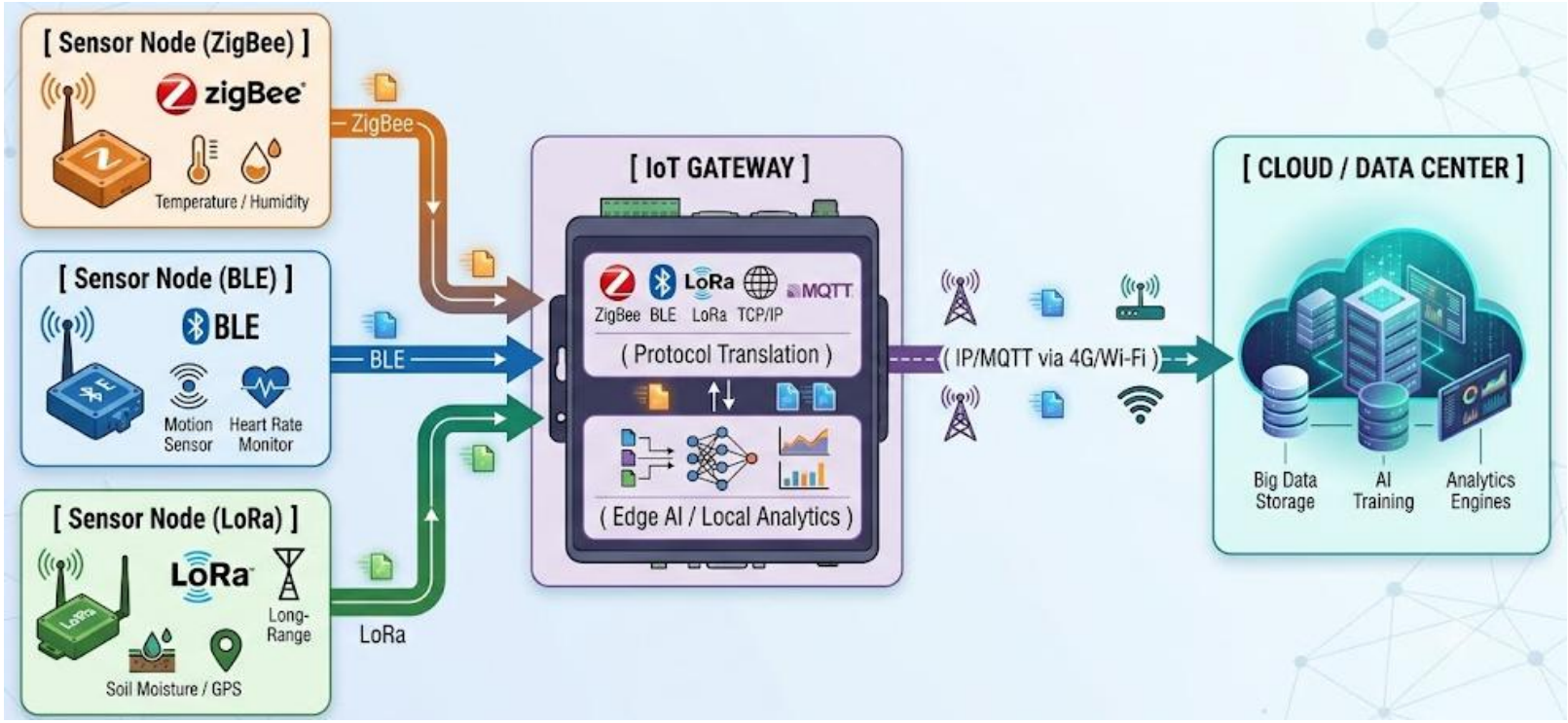
An IoT gateway acts as a bridge or a centralized hub between remote IoT devices and the internet (or central servers).



Physical design of IoT: Devices, Gateways, and Data centers

IoT Gateway capable of

- **Protocol Translation**
- **Cloud Integration**
- **Data Aggregation**
- **Device Management**
- **Edge Computing**
- **Security**

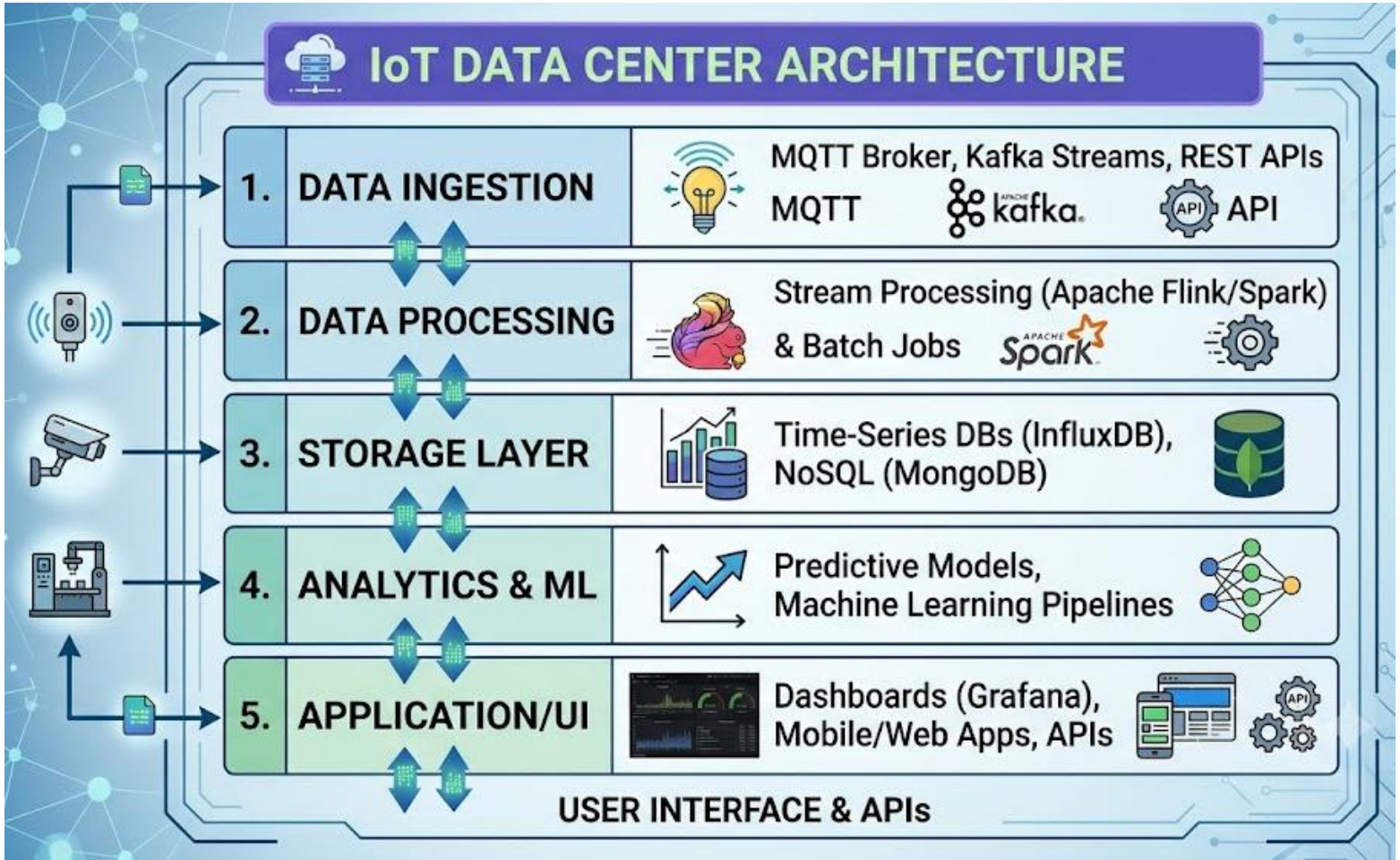


Physical design of IoT: Devices, Gateways, and Data centers

3. Data Centers & Cloud

Data centers represent the backend infrastructure where massive volumes of IoT data are stored, analyzed, and managed.

- **Massive Storage**
- **Heavy Analytics**
- **Data Management & APIs**

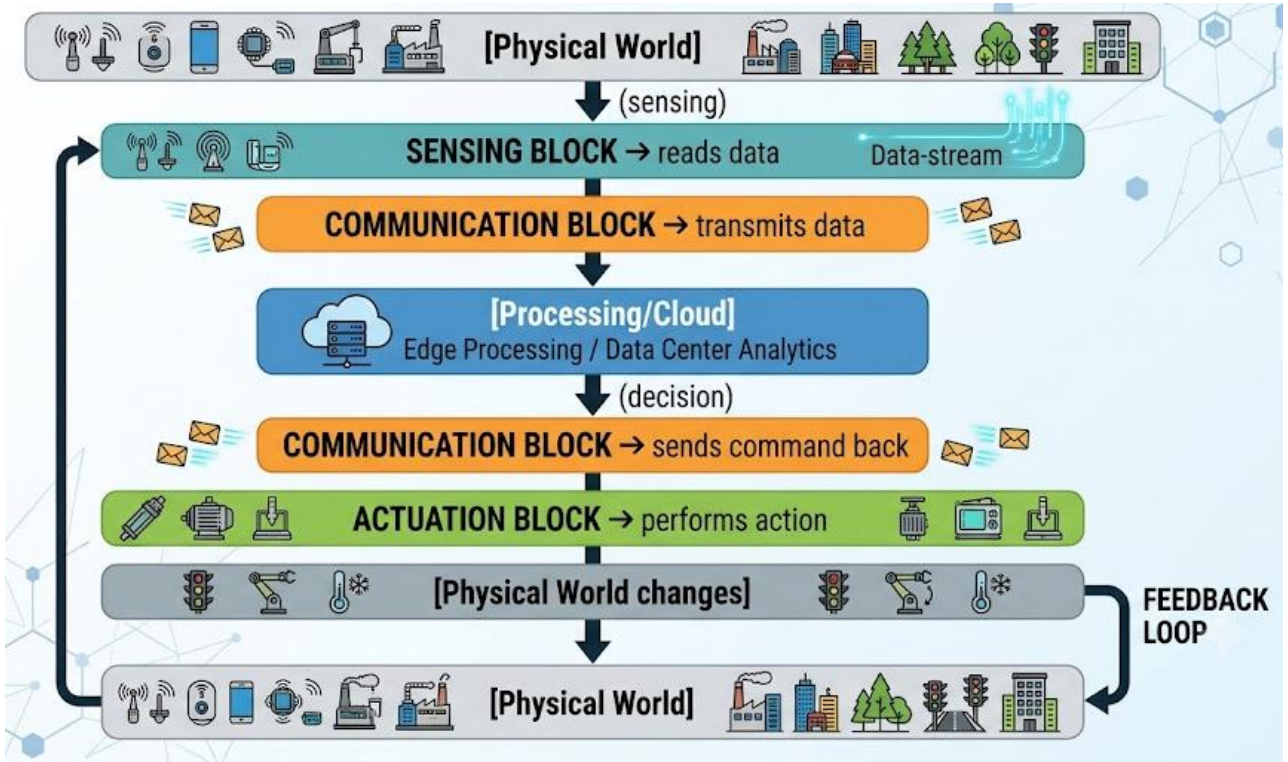


Functional Blocks of IoT: Sensing, Actuation, Communication

Introduction

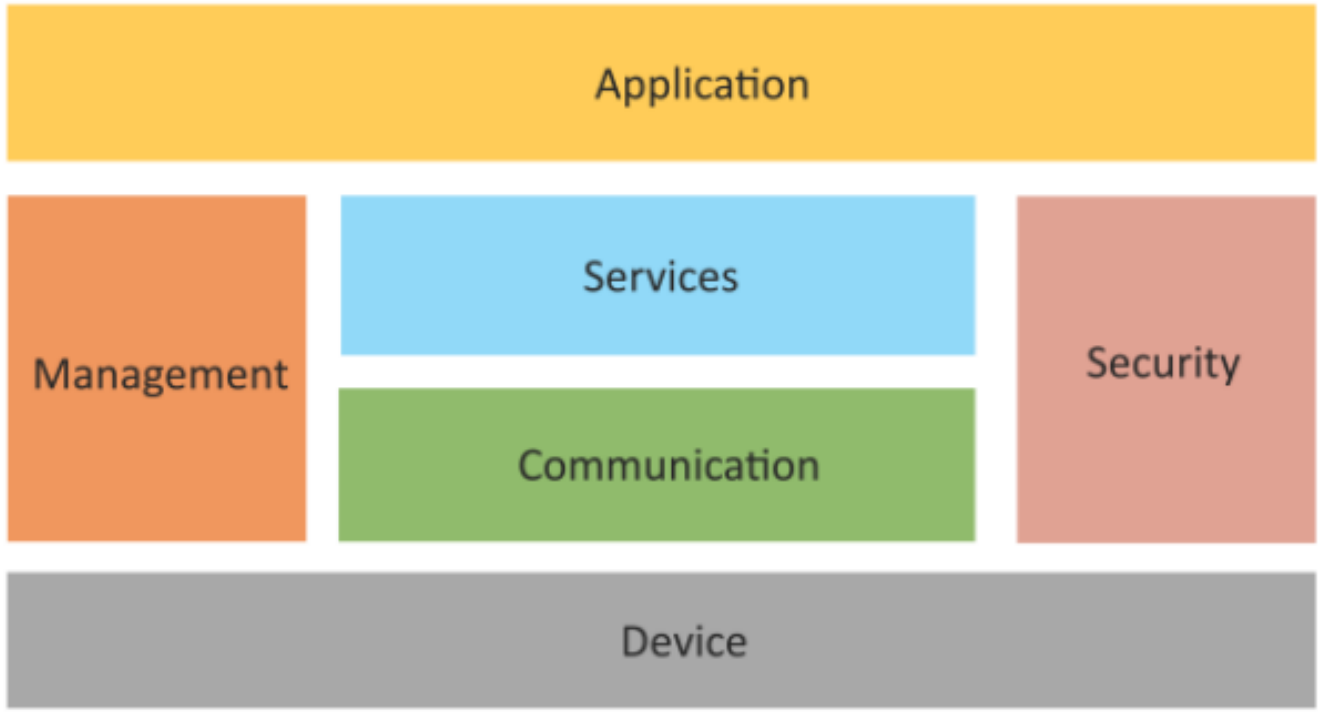
- IoT systems work by connecting the **physical world** to the **digital/cloud world**
- This connection happens through a few main **functional blocks**
- The three basic functional blocks
  - **Sensing,**
  - **Actuation,**
  - **Communication**

- These blocks together allow a device to detect something, send that info somewhere, and respond back with an action



Functional Blocks of IoT: Sensing, Actuation, Communication

Functional Block Diagram of IoT



Functional Blocks of IoT: Sensing, Actuation, Communication

Sensing

- Sensing block = **collects data from surroundings** using sensors
- Converts physical quantity (temp, light, motion, etc.) into electrical/digital signal
- This is the **input** part of the system

Examples of sensors:

- DHT11 – Temperature & Humidity
- LDR – Light
- PIR – Motion
- Ultrasonic sensor – Distance

Functional Blocks of IoT: Sensing, Actuation, Communication

Example of Sensors



Functional Blocks of IoT: Sensing, Actuation, Communication

Actuation

- Actuation block = **performs a physical action** based on a command
- Converts digital signal back into real-world action
- This is the **output** part of the system

Examples of actuators:

- Relay – switches devices ON/OFF
- Motor/Servo – movement
- Buzzer/LED – alert signals
- Water pump – used in irrigation systems

Functional Blocks of IoT: Sensing, Actuation, Communication



Electrical Actuators



Fluid Power Actuators



Mechanical Actuators



Thermal Actuators



Special Actuators

1.Electric Actuators



Servo motor

4.Hydraulic Actuators



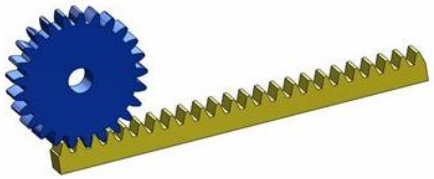
Hydraulic cylinder

2.Pneumatic Actuators



Pneumatic cylinder

5.Mechanical Actuators



Rack and pinion

Functional Blocks of IoT: Sensing, Actuation, Communication

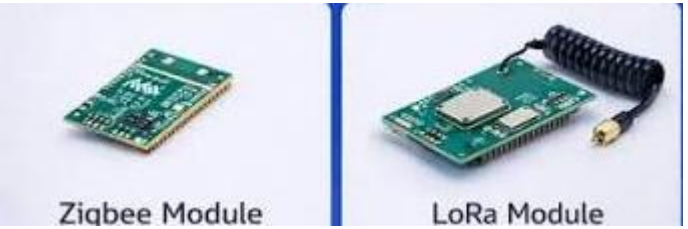
Communication

- Communication block **transfers data** between device and cloud/other devices
- Needed so that sensor data can reach the server, and commands can come back to the actuator
- Involves protocols and wireless modules

Examples:

- Wi-Fi (ESP8266/ESP32)
- Bluetooth
- Zigbee
- MQTT protocol (commonly used for sending sensor data to cloud)





Functional Blocks of IoT: Sensing, Actuation, Communication

Example:

Smart Irrigation System (mini project reference):

1.Sensor detects soil is dry → **Sensing**

2.ESP32 sends this data to cloud app → **Communication**

3.App/logic decides pump should turn ON

4.Command sent back to device → **Communication**

5.Relay switches pump ON → **Actuation**

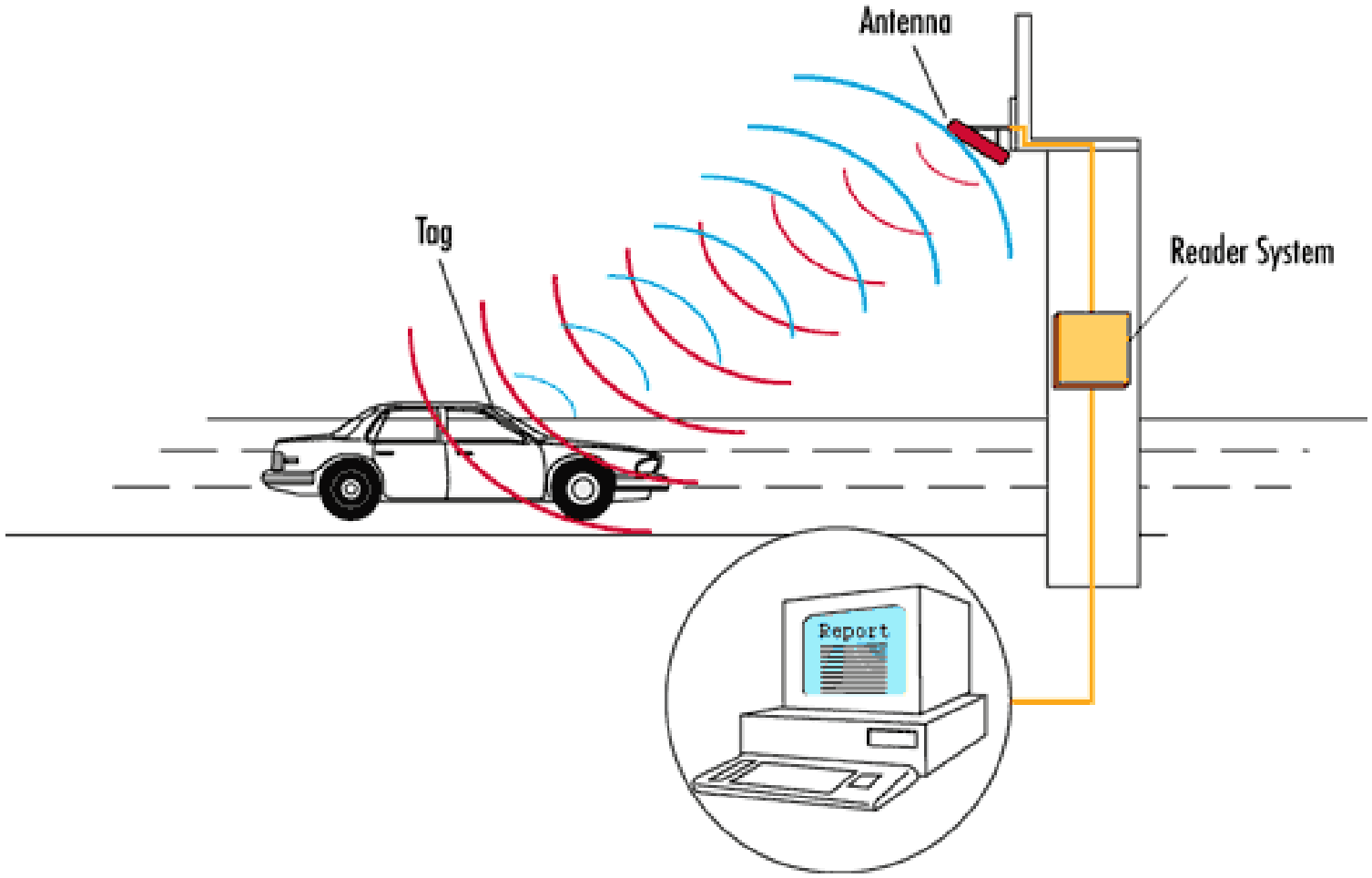
This is basically a loop: **Sense → Send → Decide → Send back → Act**

Functional Blocks of IoT: Sensing, Actuation, Communication

Block	Function	Direction	Example
Sensing	Collects data	Input	Temperature sensor
Actuation	Performs action	Output	Motor, relay
Communication	Transfers data	Both ways	Wi-Fi, MQTT

Enabling Technologies: RFID, Wireless Sensor Networks, Nanotechnology

1. Radio Frequency Identification (RFID)



Enabling Technologies: RFID, Wireless Sensor Networks, Nanotechnology

1. Radio Frequency Identification (RFID)

Operating Principles

- RFID enables contactless, automatic object identification and tracking using electromagnetic fields.
- A typical system consists of a reader (interrogator) emitting radio signals and a tag (transponder) attached to physical assets.

Passive vs. Active RFID

- Passive tags harvest energy directly from the reader's RF signal (zero onboard battery, lower cost, shorter range).
- Active tags carry onboard batteries, facilitating long-range communication (up to 100+ meters) and sensor integration.

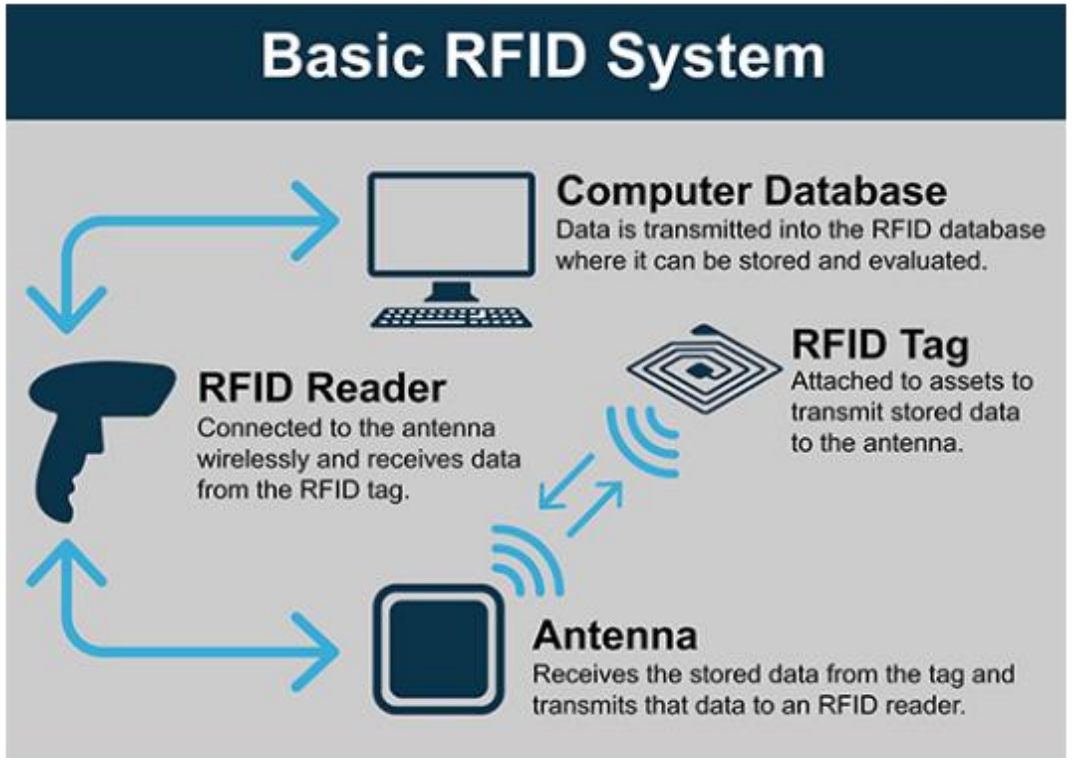
Enabling Technologies: RFID, Wireless Sensor Networks, Nanotechnology

RFID Reader & Tag Interaction

Transponder Signal Loop: The RFID reader antenna transmits a continuous RF carrier wave to query surrounding tags.

Backscatter Modulation: Passive tags modulate and reflect the reader's signal to transmit stored unique identity codes (EPC/UID).

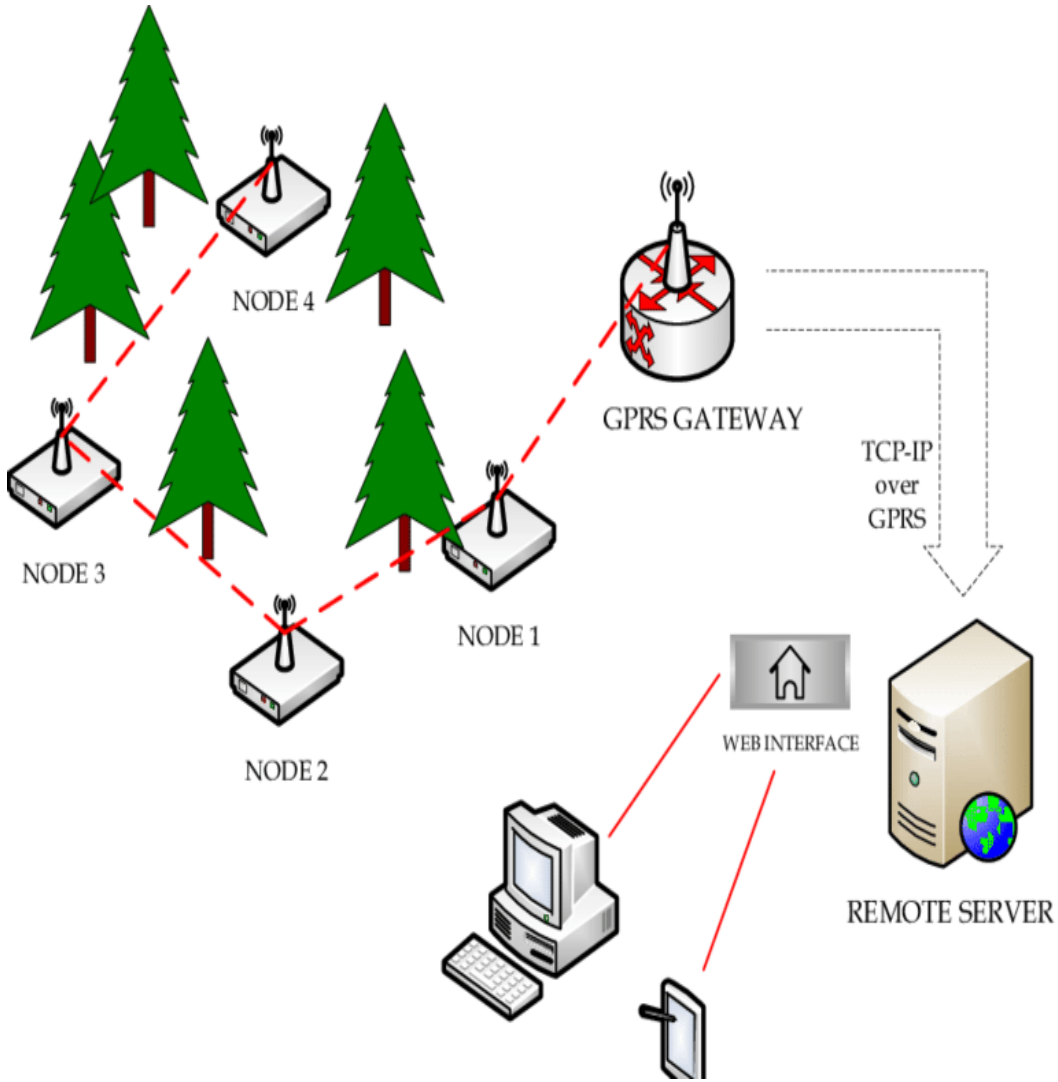
High Data Throughput: Enables simultaneous reading of hundreds of items per second without direct line-of-sight requirement.



WSN Node & Network Topology

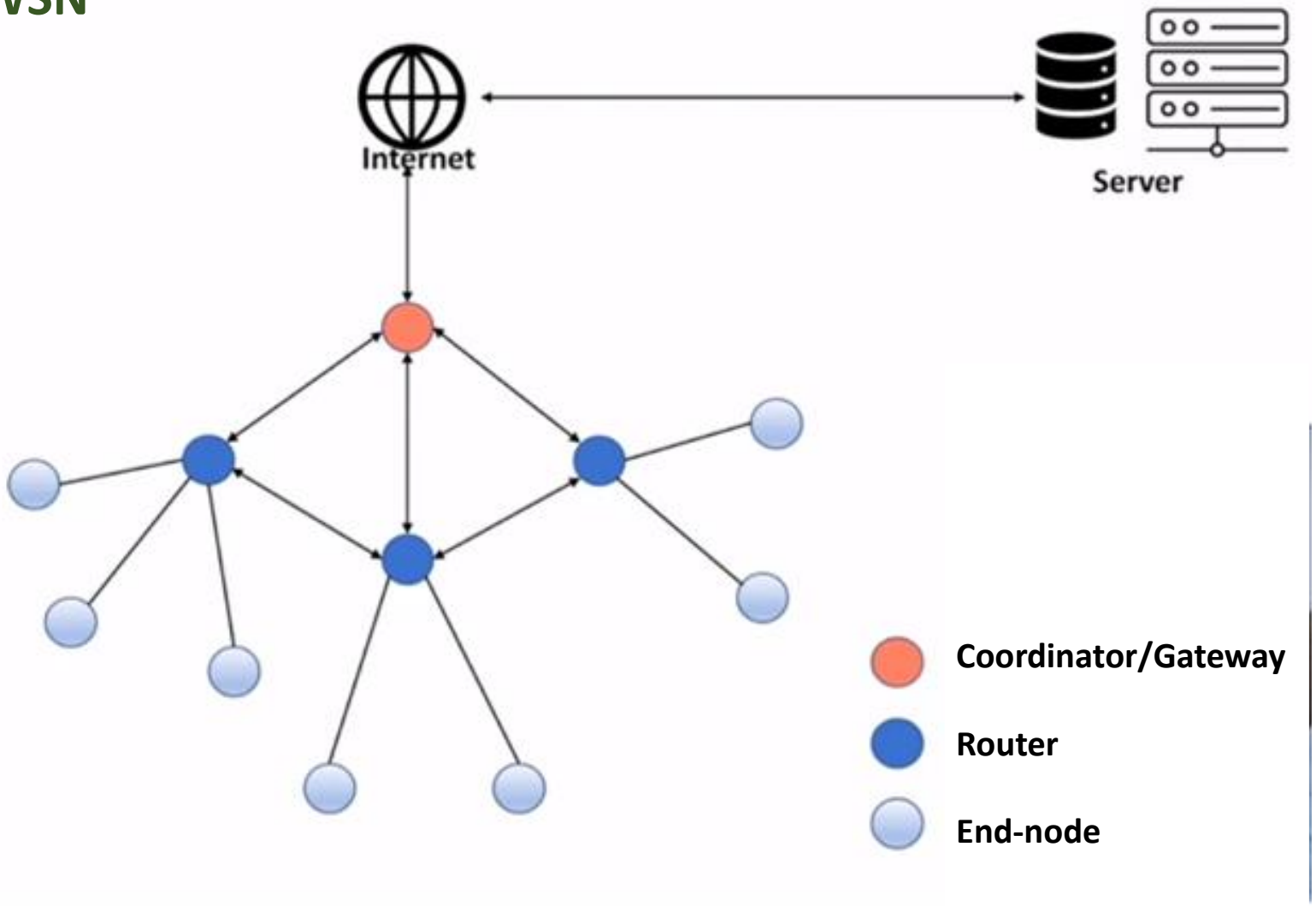
Autonomous Sensor Nodes

- A Wireless Sensor Network (WSN) consists of spatially distributed sensor nodes containing processing units, RF transceivers, power sources, and physical sensors.
- Nodes self-organize into mesh or star topologies, routing environmental metrics (temperature, vibration, pressure) across multiple hops to a central Sink Node or Gateway.



Enabling Technologies: RFID, Wireless Sensor Networks, Nanotechnology

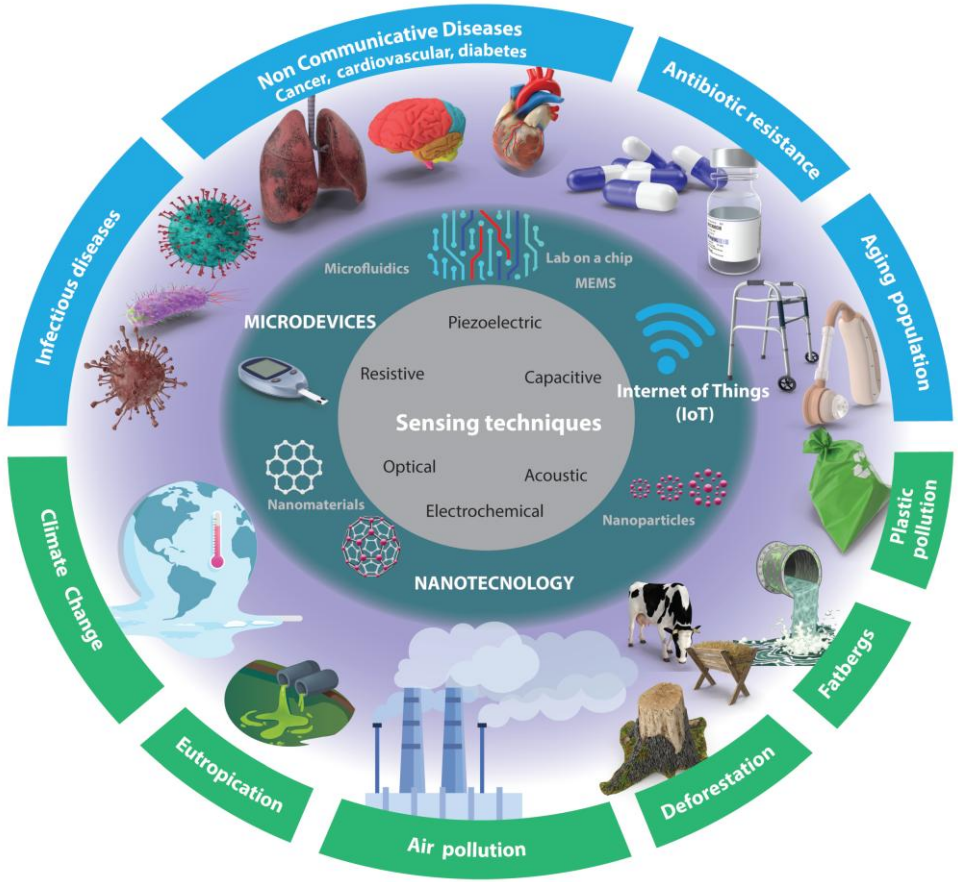
Basic Architecture of WSN



Nanomaterials in Sensing Components

Nanotechnology improves IoT mainly in these ways:

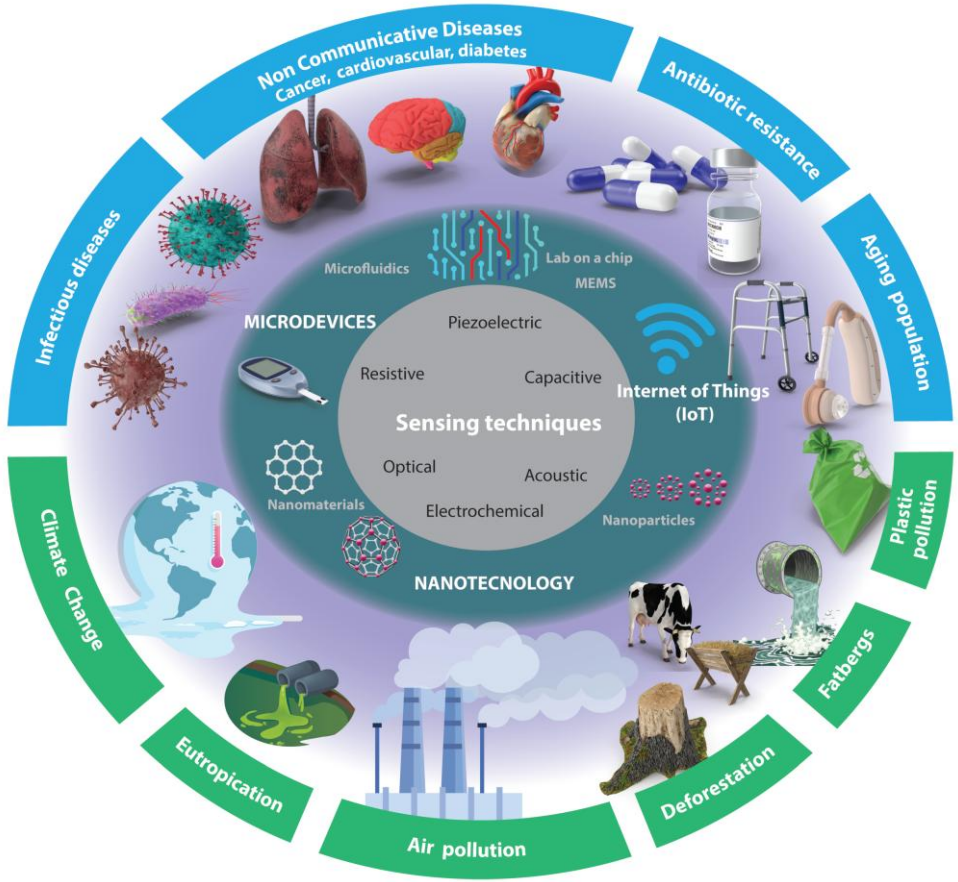
1. **Nano-sensors** – Detect gases, chemicals, biomarkers, or structural stress at very fine scales (used in health wearables, agriculture, pollution monitoring, structural monitoring).
2. **Miniaturization** – Nanoscale fabrication shrinks chips, antennas, and circuits, enabling smaller, lighter IoT devices.
3. **Energy** – Nanomaterials improve battery density and enable energy harvesting (e.g., nanogenerators from motion/heat), extending device life.



Nanomaterials in Sensing Components

Nanotechnology improves IoT mainly in these ways:

4. **Better connectivity components** – Nanomaterials (e.g., carbon nanotubes) enable smaller, more efficient antennas and RF parts.
5. **Nanocoatings** – Protect sensors from corrosion/moisture, useful in industrial or outdoor IoT.
6. **Internet of Nano Things (IoNT)** – Emerging concept of networked nanoscale devices (e.g., in-body nanosensors) using novel communication methods; still mostly research-stage.

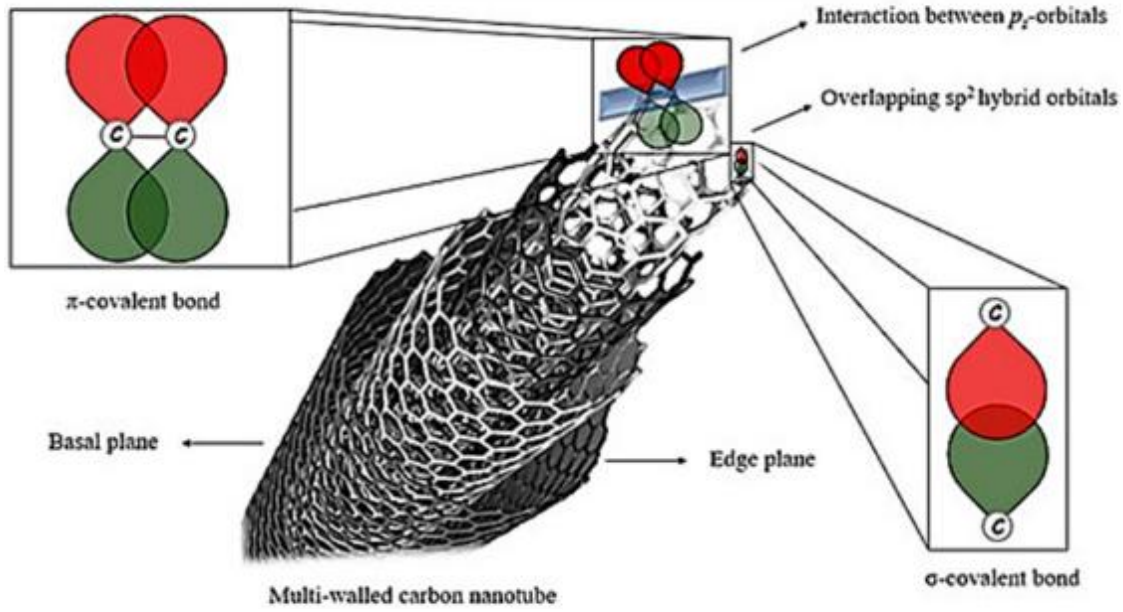


Nanomaterials in Sensing Components

Ultra-High Sensitivity: Carbon nanotubes (CNTs) and graphene offer immense surface area, detecting single gas molecules or biochemical traces.

Flexible RFID Antennas: Inkjet-printed metallic nanoparticles allow high-conductivity RF antennas on flexible substrate materials.

Energy Harvesting: Piezoelectric nanogenerators convert structural micro-vibrations into usable electrical energy for passive nodes.



Enabling Technologies: RFID, Wireless Sensor Networks, Nanotechnology

Parameter	RFID Technology	Wireless Sensor Networks	Nanotechnology Sensors
Primary Role	Identification & Tracking	Environmental Monitoring	Molecular / Trace Sensing
Network Topology	Point-to-point (Reader to Tag)	Multi-hop Mesh / Ad-hoc	Integrated on RFID/WSN nodes
Power Requirement	Passive (RF Harvest) / Active	Battery powered / Solar	Ultra-low / Self-powered
Sensing Scope	Presence, Location, ID	Physical macro-parameters	Chemical, Biological, Strain

Basics of Networking

01

Introduction

What is a network, and why does it matter?

02

Network Types

PAN, LAN, MAN and WAN classifications

03

Topologies

Bus, star, ring, and mesh arrangements

04

OSI & TCP/IP Models

Layered architecture of communication

05

Devices & Media

Hubs, switches, routers, and transmission media

06

Addressing & Protocols

IP addressing, TCP, UDP, and common protocols

01

What is a Computer Network?



A network turns isolated machines into a connected community of shared resources.

Definition

A computer network is a collection of interconnected devices — computers, servers, and peripherals — that communicate and share resources using a common set of rules called protocols.

Resource Sharing

Files, printers, and storage shared across devices

Communication

Email, messaging, and video conferencing

Data Access

Centralized databases and cloud services

Reliability

Redundant paths keep data flowing

02

Types of Networks by Scale/Range

PAN

Personal Area Network

A few metres — Bluetooth, wireless earphones, smartwatches

LAN

Local Area Network

One building or campus — office or college network

MAN

Metropolitan Area Network

A city or town — cable TV or metro-wide links

WAN

Wide Area Network

Countries or continents — the Internet itself

03

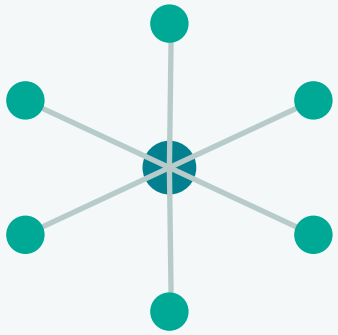
Network Topologies

The physical or logical arrangement of nodes in a network



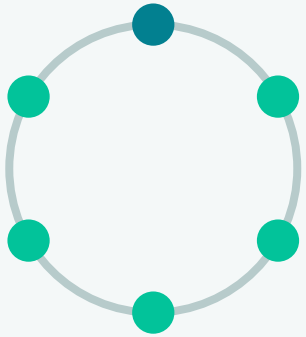
Bus

All nodes share one central cable; simple but a single break stops the network.



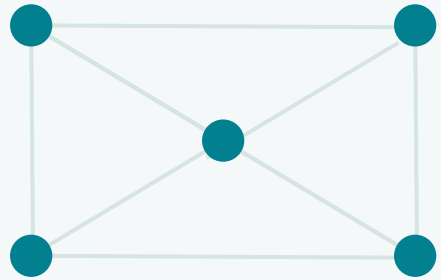
Star

Every node connects to a central hub or switch; easy to manage and isolate faults.



Ring

Each node connects to exactly two neighbours, forming a closed loop for data flow.



Mesh

Nodes are richly interconnected, giving multiple paths and high fault tolerance.

DATA TRANSMISSION AND RECEPTION USING 7-LAYER OSI MODEL



SENDER (CLIENT)

PDU (Sender)

PDU (Receiver)

RECEIVER (SERVER)



ENCAPSULATION

DECAPSULATION

		PDU (Sender)		PDU (Receiver)		
7 APPLICATION	User creates data (e.g., HTTP Request)	DATA		DATA	Delivers Data to the User/ Application	7 APPLICATION
6 PRESENTATION	Encryption, Compression, Formatting	DATA		DATA	Decrypts and Decompresses Data	6 PRESENTATION
5 SESSION	Session Establishment	DATA		DATA	Restores Session	5 SESSION
4 TRANSPORT	TCP / UDP, Segmentation, TCP/UDP, Port Numbers	SEGMENT		SEGMENT	TCP / UDP, Reassembles Segments using TCP/UDP	4 TRANSPORT
3 NETWORK	IP, Adds Source & Destination IP Addresses	PACKET		PACKET	IP, Reads IP Address and removes Network Header	3 NETWORK
2 DATA LINK	MAC, Adds MAC Addresses and Frame Header/Trailer	FRAME		FRAME	MAC, Removes Frame Header/Trailer	2 DATA LINK
1 PHYSICAL	Converts Frame into Bits (0s and 1s) and transmits over cable/wireless	BITS		BITS	Receives Bits	1 PHYSICAL

NETWORK (INTERNET)

0 1 1 0 1 0 1 1 0
1 0 0 1 1 0 0 1 0

BITS → ELECTRICAL SIGNALS / RADIO WAVES / OPTICAL SIGNALS

HTTP	Application (e.g., HTTP)	Encryption (Presentation)	Session Management	TCP / UDP	Transport Protocols	IP	IP Address (Network)	MAC	MAC Address (Data Link)	1010 0101	Binary Data (Physical)	Medium (Cable/Wireless)

04

The OSI Reference Model

Seven layers that standardize how data travels from one device to another

7	**Application**	User-facing services — browsers, email clients
6	**Presentation**	Data formatting, encryption, compression
5	**Session**	Establishes and manages communication sessions
4	**Transport**	End-to-end delivery — TCP and UDP
3	**Network**	Logical addressing and routing — IP
2	**Data Link**	Node-to-node delivery and error detection
1	**Physical**	Raw bit transmission over cables or air



Remember the order

"All People Seem To Need Data Processing" — top layer to bottom, one word per layer.

Data is wrapped in a header at each layer going down — this is called encapsulation.

05

OSI vs TCP/IP Model

TCP/IP is the practical, four-layer model that actually runs the Internet

OSI (7 Layers)	TCP/IP (4 Layers)	
Application	**Application**	HTTP, FTP, DNS, SMTP
Presentation		
Session		
Transport	**Transport**	TCP, UDP
Network	**Internet**	IP, ICMP, ARP
Data Link	**Network Access**	Ethernet, Wi-Fi
Physical		

06

Common Networking Devices



Hub

Broadcasts incoming data to every connected port — no intelligence, lots of collisions.



Switch

Learns device addresses and forwards data only to the intended port — efficient LAN traffic.



Router

Connects different networks and chooses the best path for data between them.



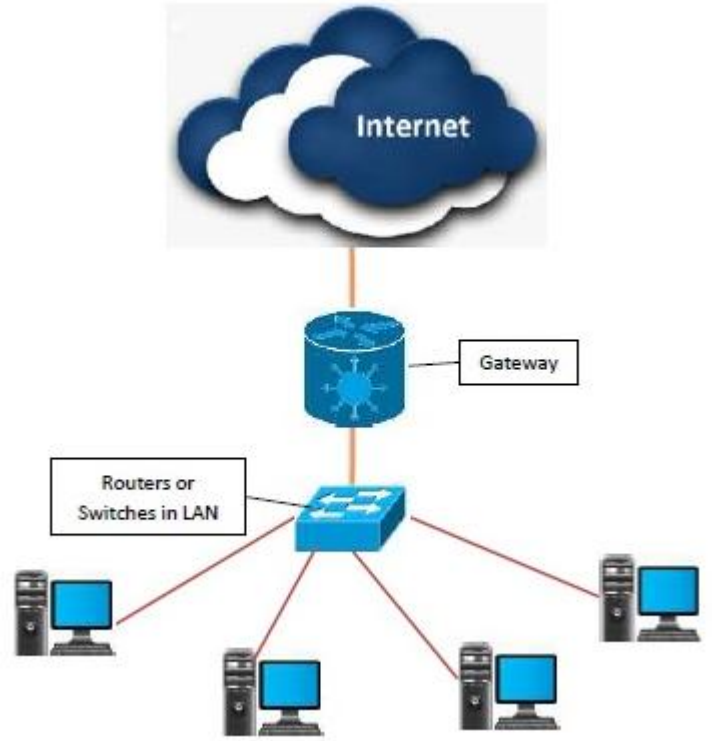
Gateway

Translates between networks using different protocols — the exit point to another system.

06

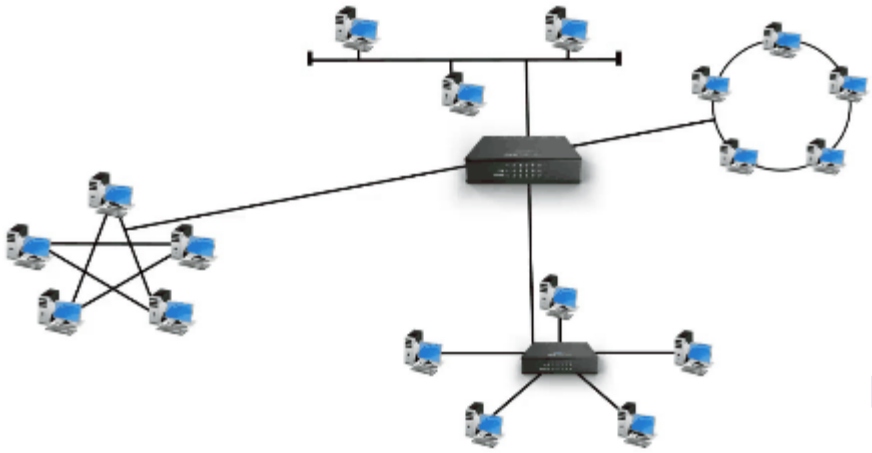
Common Networking Devices





Switch





Router

07

IP Addressing Basics

Every device on a network needs a unique address to send and receive data

IPv4

`192.168.1.10`

- 32-bit address, written as four decimal octets
- Around 4.3 billion unique addresses possible
- Still the most widely deployed addressing scheme

IPv6

`2001:db8::1a2b`

- 128-bit address, written in hexadecimal groups
- A virtually inexhaustible address space
- Designed to replace IPv4 as devices multiply

08

Transmission Media

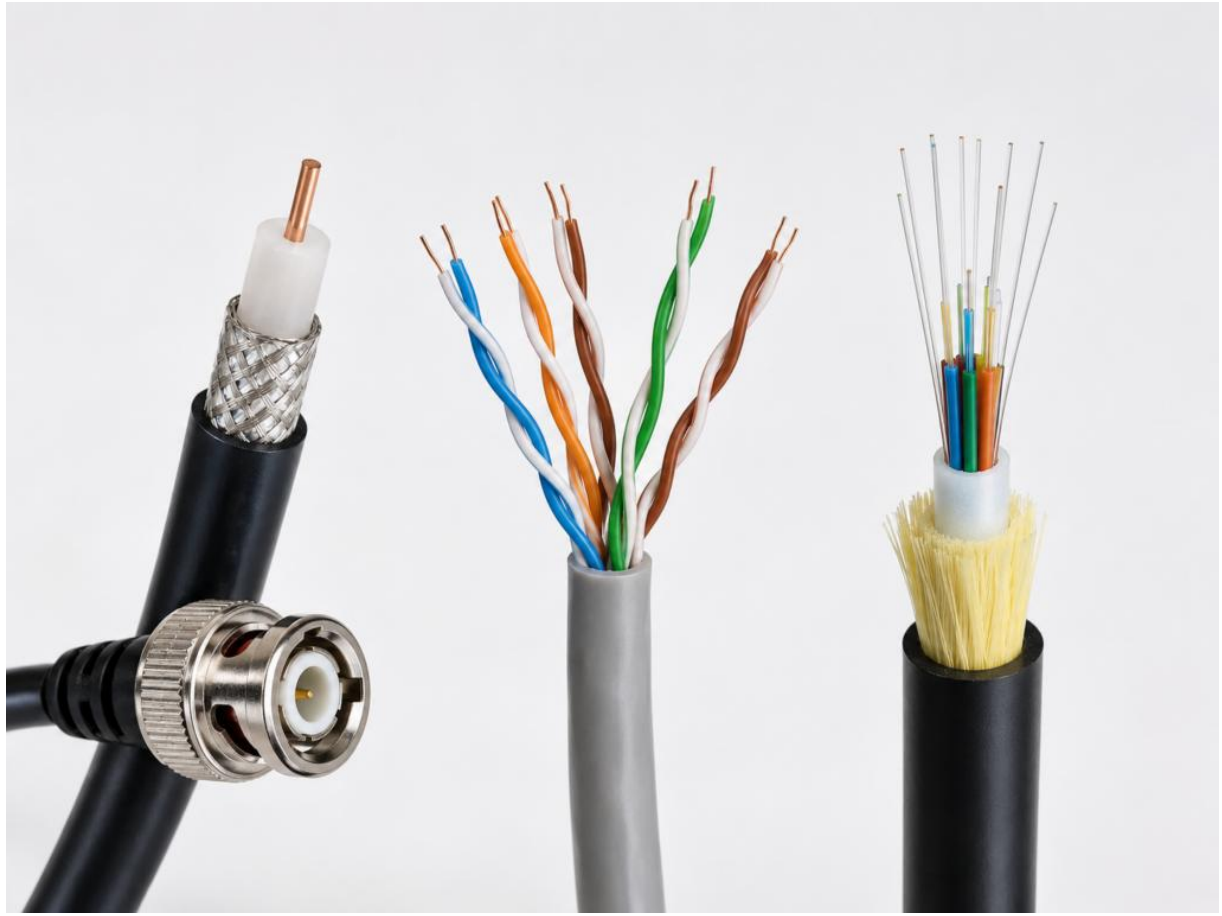
The physical or wireless path data travels between devices

Guided (Wired) Media

Twisted Pair Cable
Copper wire pairs twisted together; used in Ethernet LANs

Coaxial Cable
Shielded copper core; used for cable TV and older LANs

Fiber Optic Cable
Light pulses through glass strands; very high speed, long range



Transmission Media

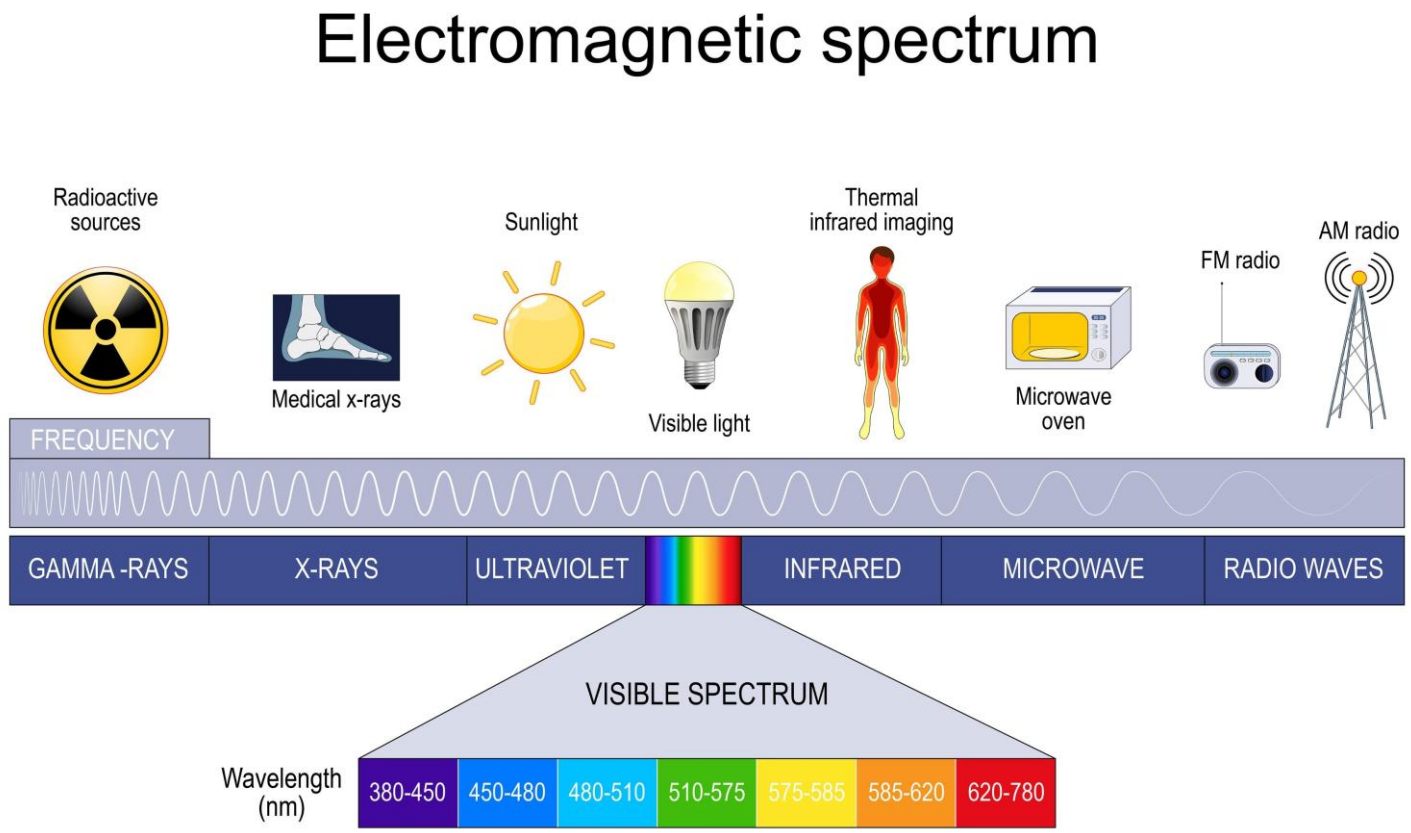
The physical or wireless path data travels between devices

Unguided (Wireless) Media

Radio Waves
Omnidirectional signals — Wi-Fi and mobile networks

Microwaves
Directional line-of-sight links — satellite and towers

Infrared
Short-range signals — remote controls, some sensors



09

Common Network Protocols

Protocols are the agreed rules that let different systems talk to each other

TCP
Transmission Control Protocol
Reliable, connection-oriented delivery with error checking

UDP
User Datagram Protocol
Fast, connectionless delivery — no guarantee, low overhead

HTTP/HTTPS
Hypertext Transfer Protocol
Loads web pages; HTTPS adds encryption for security

FTP
File Transfer Protocol
Uploads and downloads files between client and server

DNS
Domain Name System
Translates domain names into IP addresses

SMTP
Simple Mail Transfer Protocol
Sends email messages between mail servers

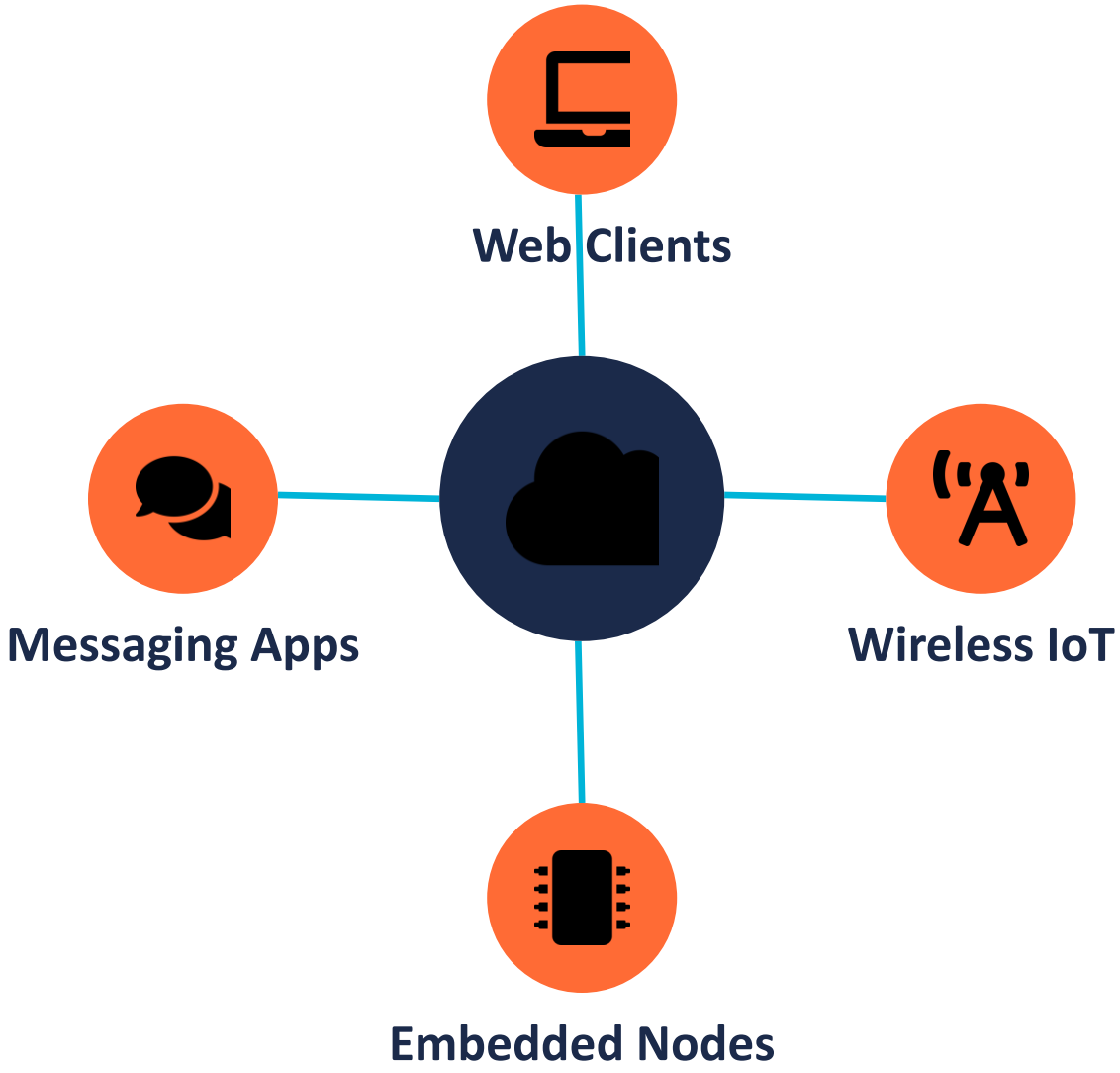
Communication Protocols: MQTT, CoAP, ZigBee, HTTP

Agenda

1. Introduction to Communication Protocols
2. HTTP — The Web’s Request/Response Protocol
3. MQTT — Lightweight Publish/Subscribe Messaging
4. CoAP — REST for Constrained Devices
5. ZigBee — Low-Power Wireless Mesh Networks
6. Comparative Analysis
7. Applications & Key Takeaways

Why Communication Protocols?

- Enable devices & systems to exchange data reliably
- Define rules for formatting, addressing & transmitting data
- Different applications demand different trade-offs
- Power, bandwidth, range & latency drive protocol choice



HTTP — HyperText Transfer Protocol

- Foundation protocol of the World Wide Web
- Follows a request–response communication model
- Stateless — each request is independent
- Runs over TCP, typically port 80 or 443

Key Facts

Layer: Application

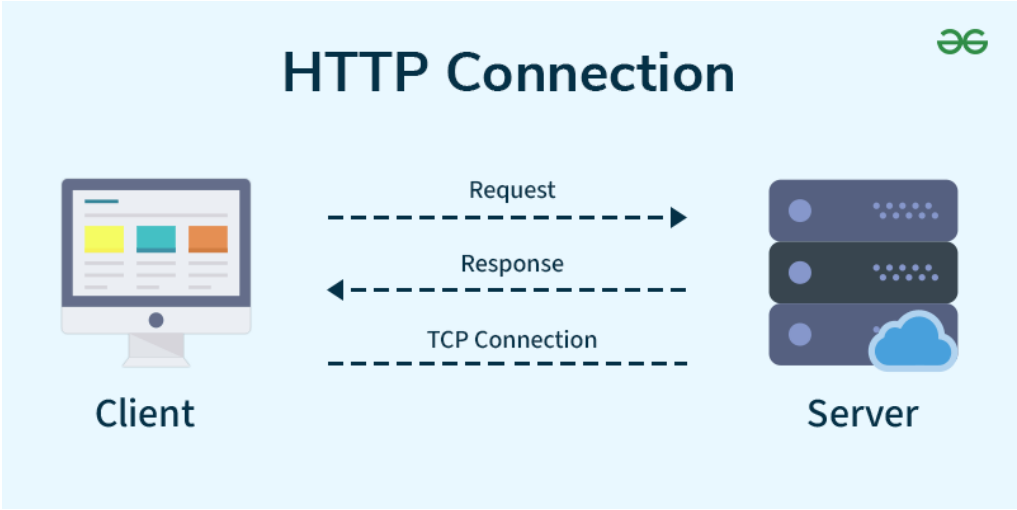
Transport: TCP

Model: Request / Response

Data Format: HTML, JSON, XML

Typical Use: Web browsing, REST APIs

HTTP — Client–Server Model



Request (GET / POST / PUT / DELETE)

Client (Browser / App)

Response (Status Code + Data)

Server

Common HTTP Methods

GET | POST | PUT | DELETE



MQTT — Message Queuing Telemetry Transport

- Lightweight publish/subscribe messaging protocol
- Designed for low-bandwidth, high-latency networks
- Devices publish & subscribe via a central broker
- Widely used for IoT telemetry & sensor data

Key Facts

Layer: Application

Transport: TCP

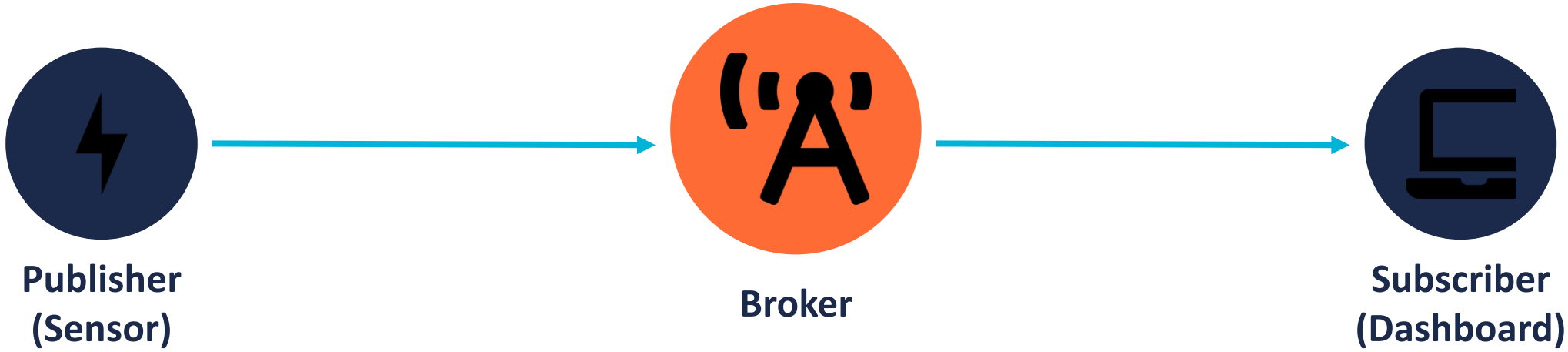
Model: Publish / Subscribe

QoS Levels: 0, 1, 2

Typical Use: IoT telemetry, remote monitoring

MQTT — Publish/Subscribe Model

Topic: "sensor/temperature"



Quality of Service (QoS)

QoS 0	QoS 1	QoS 2
At most once	At least once	Exactly once

CoAP — Constrained Application Protocol

- RESTful protocol built for constrained devices
- Uses UDP for minimal overhead & fast exchange
- Mirrors HTTP methods in a compact binary format
- Ideal for low-power, low-memory networks

Key Facts

Layer: Application

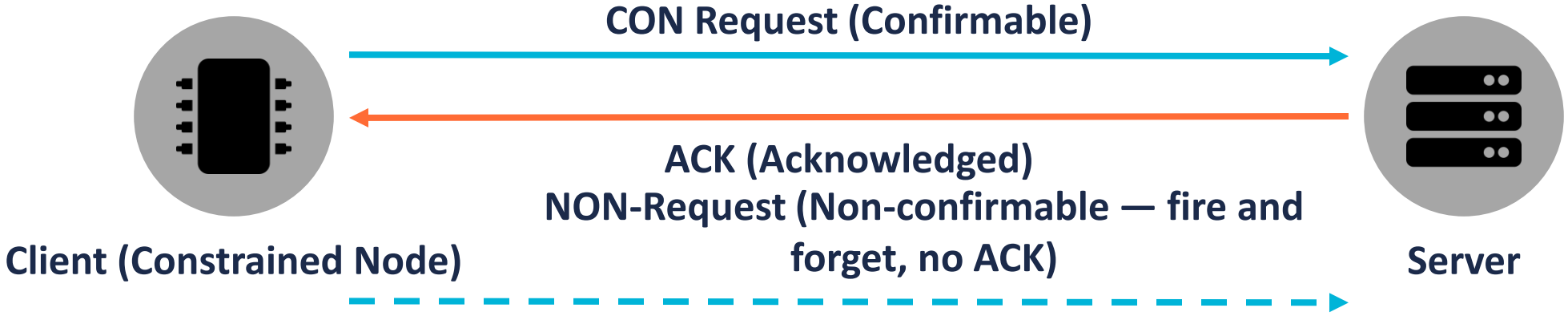
Transport: UDP

Model: Request / Response (REST)

Msg Types: Confirmable, Non-confirmable

Typical Use: Constrained IoT nodes, WSNs

CoAP — Request/Response over UDP



CoAP Methods (Binary-Encoded)

GET | POST | PUT | DELETE



ZigBee — Low-Power Wireless Mesh

- Wireless mesh networking standard (IEEE 802.15.4)
- Optimized for low power & low data rate
- Self-healing mesh extends coverage & reliability
- Popular in home automation & smart lighting

Key Facts

Layer: Network / Data Link

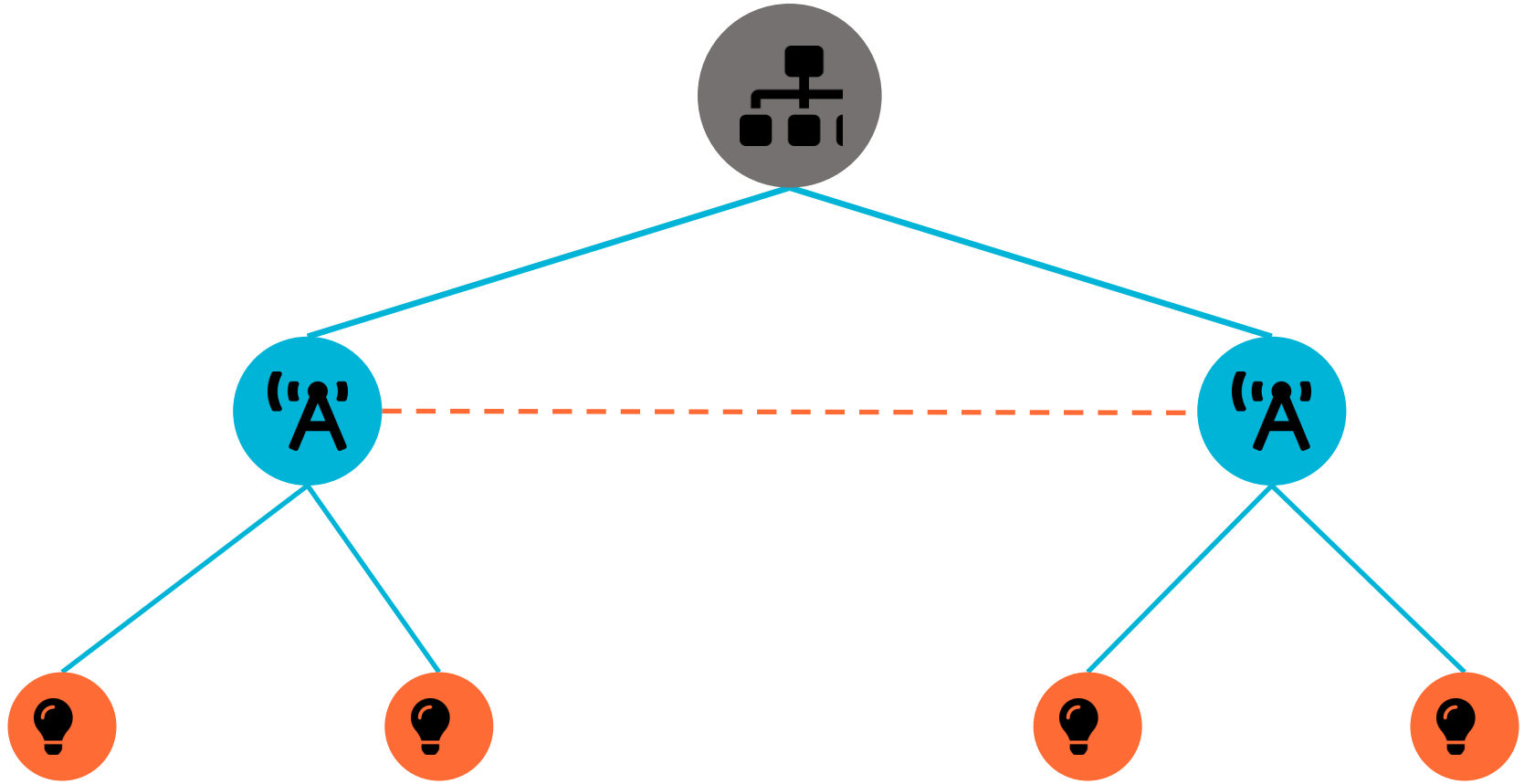
Frequency: 2.4 GHz (also 900/868 MHz)

Model: Mesh (Coordinator–Router–End Device)

Range: ~10–100 m per hop

Typical Use: Smart home, building automation

ZigBee — Mesh Network Topology



Coordinator

Router

End Device

Comparative Analysis

Parameter	HTTP	MQTT	CoAP	ZigBee
Layer	Application	Application	Application	Data Link / Network
Transport	TCP	TCP	UDP	IEEE 802.15.4
Model	Request/Response	Publish/Subscribe	Request/Response	Mesh Networking
Power Usage	High	Low	Very Low	Very Low
Overhead	High	Low	Very Low	Low
Typical Use	Web & REST APIs	IoT Messaging	Constrained IoT	Home Automation

Real-World Applications



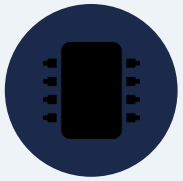
Smart Home Automation (ZigBee)

Lighting, thermostats & sensors communicating over a low-power mesh.



Industrial IoT Messaging (MQTT)

Publish/subscribe telemetry from factory sensors to cloud dashboards.



Constrained Sensor Networks (CoAP)

Battery-powered nodes exchanging RESTful data over UDP.



Web & Cloud Applications (HTTP)

Browsers, REST APIs & cloud services communicating over reliable TCP.

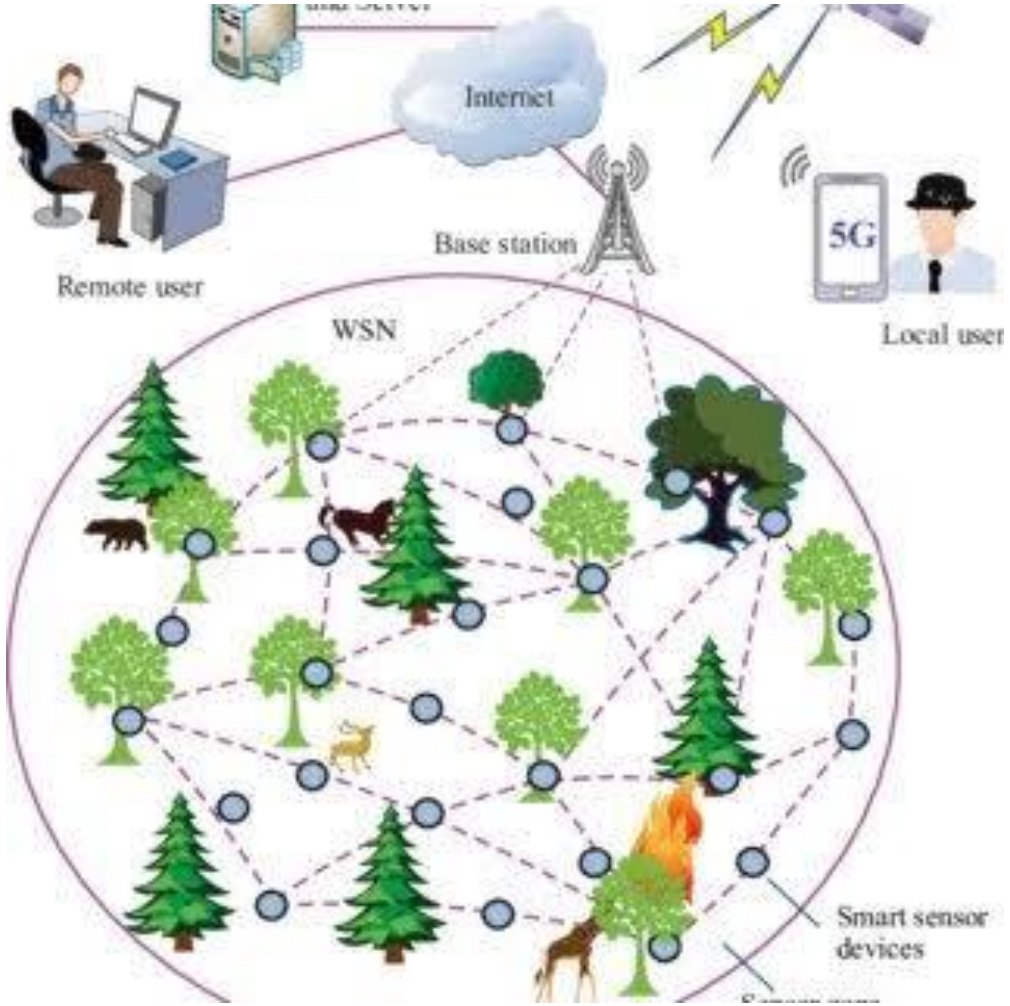
Communication Protocols: MQTT, CoAP, Zigbee, and HTTP

- **HTTP is best for traditional web communication.**
- **MQTT is best for event-driven IoT messaging.**
- **CoAP is best for very limited devices.**
- **Zigbee is best for wireless sensor and control networks.**

- Choice of protocol depends on device capability, power, range, and application.

INTRODUCTION

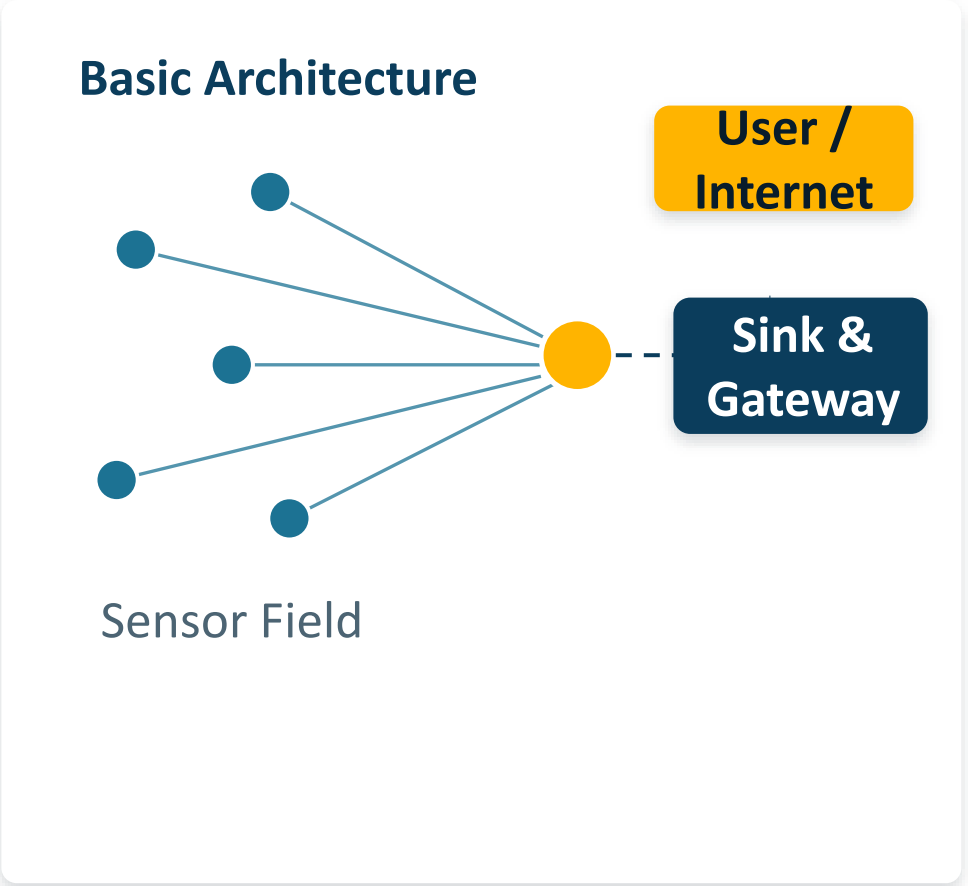
What Is a Sensor Network?



INTRODUCTION

What Is a Sensor Network?

- A collection of spatially distributed, low-power sensor nodes
- Nodes sense physical phenomena: temperature, motion, pressure, humidity
- Nodes cooperate and communicate over wired/wireless links
- Sensed data is routed toward a central sink or base station
- Enables remote monitoring, control, and decision-making



BUILDING BLOCKS

Core Components

1. **Sensor Node**

   Senses data, processes it locally, and transmits over the wireless link

2. **Sink / Base Station**

   Collects data from all nodes and forwards it beyond the network

3. **Gateway**

   Bridges the sensor network to external networks such as the Internet

4. **Task Manager**

   The end-user application that issues queries and analyzes results

CLASSIFICATION

Types of Sensor Networks

1

Terrestrial

Dense node deployment on land

2

Underground

Buried nodes monitor soil & mines

3

Underwater

Acoustic links for ocean sensing

4

Multimedia

Captures audio, video & imagery

5

Mobile

Nodes move and self-organize

CLASSIFICATION

Types at a Glance

Type	Medium	Typical Application
Terrestrial	Radio waves	Environment & agriculture monitoring
Underground	Radio (attenuated)	Soil condition & mine safety
Underwater	Acoustic waves	Ocean & pipeline monitoring
Multimedia	Radio (high bandwidth)	Surveillance & traffic monitoring
Mobile	Radio, ad-hoc routing	Target tracking & disaster response

NETWORK TOPOLOGIES

How Nodes Are Arranged

Topology shapes energy use, fault tolerance, and scalability of the network.

1

Star

All nodes connect directly to one central hub

2

Mesh

Nodes relay data through multiple neighbors

3

Tree / Cluster

Hierarchical grouping with cluster heads

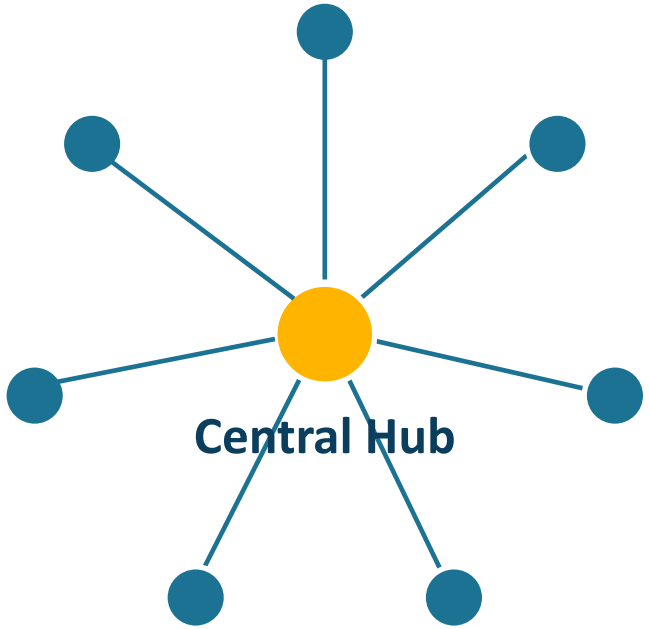
4

Hybrid

Combines two or more topologies together

TOPOLOGY 1 OF 3

Star Topology



Advantages

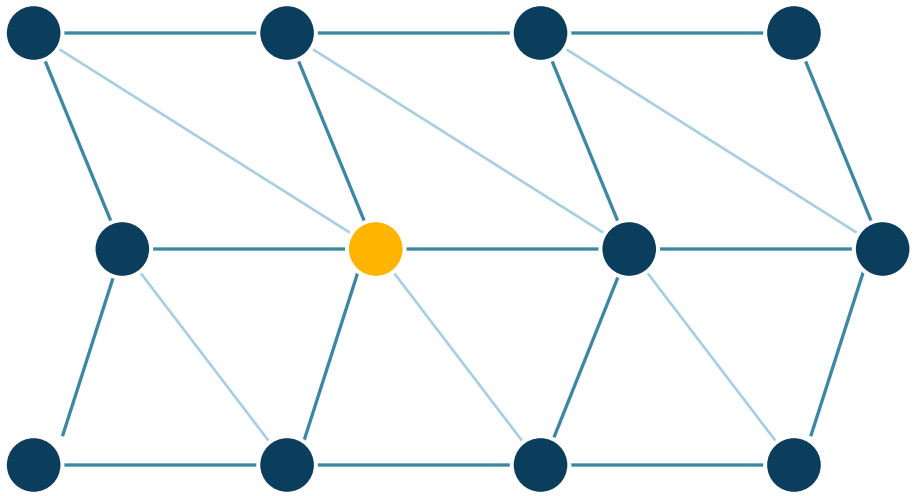
- Simple to set up and manage
- Easy fault isolation per node
- Low routing complexity

Limitations

- Hub failure disables the network
- Limited coverage range
- Hub can become a bottleneck

TOPOLOGY 2 OF 3

Mesh Topology



Every node can relay data through multiple paths

Advantages

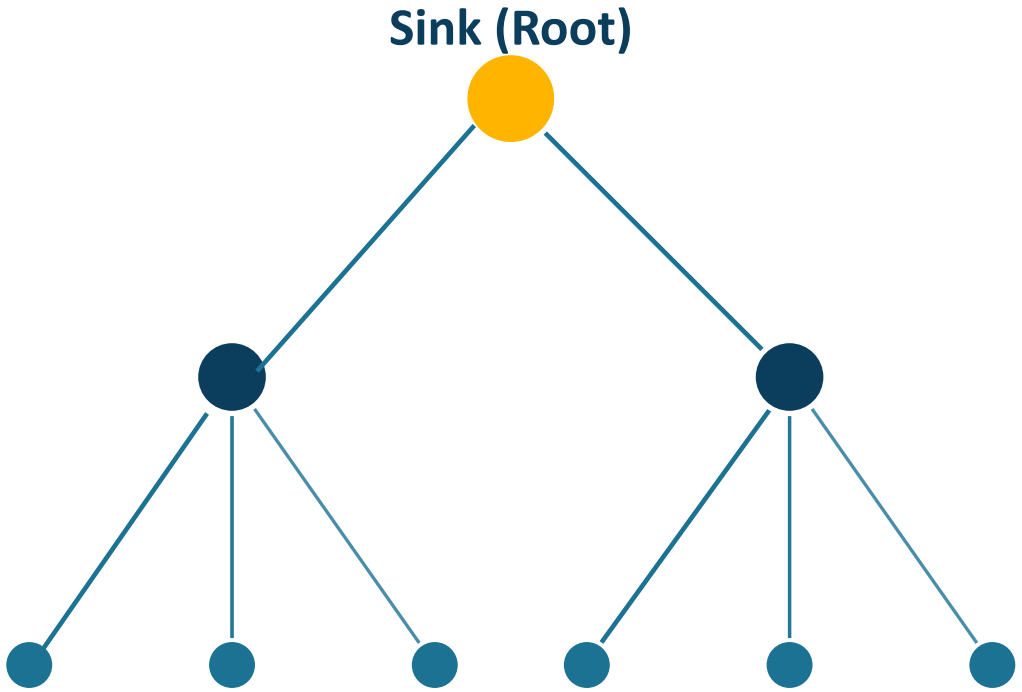
- Highly fault tolerant, self-healing
- Multiple redundant paths for data
- Scales well to more nodes

Limitations

- Higher energy consumption
- Complex routing decisions
- More overhead to maintain routes

TOPOLOGY 3 OF 3

Tree / Cluster-Tree Topology



Cluster heads aggregate data from member nodes below them

Advantages

- Efficient data aggregation at cluster heads
- Reduces long-distance transmissions
- Organized, scalable structure

Limitations

- Cluster head failure affects its branch
- Head nodes drain energy faster
- Depth adds latency to root

BRINGING IT TOGETHER

Hybrid Topology & Comparison

Hybrid topologies combine star, mesh, and tree structures to balance efficiency, reliability, and coverage in real deployments.

Criteria	Star	Mesh	Tree	Hybrid
Energy Efficiency	Medium	Low	High	High
Fault Tolerance	Low	High	Medium	High
Scalability	Low	High	High	High
Routing Complexity	Low	High	Medium	Medium

PROTOCOLS

The Sensor Network Protocol Stack

Application Layer	Delivers sensed data & queries to the user
Transport Layer	Ensures reliable end-to-end data delivery
Network Layer	Routes data from nodes toward the sink
MAC / Data Link Layer	Controls channel access between nodes
Physical Layer	Handles signal transmission over radio

PROTOCOLS

MAC Layer Protocols

S-MAC

Nodes sleep and wake on a fixed schedule to save energy

T-MAC

Adapts the wake time to traffic load for efficiency

IEEE 802.15.4

Standard defining low-rate wireless PAN access

TRAMA

Schedules time slots to avoid transmission collisions

PROTOCOLS

Routing Protocols

LEACH

Rotates cluster-head duty to balance energy use

Directed Diffusion

Data-centric queries pull data from sources

AODV

Builds on-demand routes only when needed

Flooding / Gossiping

Broadcasts data through neighbor nodes

OPEN ISSUES

Key Design Challenges

1. **Energy Constraints**
   Nodes run on limited, often irreplaceable batteries

2. **Scalability**
   Protocols must handle networks of thousands of nodes

3. **Fault Tolerance**
   The network must survive individual node failures

4. **Security**
   Wireless links are exposed to interception and attack

5. **Data Aggregation**
   Redundant readings must be combined efficiently

6. **Deployment**
   Nodes are often placed in remote, harsh environments

Introduction to IoT Security and Privacy

1. Define IoT and describe its layered architecture
2. Identify the expanded attack surface introduced by connected devices
3. Recognize common IoT vulnerabilities through real-world cases
4. Apply security-by-design and privacy-by-design principles

What Is the Internet of Things?

A network of physical objects embedded with sensors, software, and connectivity, enabling them to collect and exchange data over the internet without direct human input.

- Smart thermostats
- Wearable health monitors
- Connected vehicles
- Industrial sensors

40B+

connected IoT devices
projected worldwide by 2030

More devices means a larger attack surface

The Three-Layer IoT Architecture

1

Perception Layer

Sensors and actuators that sense and collect physical data

2

Network Layer

Gateways and protocols that transmit data between devices and systems

3

Application Layer

Software and cloud platforms that process data and deliver services

The Expanding IoT Attack Surface

1

Device

Firmware, sensors, physical access

2

Network

Wireless protocols, gateways

3

Cloud

Storage, APIs, backend services

4

Application

Mobile apps, dashboards, user interface

Every layer is a potential entry point for an attacker.

Common IoT Vulnerabilities

- Weak or default credentials left unchanged after deployment
- Unpatched firmware with no reliable update mechanism
- Insecure or undocumented APIs exposing backend systems
- Unencrypted data transmission across the network
- Lack of physical tamper protection on the device itself

Case Study: The Mirai Botnet (2016)

Malware scanned the internet for IoT devices such as cameras and routers still using factory-default passwords, then enlisted them into a botnet used to launch massive denial-of-service attacks.

- Default credentials are the single most exploited IoT weakness
- A single vulnerable device class can threaten global infrastructure
- Manufacturers share responsibility for baseline device security

600K+

devices infected worldwide

1.2 Tbps

peak traffic against a major DNS provider

The CIA Triad in an IoT Context

Confidentiality

Preventing unauthorized access to sensor data and personal information

Integrity

Ensuring data and commands are not altered in transit or storage

Availability

Keeping devices and services operational despite attack or failure

Privacy Concerns in IoT

Continuous Data Collection

Always-on sensors gather far more data than users realize

Behavioral Profiling

Aggregated data can reveal habits, routines, and location

Third-Party Sharing

Data often flows to advertisers and analytics partners

Weak User Consent

Interfaces rarely give meaningful, informed control

Security-by-Design Principles

1 Least Privilege

Grant devices and users only the access they need

2 Defense in Depth

Layer multiple safeguards so no single failure is fatal

3 Secure Boot

Verify firmware authenticity before a device starts

4 Encryption Everywhere

Protect data both at rest and in transit

Privacy-by-Design Principles

1	**Data Minimization**	Collect only what is strictly necessary for the service
2	**Purpose Limitation**	Use data only for the reason it was originally collected
3	**Anonymization**	Strip identifying details wherever possible
4	**Transparency & Consent**	Give users clear, meaningful control over their data

Standards & Regulations

GDPR

EU regulation governing collection and processing of personal data

NIST IR 8259

US federal baseline for IoT device cybersecurity capabilities

ETSI EN 303 645

European consumer IoT security standard, e.g. banning default passwords

IoT Cybersecurity Improvement Act

US law setting minimum security standards for federal IoT purchases